

FIITJEE
ALL INDIA INTEGRATED TEST SERIES
JEE (Advanced)-2024
PART TEST – III
PAPER –1
TEST DATE: 20-10-2023

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 06, 19 – 24, 37 – 42): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A (07 – 10, 25 – 28, 43 – 46): This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

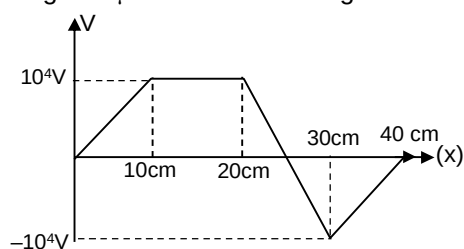
Physics

PART – I

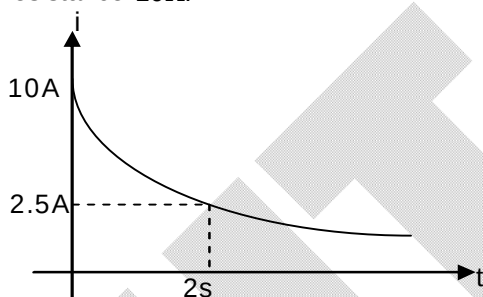
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

1. For the given potential versus distance graph for one dimensional electric field and a particle of charge $20 \mu\text{C}$ and mass 0.2 kg . Choose the correct option(s):

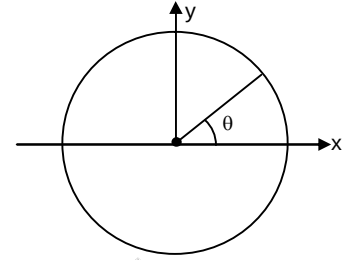


- (A) force acting on the particle at $x = 25 \text{ cm}$ is zero
 (B) Force acting on the particle at $x = 25 \text{ cm}$ is $4\hat{i}$
 (C) minimum speed at origin required by the particle to reach $x = 25 \text{ cm}$ is 2 m/s
 (D) minimum speed at origin required by the particle to reach $x = 25 \text{ cm}$ is $\sqrt{2} \text{ m/s}$
2. The figure, a graph of the current in a discharging circuit of a capacitor through a resistor of resistance 10Ω .

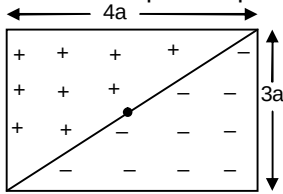


- (A) The initial potential difference across the capacitor is 100 volt .
 (B) The capacitance of the capacitor is $\frac{1}{10\ln 2} \text{ F}$.
 (C) The total heat produced in the circuit will be $\frac{500}{10\ln 2} \text{ joules}$.
 (D) The thermal power in the resistor will decrease with a time constant $\frac{1}{2\ln 2} \text{ second}$.
3. Electric potential in a region varies only with x-coordinate and is given by $v = 2x^3$. Then choose the correct option(s).
- (A) electric field in the region is $6x^2 \text{ N/C}$ along positive x-axis.
 (B) electric field in the region is $6x^2 \text{ N/C}$ along negative x-axis.
 (C) charge density in the region is constant with x-coordinate
 (D) charge density varies linearly with x-coordinate

4. On a non-conducting ring of radius R linear charge density ' λ ' given by $\lambda = \lambda_0 \cos \theta$ is distributed. Choose the correct option(s).
- (A) Electric potential at center is zero
 (B) Electric field at center is zero
 (C) Electric field at center is $\frac{\lambda_0}{4\epsilon_0}$
 (D) Dipole moment of ring is $\lambda_0 \pi R^2$



5. A non-conducting rectangular plate of M is given two charge $+Q$ and $-Q$ in two halves which is also distributed uniformly. This plate is free to rotate in its plane about an axis passing through its center. This plate is put in uniform electric field. $\vec{E}$, and it is in equilibrium.

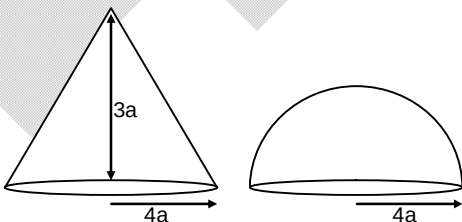


- (A) Dipole moment of plate is $\frac{5Qa}{3}$
 (B) Time period of small oscillation if disturbed from mean position is $\pi \sqrt{\frac{5ma}{QE}}$
 (C) Torque acting on plate when it is rotated through 37° is QaE
 (D) Charge in potential energy when it is rotated through 53° is QaE
6. A uniformly charged small sphere placed at point $A(h, k)$ so that at point $B(8, 7)$ electric field strength is $\vec{E} = (54\hat{i} + 72\hat{j}) \frac{N}{C}$ and potential is $+900$ volt. Then choose the correct option(s).
- (A) Charge on the sphere is $1\mu C$
 (B) Charge on the sphere is $4\mu C$
 (C) Coordinates of point A is $(2, -1)$
 (D) Coordinates of point A is $(4, -2)$

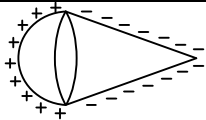
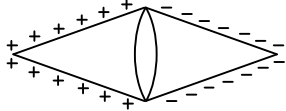
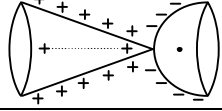
Section – A (Maximum Marks: 12)

This section contains **FOUR (04) Matching List Type Questions**. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

7. Consider a hollow cone of radius $4a$, and height $3a$, and a hemisphere shell of radius $4a$. Various combination of them are made by giving them equal and opposite charge Q . Match their dipole moment given in List II.



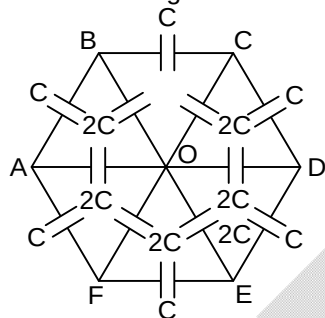
List - I		List - II	
(I)		(P)	$4Qa$

(II)		(Q)	5Qa
(III)		(R)	2Qa
(IV)		(S)	3Qa
		(T)	Zero

Codes:

	I	II	III	IV
(A)	P	R	Q	P
(B)	T	Q	R	S
(C)	P	S	R	P
(D)	Q	R	T	P

8. Match List-I against List-II referring to the given figure.

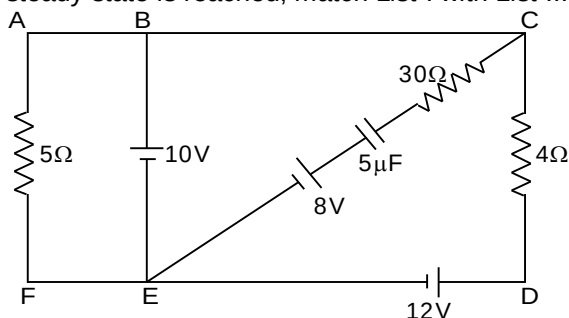


List - I		List - II	
(I)	Effective capacitance between A and D	(P)	$\frac{45}{13}C$
(II)	Effective capacitance between F and E	(Q)	$\frac{15}{8}C$
(III)	Effective capacitance between A and E	(R)	$\frac{45}{19}C$
(IV)	Effective capacitance between A and O	(S)	$\frac{9}{5}C$
		(T)	$\frac{3}{5}C$

Codes:

	I	II	III	IV
(A)	P	T	R	S
(B)	Q	P	R	S
(C)	S	R	Q	P
(D)	S	Q	R	T

9. A network consisting of three resistors, three batteries and a capacitor is shown in figure. After steady state is reached, match List-I with List-II.



List - I		List - II	
(I)	Current in branch EB is	(P)	10 μC
(II)	Current in branch CB is	(Q)	0.5 A
(III)	Current in branch ED is	(R)	1.5 A
(IV)	Charge on capacitor is	(S)	5 μC
		(T)	None of these

Codes:

	I	II	III	IV
(A)	P	Q	R	S
(B)	R	Q	Q	P
(C)	T	P	R	S
(D)	Q	R	T	P

10. A particle of charge 'q' and mass 'm' is projected in uniform magnetic field $\vec{B} = B_0(4\hat{i} + 3\hat{j})$ with velocity $V_0\hat{i}$ at $t=0$

List - I		List - II	
(I)	Velocity of particle at $t = \frac{\pi m}{5qB_0}$ is	(P)	$V_0 \left(\frac{7\hat{i} + 24\hat{j}}{25} \right)$
(II)	Velocity of particle at $t = \frac{\pi m}{10qB_0}$ is	(Q)	$V_0 \left(\frac{16\hat{i} + 12\hat{j} + 15\hat{k}}{25} \right)$
(III)	Velocity of particle at $t = \frac{3\pi m}{10qB_0}$ is	(R)	$V_0 \left(\frac{16\hat{i} + 12\hat{j} - 15\hat{k}}{25} \right)$
(IV)	Velocity of particle at $t = \frac{2\pi m}{5qB_0}$ is	(S)	$V_0\hat{i}$
		(T)	$V_0\hat{j}$

Codes:

	I	II	III	IV
(A)	Q	R	T	P
(B)	P	Q	R	S
(C)	P	R	S	Q
(D)	R	T	Q	P

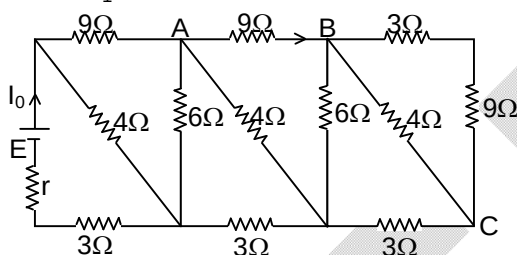
Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

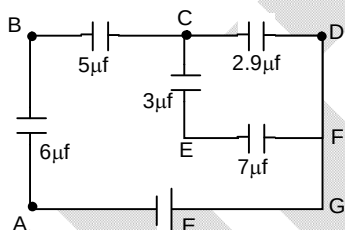
11. A rod of length ' ℓ ' and area ' A ' has resistivity which varies as $\rho = \rho_0 \left(1 + \frac{x}{\ell}\right)$. Rod is divided into three equal lengths of $\frac{\ell}{3}$ each. If resistance of first part $R_1 = 4.2\Omega$. Find resistance of middle part R_2 (in Ω).



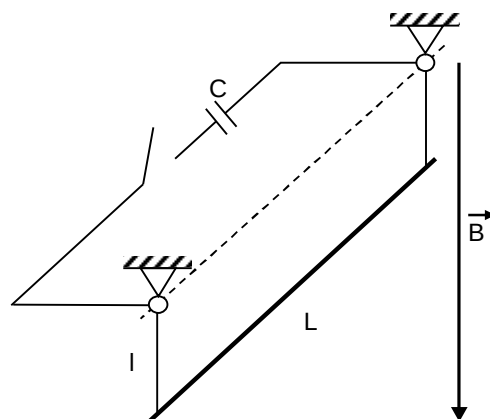
12. In the circuit shown if current flowing through battery is I_0 and current in branch AB is I_1 . Find the ratio of $\frac{I_0}{I_1}$.



13. Consider the circuit shown. If in steady state $V_A - V_C = 11$ volt. Find $V_E - V_G$ in steady state.

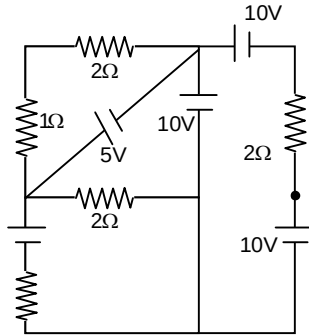


14. Metal rod with a mass $m = 10g$, and length $L = 0.2m$ is suspended by two light wires $l = 1cm$ in a magnetic field induction $B = 1T$ which is directed vertically downwards (figure). A capacitor of capacitance $C = 100\mu F$ charged to a voltage of $100V$ is connected as shown. Determine the maximum deflection (in degree) of the rod from the initial position after the switch is closed. Resistance of wire and rod is not taken into account. (Neglect induction effect)

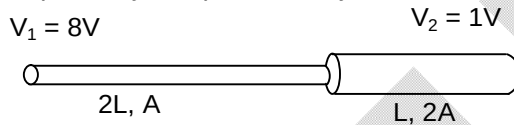


15. A cylindrical conductor of length L and Area A has variable electrical resistivity $\sigma = \sigma_0 \left(1 + \frac{x}{L}\right)$. Where x is distance from one end. The end at $x = 0$ is given a potential V_0 where end at $x = L$ is grounded. If electric field at $x = \frac{L}{2}$ is given by $E = \frac{V_0}{\alpha L \cdot \ln \beta}$, find $(\alpha + \beta)$.

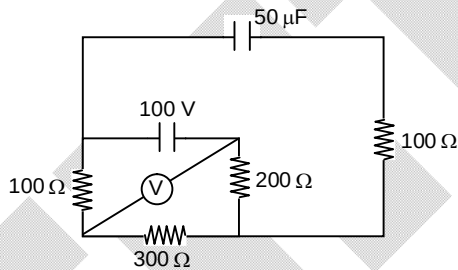
16. In the circuit shown, find the current through 1Ω resistor.



17. Two wires of same material but one of length $2L$ and area A whereas other of length L and area $2A$ are joined as shown in figure. If two ends are maintained at potential $V_1 = 8V$ and $V_2 = 1V$ respectively find potential of junction.



18. In the given figure resistance of voltmeter is 100Ω . In steady state potential difference across capacitor is $\frac{200}{n}$ volts. Find n .



Chemistry

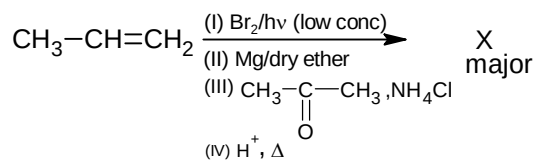
PART – II

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

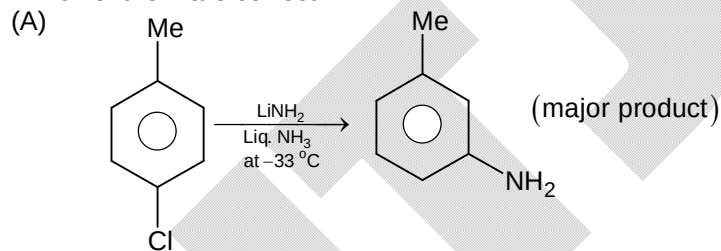
19. Monomer unit of Nylon – 6,6 are
 (A) Adipic acid (B) Caprolactum
 (C) Terylene (D) 1, 6 – Hexamethylene diamine

20. About the structure in the reaction of the product X

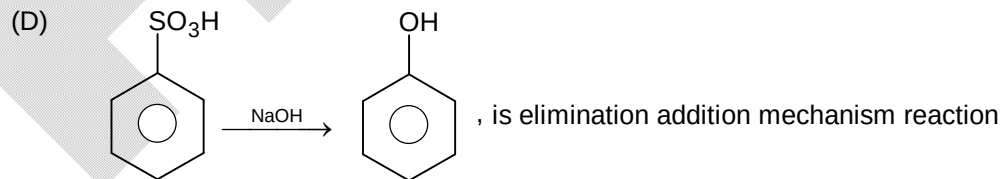
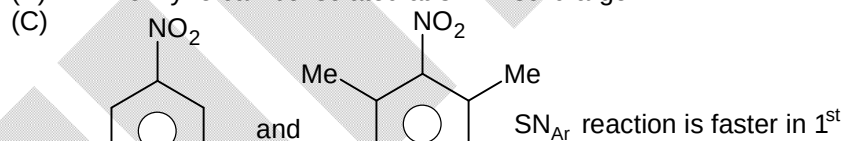


X will

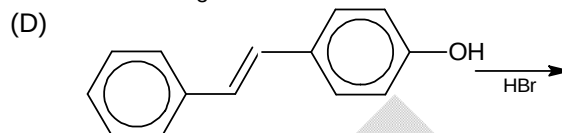
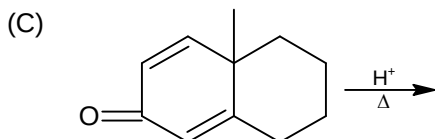
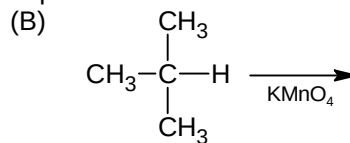
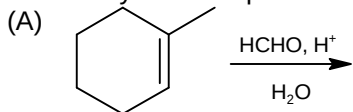
- (A) Total geometrical isomers by x = 4
 (B) Forms 3 products after reductive ozonolysis
 (C) Forms 4 different products after reaction with Baeyer's reagent
 (D) After reduction with 1 eq. H₂ and Wilkinson catalyst, the product will not show geometrical isomerism.
21. Which of them are correct?



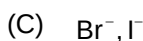
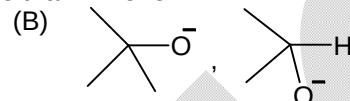
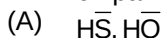
- (B) Benzyne can be isolated at 8 K in solid argon



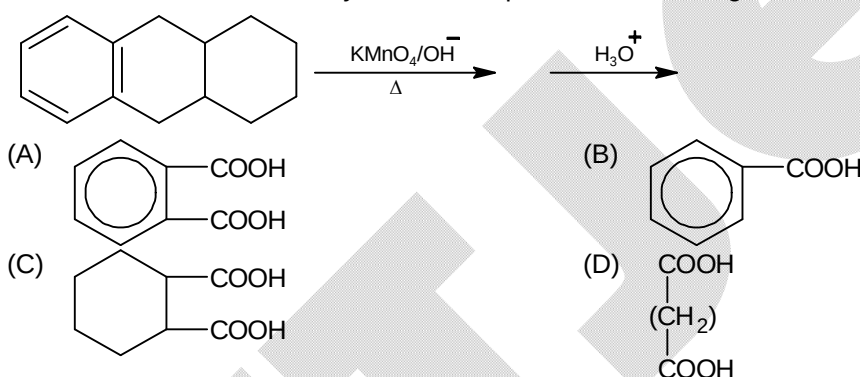
22. How many will form product having one OH groups?



23. In which pair the first one is weaker nucleophile than 2nd one?



24. Number of different dicarboxylic acids are possible in following reactions



Section – A (Maximum Marks: 12)

This section contains **FOUR (04) Matching List Type Questions**. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

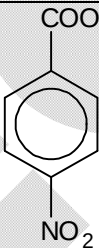
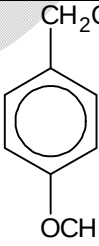
25. Match List-I with List-II:

List - I		List - II	
(I)	Aniline	(P)	Sodium extract of compound give colour with sodium nitro pruside
(II)	Crystein	(Q)	Bromo substitution with $\text{Br}_2/\text{AlCl}_3$
(III)	Terylene	(R)	Can react with CH_3MgBr
(IV)	Xylene	(S)	Formed from propyne
		(T)	Gives white ppt with Br_2/water

Codes:

	I	II	III	IV
(A)	SP	P	RS	QS
(B)	T	P	RS	QS
(C)	SR	RS	P	QS
(D)	T	P	P	QS

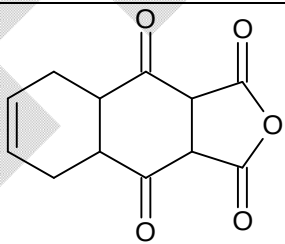
26. Match List-I with List-II:

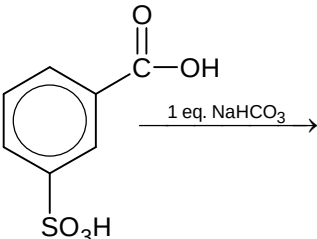
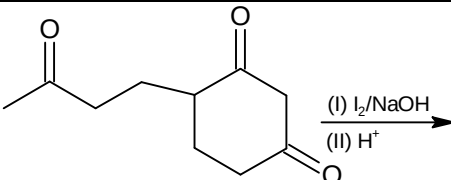
List - I		List - II	
(I)	$\text{F}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H} + \text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H} \xrightarrow[\text{H}_3\text{O}^+]{\text{(i) conc. NaOH}} \text{cross product}$	(P)	HCOOH
(II)	$\text{C}_6\text{H}_5\text{CHO} + \text{HCHO} \xrightarrow[\text{H}_3\text{O}^+]{\text{conc. NaOH}} \text{cross product}$	(Q)	C ₆ H ₅ COOH
(III)	$\text{C}_6\text{H}_5\text{CHO} + \text{p-NO}_2\text{C}_6\text{H}_4\text{CHO} \xrightarrow[\text{H}_3\text{O}^+]{\text{Conc. NaOH}} \text{Cross product}$	(R)	C ₆ H ₅ CH ₂ OH
(IV)	$\text{p-NO}_2\text{C}_6\text{H}_4\text{CHO} + \text{p-MeOC}_6\text{H}_4\text{CHO} \xrightarrow[\text{H}_3\text{O}^+]{\text{Conc NaOH}} \text{cross product}$	(S)	
		(T)	

Codes:

	I	II	III	IV
(A)	P	PR	Q	S
(B)	P	PR	S	Q
(C)	P	PR	QS	ST
(D)	RP	QPR	S	T

27. The product in List I will match with List II:

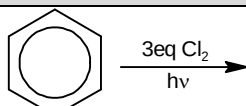
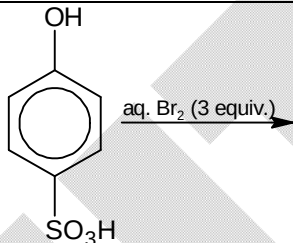
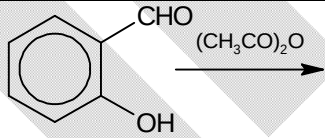
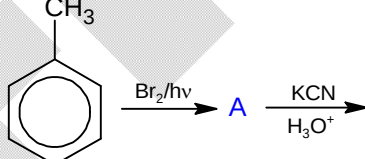
List - I		List - II	
(I)	$\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH} \xrightarrow{\text{heat}} \text{A} \xrightarrow[\text{H}^+]{\text{I}_2/\text{NaOH}} \text{B}$	(P)	Loss in weight of the reactant in complete reaction
(II)	 $\xrightarrow[\text{(II) O}_3, \text{H}_2\text{O}_2]{\text{(I) H}_3\text{O}^+, \Delta}$	(Q)	Product gives effervescence with NaHCO ₃

(III)		(R)	Yellow ppt. forms
(IV)		(S)	Product has more than one stereoisomers.
		(T)	Gas is evolved in reaction

Codes:

	I	II	III	IV
(A)	PQ	PQ	PQT	ST
(B)	PQRT	PQS	PQS	QST
(C)	PQRT	PQST	QT	PQRS
(D)	PQ	QT	QT	ST

28. Match List-I with List-II:

List - I		List - II	
(I)		(P)	Electrophilic aromatic substitution
(II)		(Q)	Addition reaction
(III)		(R)	Nucleophilic addition elimination
(IV)		(S)	Increase in molecular weight of product than reactant
		(T)	Product has more degree of unsaturation than reactant

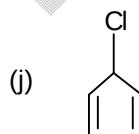
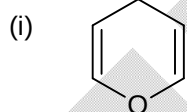
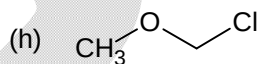
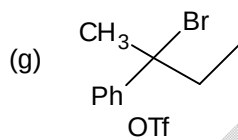
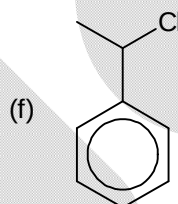
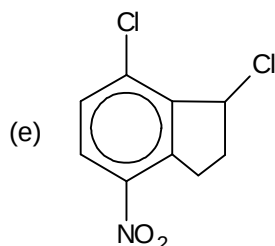
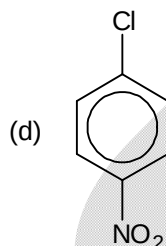
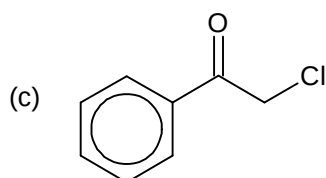
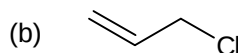
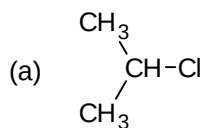
Codes:

	I	II	III	IV
(A)	QS	PS	RST	ST
(B)	QST	P	RT	S
(C)	Q	PS	RST	T
(D)	QT	PST	RS	ST

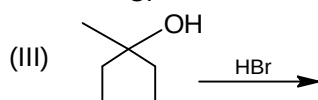
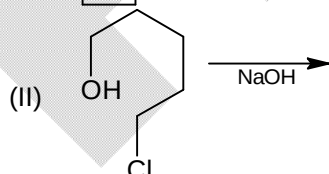
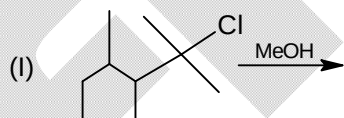
Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

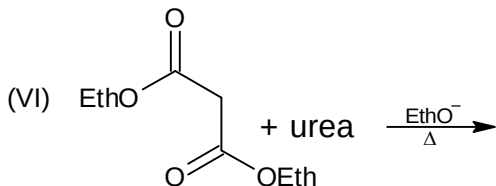
29. If X is number of molecules giving SN^2 , then $\frac{X}{3}$ will be:



30. Number of reactions giving cyclic product as major product is X, then $\frac{3X}{5}$ is



(V) Malonic acid $\xrightarrow{\Delta}$



31. Find sum of $\left(\frac{X+Y+Z}{3}\right)$ if

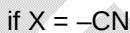
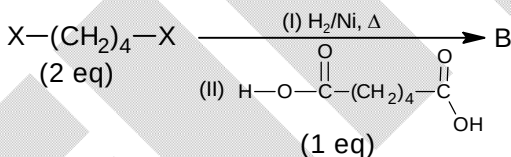
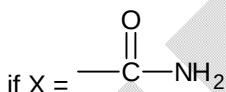
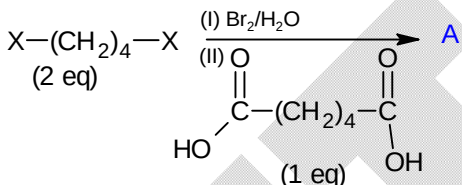
(a) X is number of chiral carbon in β -D glucose

(b) Y is number of molecules of phenyl hydrazine used in Osazone formation from glucose

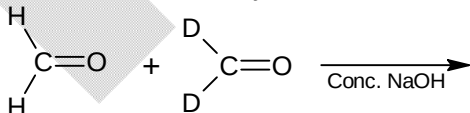
(c) Z is number of isomeric alkenes formed if $\text{Ph}-\overset{\text{CH}_3}{\underset{\text{OH}}{\text{CH}}}-\text{CH}-\text{CH}_3$ is heated with conc. H_2SO_4

32. In Duma's method for estimation of N_2 , 0.25 gm of an organic compound gave 40 ml of nitrogen collected at 300 K temperature and 750 mm pressure. If aqueous tension at 300 K is 50 mm, the percentage of nitrogen in the compound is

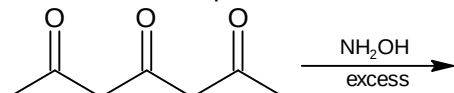
33. Find the ratio of $\frac{3}{2} \left(\frac{\text{Mw of A}}{\text{Mw of B}} \right)$



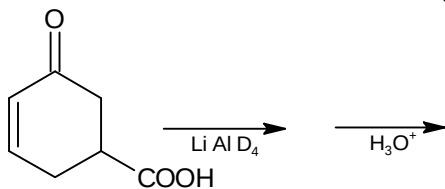
34. Find value of $\frac{X}{4}$, if X is the molecular weight of alcohol with highest molecular weight in following Cannizaro reaction by



35. Find ratio of lone pairs to number of isomers of the product formed.



36. Find ratio of number of deuterium to degree of unsaturation in product of



Mathematics**PART – III****Section – A (Maximum Marks: 24)**

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

37. If $S = x^2 + y^2$ where $3x^2 + 4xy + 3y^2 = 1$ then
 (A) $S_{\max} = 1$ (B) $S_{\max} = 5$
 (C) $S_{\min} = \frac{1}{5}$ (D) $S_{\min} = \frac{1}{\sqrt{5}}$
38. Let $f(x) = \frac{1-x^{n+1}}{1-x}$ and $g(x) = 1 + \frac{2}{x} + \frac{3}{x^2} + \dots + (-1)^n \frac{n+1}{x^n}$ then the constant term in $f'(x) \times g(x)$ is equal to
 (A) $\frac{n(n^2-1)}{6}$ when n is even (B) $\frac{n(n+1)}{2}$ when n is odd
 (C) $-\frac{n}{2}(n+1)$ when n is even (D) $\frac{n(n-1)}{2}$ when n is odd
39. $f(x) = \frac{\left[\frac{1}{2} + x\right] - \left[\frac{1}{2}\right]}{x}$, $-1 \leq x \leq 2$ has ([.] denotes greatest integer function)
 (A) discontinuity at $x = 0$ (B) discontinuity at $x = \frac{1}{2}$
 (C) discontinuity at $x = 1$ (D) discontinuity at $x = \frac{3}{2}$
40. The value of α which satisfy $\int_{\frac{\pi}{2}}^{\alpha} \sin x dx = \sin 2\alpha$ ($\alpha \in [0, 2\pi]$) are equal to
 (A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{2}$
 (C) $\frac{7\pi}{6}$ (D) $\frac{11\pi}{6}$
41. If $\int \frac{\cos 4x + 1}{\cot x - \tan x} dx = Af(x) + B$, then
 (A) $Ax^2 + Bx + 1 = 0$ must have real and distinct roots
 (B) $f(x)$ is an even function
 (C) $A = -\frac{1}{8}$
 (D) $\frac{1}{8}$

42. Let $f(x)$ be a differentiable function and $f(a) = f(b) = 0$ ($a < b$), then in the interval (a, b) .
- (A) $f(x) + f'(x) = 0$ has atleast one root.
- (B) $f(x) - f'(x) = 0$ has atleast one root.
- (C) $f(x) \times f'(x) = 0$ has atleast one root
- (D) None of these

Section – A (Maximum Marks: 12)

This section contains **FOUR (04) Matching List Type Questions**. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

43. Match List-I with List-II:

List - I		List - II	
(I)	If $\lim_{x \rightarrow 0} \frac{4 + \sin 2x + A \sin x + B \cos x}{x^2}$ exists, then $A - B$ is equal to	(P)	-1
(II)	$\lim_{n \rightarrow \infty} \int_0^2 \frac{\left(1 + \frac{t}{n+1}\right)^n}{n+1} dt$ is equal to $e^A - B$, then $A - B$ is equal to	(Q)	0
(III)	If $f(x) = [2 + 3 \sin x]$, $0 < x < \pi$, where $[.]$ denotes GIF, then number of points at which function is not differentiable, is	(R)	2
(IV)	$\lim_{x \rightarrow -\infty} \left(x + \sqrt{x^2 + 2x}\right)$ is equal to	(S)	1
		(T)	5

Codes:

	I	II	III	IV
(A)	R	Q	S	T
(B)	R	S	T	P
(C)	S	Q	R	P
(D)	P	R	T	Q

44. Match List-I with List-II:

List - I		List - II	
(I)	$\lim_{x \rightarrow \infty} \frac{\ln x}{x} \int_{e^3}^x \frac{2dt}{\ln t}$ is	(P)	1
(II)	The number of solution of the equation $\sin^{-1} x - 2\cos^{-1}(1+x) = 0$ is	(Q)	0
(III)	If $f(x) = x - 2\sin x$, $0 \leq x \leq 2\pi$ is increasing in the interval $[a\pi, b\pi]$ then $(a+b)^2$ is	(R)	2
(IV)	The intercept of the common tangent to the curves $y^2 = 8x$ and $xy = -1$ on the axis of y is equal to k, then $k - 2$ is	(S)	4
		(T)	3

Codes:

	I	II	III	IV
(A)	R	P	S	Q
(B)	R	S	T	Q
(C)	T	R	S	P
(D)	S	T	P	R

45. Match List-I with List-II:

List - I		List - II	
(I)	If $x^2 + y^2 + z^2 - 2xyz = 1$, then the value of $\frac{dx}{\sqrt{1-x^2}} + \frac{dy}{\sqrt{1-y^2}} + \frac{dz}{\sqrt{1-z^2}}$	(P)	0
(II)	The length of longest interval in which the function $f(x) = 3\sin x - 4\sin^3 x$ is increasing is $\frac{a\pi}{6}$, then value of a is	(Q)	8
(III)	$\int_{-2}^2 \frac{3x^2}{1+e^x} dx$ is equal to	(R)	2
(IV)	If $I = \frac{2}{\pi} \int_{-\pi/4}^{\pi/4} \frac{dx}{(1+e^{\sin x})(2-\cos 2x)}$, then $27(I)^2$ equals	(S)	3
		(T)	4

Codes:

	I	II	III	IV
(A)	P	T	S	R
(B)	S	Q	R	T
(C)	P	R	Q	T
(D)	T	R	Q	P

46. Match List-I with List-II:

List - I		List - II	
(I)	Let $f(x)$ be a function on $(-\infty, \infty)$ and $f(x+2) = f(x-2)$. If $f(x) = 0$ has only three real roots in $[0, 4]$ and one of them is 4, then the number of real roots of $f(x) = 0$ in $(-8, 10]$, is	(P)	4
(II)	Let $r_1, r_2, r_3, \dots, r_n$ be n positive integers, not necessarily distinct, such that $(x+r_1)(x+r_2)\dots(x+r_n) = x^n + 56x^{n-1} + \dots + 2009$. The possible value of n is	(Q)	5
(III)	If x and y are positive integers and $2xy = 2009 - 3y$, then the number of ordered pairs of (x, y) , is	(R)	8
(IV)	If $x, y \in \mathbb{R}$, satisfying the equation $\frac{(x-4)^2}{4} + \frac{y^2}{9} = 1$, then the difference between the largest and smallest value of the expression $\frac{x^2}{4} + \frac{y^2}{9}$, is	(S)	9
		(T)	6

Codes:

	I	II	III	IV
(A)	R	P	Q	S
(B)	R	P	T	Q
(C)	S	Q	P	R
(D)	S	P	Q	R

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

47. If $f(x) = \begin{cases} \frac{36^x - 9^x - 4^x + 1}{\sqrt{2} - \sqrt{1 + \cos x}}, & x \neq 0 \\ \lambda, & x = 0 \end{cases}$ is continuous at $x = 0$, then $\lambda = \sqrt{\mu} \ln 2 \cdot \ln 3$ then the value of μ must be:
48. If $y = \sin x - \cos x$ and if $f_k(x) = \frac{d^k y}{dx^k}$ then the value of $\left(\frac{2024}{\sqrt{2}}\right) f_{2025}(x)$ must be.
49. If the parabola $y = ax^2 + bx + c$ has vertex at $(4, 2)$ and $a \in [1, 3]$. Then difference between the extreme values of abc is equal to
50. The number of points of extreme of the function $F(x) = \int_0^x \frac{\sin t}{t} dt$ in the interval $(0, 2023)$ must be
51. If $\int \frac{\ln x}{x^3} dx = -\frac{\ln x}{ax^2} - \frac{1}{bx^2} + c$, then the value of $a^5 + b^4 + a^3 + b^2 + a + 1$ is
52. If $f(x) = x - x^2 + 1$ and $g(x) = \max\{f(t) : 0 \leq t \leq x\}$, then $\int_0^1 g(x) dx = k$, then $6k$ is equal to
53. Let $f(x)$ denote the sum of the infinite trigonometric series,

$$f(x) = \sum_{n=1}^{\infty} \sin \frac{2x}{3^n} \sin \frac{x}{3^n}.$$
The sum of the solutions of the equation $f(x) = 0$ lying in the interval $(0, 629)$ is $K\pi$ then $K = ?$
54. Given a relation $R = \{(1,2), (2,3)\}$ on the set of natural numbers. Minimum number of ordered pairs required so that the enlarged relation is symmetric, transitive & reflexive is K , then K is:

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JEE (Advanced)-2024
PART TEST – III
PAPER –1
TEST DATE: 20-10-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. BD

Sol. $F = qE = -q \frac{dv}{dx}$

$$\frac{1}{2}mv_0^2 = q(10^4)$$

2. ABCD

Sol. $V_0 = I_0 R = 10 \times 10 = 100$ volts (since, $I_0 = 10$ amp from figure).

Hence (A) is correct.

Also: $I = I_0 e^{-t/RC}$

Taking log; $\ln\left(\frac{I_0}{I}\right) = \frac{t}{RC}$

$$\Rightarrow C = \frac{1}{R \ln\left(\frac{I_0}{I}\right)}$$

At $t = 2$ sec, $I = 2.5$ A

$$C = \frac{2}{10 \ln\left(\frac{10}{2.5}\right)}$$

$$C = \frac{2}{10 \ln 4} = \frac{2}{10 \times 2 \ln 2} = \frac{1}{10 \ln 2}$$

Hence (B) is correct.

$$C = \frac{1}{10 \ln 2}$$

$$\text{Heat produced} = \frac{1}{2} CV^2 = \frac{1}{2} \left(\frac{1}{10 \ln 2} \right) (100)^2$$

$$= \frac{500}{\ln 2} \text{ joule}$$

Hence, (C) is correct

Thermal power in the resistor will decrease with a time constant $\frac{1}{2\ln 2}$ second.

Hence, (D) is correct.

3. BD

Sol. $E = -\frac{dV}{dx} = -6x^2 \hat{i}$

4. ACD

Sol. $E_x = \frac{k}{R^2} \int_0^{2\pi} (\lambda_0 \cos \theta) R d\theta \cdot \cos \theta = \frac{k\lambda_0}{R} \times \pi$

5. ABCD

Sol. Dipole moment $Q \times 2 \times \frac{5a}{2} \times \frac{1}{3} = \frac{5Qa}{3}$

$$\vec{\tau} = \vec{p} \times \vec{E} \Rightarrow \tau = pE \sin \theta$$

$$U = -\vec{p} \cdot \vec{E}$$

6. AC

Sol. $\vec{E} = \frac{k \cdot q (\vec{r}_2 - \vec{r}_1)}{|\vec{r}_2 - \vec{r}_1|^3}$

$$V = \frac{k \cdot q}{|\vec{r}_2 - \vec{r}_1|}$$

7. C

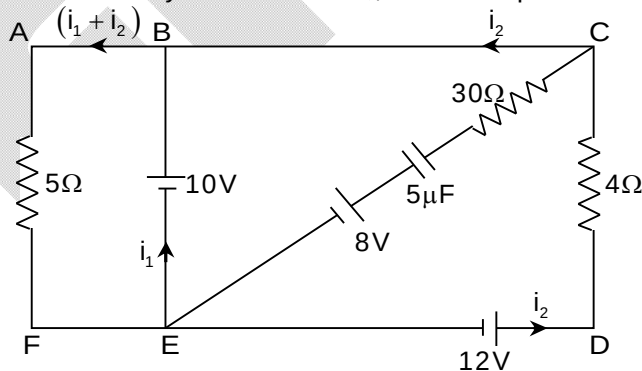
Sol. Basic concept.

8. C

Sol. Basic concept of combination and symmetry.

9. B

Sol. When a steady state is reached, no current passes through the capacitor or the branch CE.



Considering the loop BCDEB $4i_2 = 12 - 10 = 2$

$$\Rightarrow i_2 = 0.5A \text{ So } i_1 = 2 - 0.5 = 1.5A$$

To find the charge on capacitors, we must know potential difference across the plates.

Consider the loop CEDC:

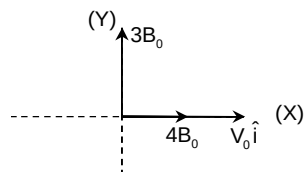
$$-12 + 4i_2 + 3 \times 0 - V_C + 8 = 0$$

$$\Rightarrow V_C = -2V$$

So charge on capacitor $Q = CV = 10\mu\text{C}$

10. B

Sol.



$$\vec{V}_{\parallel} = \text{constant}$$

$\vec{V}_{\perp}$ changes with time.

Section - B

11. 5.40

Sol.

$$R = \int \frac{\rho dx}{A}$$

$$= \int \frac{\rho_0}{A} \left(1 + \frac{x}{\ell}\right) dx$$

$$R_1 = \frac{\rho_0}{A} \left(x + \frac{x^2}{2\ell} \right)_0^{\ell}$$

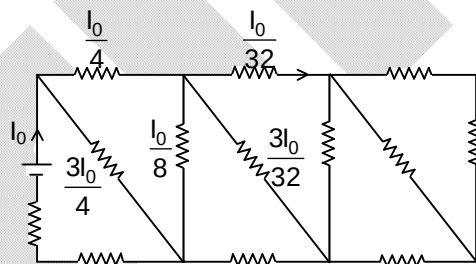
$$R_2 = \frac{\rho_0}{A} \left(x + \frac{x^2}{2\ell} \right)_{\ell/3}^{2\ell/3}$$

12. 32.00

Sol.

$$R_{\text{eq}} = 6\Omega + r$$

$$I_0 = \frac{E}{6 + r}$$



13. 1.80

$$\text{Sol. } = \frac{\frac{30}{11} \times 5}{\frac{30}{11} + 5} = \frac{150}{85}$$

$$Q = \frac{150}{85} \cdot E ; V_A - V_C = \frac{Q}{\left(\frac{30}{11}\right)} = \frac{150E \times 11}{85 \times 30} = 11 \text{ volt}$$

$$E = 17 \text{ Volt.}$$

$$\frac{Q-q}{2.9} = \frac{q}{2.1} \Rightarrow V_I - V_F = \frac{q}{7}$$

14. 37.00

Sol. $F = iLB$

$$mv = \int F dt = \int (idt)LB$$

$$m \cdot \sqrt{2gh} = QLB$$

$$m\sqrt{2gl(1 - \cos\theta)} = QLB$$

$$\theta = 37^\circ$$

15. 3.50

$$\text{Sol. } R = \int_0^L \frac{dx}{\sigma A} = \frac{L}{\sigma_0 A} \ln 2$$

$$j = \frac{V_0}{A \cdot R} = \frac{V_0 \sigma_0}{L \cdot \ln 2}$$

$$E = \frac{j}{\sigma} = \frac{V_0}{L \left(1 + \frac{x}{L}\right) \ln 2}; \text{ at mid-point } E = \frac{V_0}{1.5L \cdot \ln 2}$$

16. 2.50

Sol. Applying Kirchhoff's law we can find current.

17. 2.40

$$\text{Sol. } \frac{V_1 - V}{\left(\frac{\rho \cdot 2L}{A}\right)} = \frac{V - V_2}{\left(\frac{\rho \cdot L}{2A}\right)}$$

$$\frac{V_1 - V}{2} = 2(V - V_2) \Rightarrow V = \frac{V_1 + 4V_2}{5} = \frac{8 + 4}{5} = 2.4$$

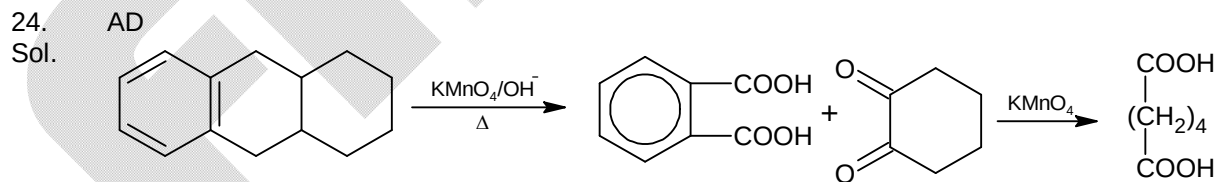
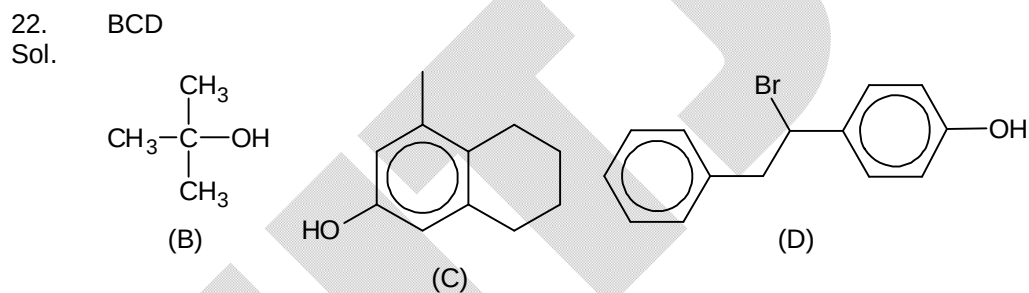
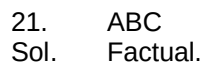
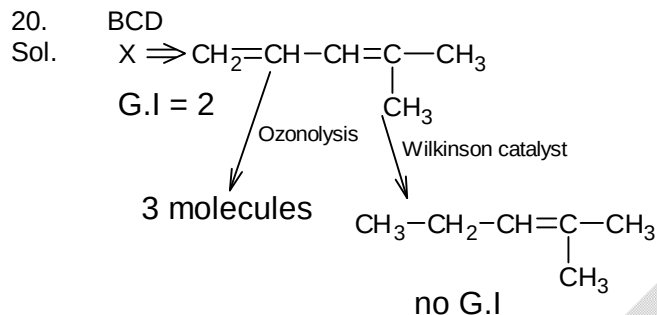
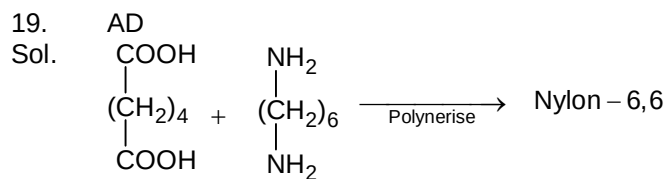
18. 11.00

Sol. In steady state no current flow through capacitor.

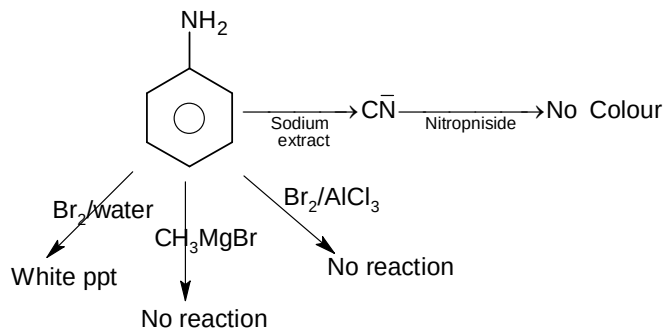
Chemistry

PART – II

Section – A

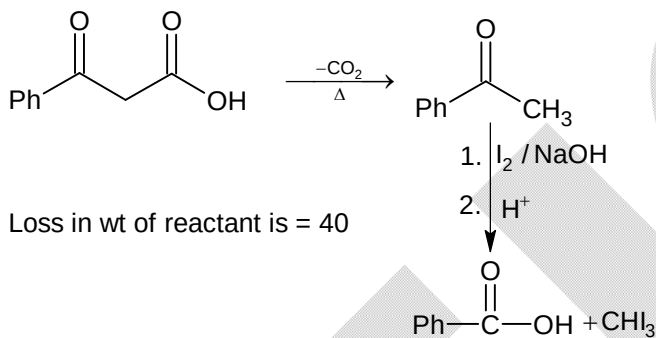


25. B
Sol.

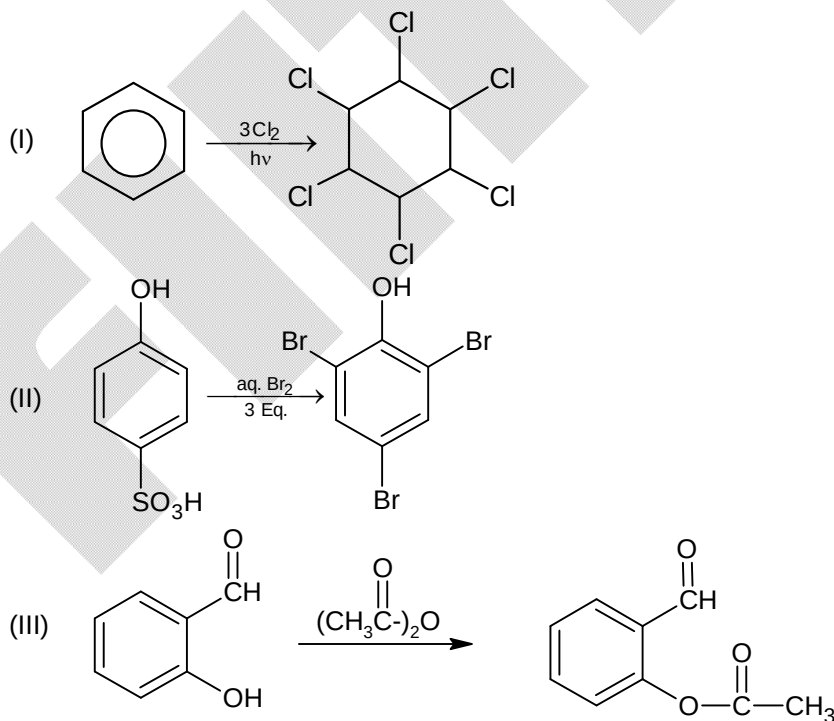


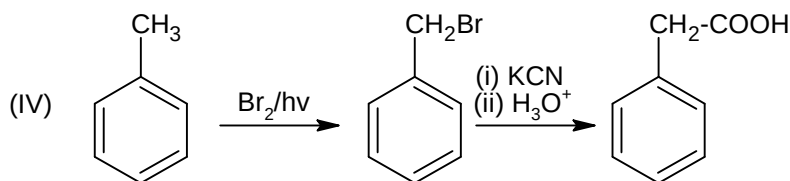
26. C
Sol. By property of inter and intra Cannizzaro.

27. C
Sol.



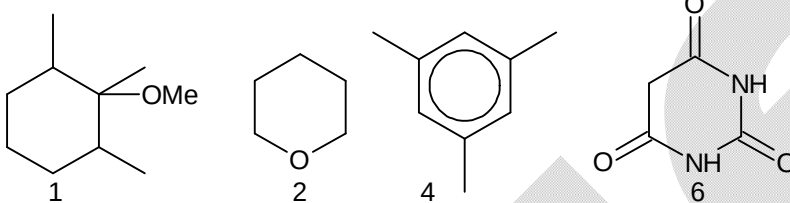
28. A
Sol.





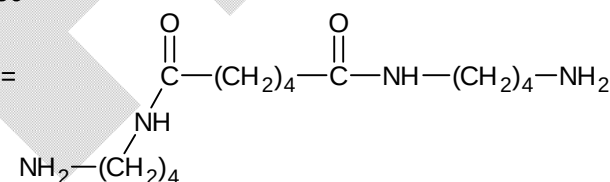
Section – B

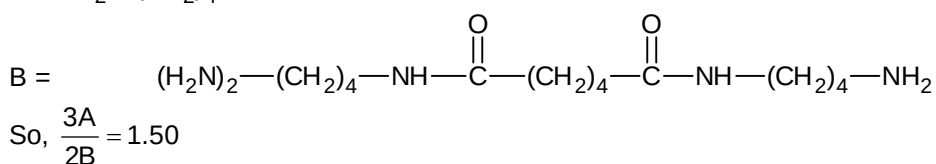
29. 2.66
Sol. SN^2 by all except = d,g
So, $X = 8$ so, $\frac{X}{3} = 2.66$

30. 2.40
Sol. 
So, $\frac{3x}{5} = \frac{12}{5} = 2.40$

31. 4.33
Sol. $X = 5$ $Y = 3$ $Z = 5$
So, $\frac{X+Y+Z}{3} = \frac{13}{3} = 4.33$

32. 16.73
Sol. Moles of $N_2 = \frac{PV}{RT}$
 $= \frac{700 \times 40 \times 10^{-3}}{760 \times 0.0821 \times 300} = 1.4 \times 10^{-3}$
Wt. of $N_2 = 28 \times 1.4 \times 10^{-3}$
 $= 0.041 \text{ gm}$
 $\% N = \frac{0.041}{.25} \times 100 = 16.73$

33. 1.50
Sol. $A =$ 



34. 8.75

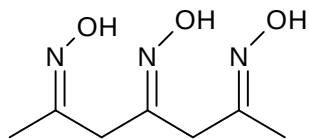
Sol. CD_3OH is the alcohol with highest M.W

So, $X = 35$

$$\frac{X}{4} = 8.75$$

35. 1.50

Sol.



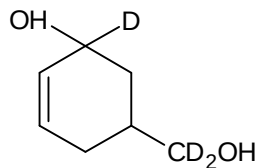
So, lone pair = 9

Isomers = 6

36. 1.50

Sol.

Product is



42. ABC

Sol. Consider $g(x) = e^x f(x)$

Then $g(a) = g(b)$

From Rolle's theorem there exists at least one $x = C$ in (a, b)

Such that $g'(c) = 0$

$$\Rightarrow e^c [f(c) + f'(c)] = 0$$

Similarly consider $g(x) = e^{-x} f(x)$ & $g(x) = (f(x))^2$

Use Rolle's theorem.

43. B

Sol. (I) $\lim_{x \rightarrow 0} \frac{4 + \sin 2x + A \sin x + B \cos x}{x^2}$ exists

Then, $4 + B = 0$, $B = -4$

Using L-Hopital rule,

$$\lim_{x \rightarrow 0} \frac{2 \cos 2x + A \cos x - B \sin x}{2x}$$

$$1 + \frac{A}{2} - \frac{B}{2} = 0$$

$$1 + \frac{A}{2} + 2 = 0$$

$$\frac{A}{2} = -1$$

$$A = -2$$

$$A - B = -2 - (-4) = 2$$

$$(II) \lim_{n \rightarrow \infty} \left[\frac{\int_0^2 \left(1 + \frac{t}{n+1}\right)^n dt}{n+1} \right] = \lim_{n \rightarrow \infty} \left[\left(1 + \frac{t}{n+1}\right)^{n+1} \right]_0^2$$

$$= \lim_{n \rightarrow \infty} \left(1 + \frac{2}{n+1}\right)^{n+1} - 1$$

$$= e^2 - 1$$

(III) $0 < 3 \sin x \leq 3$ for $x \in (0, \pi)$

Then, $2 < 2 + 3 \sin x \leq 5$

Hence, $[2 + 3 \sin x]$ is discontinuous at 5 points.

$$(IV) \lim_{x \rightarrow -\infty} (x + \sqrt{x^2 + 2x}) = \lim_{x \rightarrow -\infty} \frac{x^2 - (x^2 + 2x)}{x - \sqrt{x^2 + 2x}}$$

$$= \lim_{x \rightarrow -\infty} \frac{-2x}{x - |x| \sqrt{1 + \frac{2}{x}}} = -1$$

44. A

Sol. (I) $\lim_{x \rightarrow \infty} 2 \left[\frac{\int_{e^3}^x \frac{1}{\ln t} dt}{\frac{x}{\ln x}} \right]$

From L-Hospital rule,

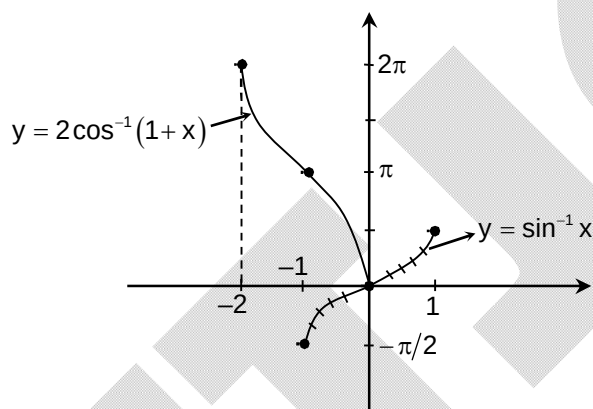
$$= \lim_{x \rightarrow \infty} 2 \frac{\frac{1}{\ln x}}{\frac{\ln x - x \cdot \frac{1}{x}}{(\ln x)^2}}$$

$$\Rightarrow \lim_{x \rightarrow \infty} 2 \frac{\ln x}{\ln x - 1} = 2 \times 1 = 2$$

(II) $\sin^{-1} x - 2\cos^{-1}(1+x) = 0$

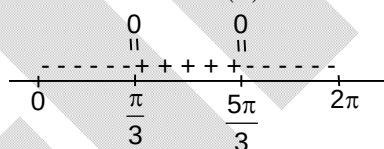
Only one solution

$$\sin^{-1} x = 2\cos^{-1}(1+x)$$



(III) $f(x) = x - 2\sin x$

$$f'(x) = 1 - 2\cos x$$

Sign scheme of $f'(x)$ 

Hence, $a = \frac{1}{3}$ & $b = \frac{5}{3}$

So, $(a+b)^2 = 4$

(IV) Length of intercept is 2.

45. C

Sol. (I) $x^2 + y^2 + z^2 - 2xyz = 1$

Differentiate

$$\Rightarrow 2xdx + 2ydy + 2zdz - 2yzdx - 2xzdy - 2xydz = 0$$

$$\Rightarrow xdx + ydy + zdz - yzdx - xzdy - xydz = 0$$

$$\Rightarrow (x - yz)dx + (y - xz)dy + (z - xy)dz = 0 \quad \dots\dots\dots (i)$$

$$\text{As } (x - yz)^2 = x^2 - 2xyz + y^2z^2 = 1 - y^2 - z^2 + y^2z^2$$

$$(x - yz)^2 = (1 - y^2)(1 - z^2)$$

$$\text{Similarly, } (y - xz)^2 = (1 - x^2)(1 - z^2)$$

$$\& (z - xy)^2 = (1 - x^2)(1 - y^2)$$

Put and find.

$$(II) \quad f(x) = 3 \sin x - 4 \sin^3 x = \sin 3x$$

Largest interval in which it increases is $\frac{\pi}{3}$.

So, $a = 2$

$$(III) \quad \text{If } I = \int_{-2}^2 \frac{3x^2}{(1+e^x)} dx = \int_{-2}^2 \frac{3(-x)^2}{1+e^{-x}} dx$$

$$2I = \int_{-2}^2 3x^2 dx = 16$$

$$I = 8$$

$$(IV) \quad I = \frac{2}{\pi} \int_{-\pi/4}^{\pi/4} \frac{dx}{(1+e^{\sin x})(2-\cos 2x)} = \frac{2}{\pi} \int_{-\pi/4}^{\pi/4} \frac{dx}{(1+e^{-\sin x})(2-\cos 2x)}$$

$$2I = \frac{2}{\pi} \int_{-\pi/4}^{\pi/4} \frac{dx}{2-\cos 2x}$$

$$I = \frac{2}{\pi} \int_0^{\pi/4} \frac{dx}{2-\cos 2x} = \frac{2}{3\sqrt{3}}$$

46. A

$$\text{Sol. (I) } f(x+2) = f(x-2)$$

$$\Rightarrow f(x+4) = f(x)$$

$f(x)$ is periodic with period 4.

Hence roots in $[0, 4)$ are $0, \alpha, 4$, where $\alpha \in (0, 4)$.

$$(II) \quad \text{Here, } r_1 + r_2 + \dots + r_n = 56$$

$$\text{and } r_1 r_2 \dots r_n = 2009 = 1 \times 7 \times 7 \times 41$$

$$\text{So, } n = 4$$

$$(III) \quad 2xy = 2009 - 3y$$

$$y(2x+3) = 2009 = 1 \times 7 \times 7 \times 41$$

Ordered pairs are $(1003, 1), (142, 7), (19, 49), (23, 41), (2, 287)$.

Hence, 5 number of ordered pairs.

$$(IV) \quad \text{Put } x = 4 + 2\cos\theta$$

$$\text{and } y = 3\sin\theta$$

Section - B

47. 512.00

Sol. f is continuous at $x = 0$ then $f(0) = \text{RHL}$

$$\Rightarrow \lambda = \lim_{h \rightarrow 0} \left\{ \frac{36^h - 9^h - 4^h + 1}{\sqrt{2} - \sqrt{1 + \cosh h}} \right\}$$

$$\lambda = \lim_{h \rightarrow 0} \frac{\left(\frac{9^h - 1}{h}\right) \left(\frac{4^h - 1}{h}\right) (\sqrt{2} + \sqrt{1 + \cosh})}{\frac{1 - \cosh}{h^2}}$$

$$\lambda = \frac{\ln 9 \cdot \ln 4 \cdot 2\sqrt{2}}{\frac{1}{2}}$$

$$\lambda = \sqrt{512} \ln 2 \cdot \ln 3$$

48. 2024.00

Sol. Differentiation of $\sin x$ and $\cos x$ repeat after fourth order then $f_{2025}(x) = f_1(x) = \cos x + \sin x$

$$f_1\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}$$

49. 3456.00

Sol. $y = ax^2 + bx + c$

$$\text{vertex} = \left(-\frac{b}{2a}, \frac{4ac - b^2}{4a}\right) = (4, 2)$$

$$\Rightarrow -b = 8a \text{ and } 4ac - b^2 = 8a$$

$$c = 2 + 16a$$

$$\Delta = abc = -16(a^2 + 8a^3)$$

$$\Delta_{\max} = -144 \quad \Delta_{\min} = -3600$$

$$\text{Difference} = 3456$$

50. 644.00

Sol. $F'(x) = \frac{\sin x}{x}$

$$\Rightarrow F'(x) = 0 \Rightarrow \sin x = 0, \quad x \neq 0$$

$$\therefore x = \pi, 2\pi, 3\pi, \dots, 644\pi, \text{ all lies between 0 and 2023}$$

$$\text{Number of point} = 644$$

51. 315.00

Sol. $I = \int \frac{\ln x}{x^3} dx = \int (\ln x) \cdot \left(\frac{1}{x^3}\right) dx = \frac{-\ln x}{2x^2} - \frac{1}{4x^2} + c$ (using by parts),

$$a = 2 \quad b = 4.$$

52. 7.25

Sol. $f(x)$ is increasing in $\left(0, \frac{1}{2}\right)$ and decreasing in $\left(\frac{1}{2}, 1\right)$

$$\text{So, } g(x) = \begin{cases} x - x^2 + 1 & 0 \leq x < \frac{1}{2} \\ \frac{1}{2} - \frac{1}{4} + 1 & \frac{1}{2} \leq x \leq 1 \end{cases}$$

$$g(x) = \begin{cases} x - x^2 + 1 & 0 \leq x < 1 \\ \frac{5}{4} & \frac{1}{2} \leq x \leq 1 \end{cases}$$

$$\int_0^1 g(x) dx = \int_0^{\frac{1}{2}} (x - x^2 + 1) dx + \int_{\frac{1}{2}}^1 \frac{5}{4} dx = \frac{29}{24}$$

53. 10100.00

Sol. $f(x) = \sum_{n=1}^{\infty} \sin \frac{2x}{3^n} \cdot \sin \frac{x}{3^n}$

$$f(x) = \frac{1}{2} \sum_{n=1}^{\infty} \left[\cos \frac{x}{3^n} - \cos \frac{x}{3^{n-1}} \right]$$

$$f(x) = \lim_{n \rightarrow \infty} \frac{1}{2} \left[\cos \frac{x}{3^n} - \cos x \right] = \frac{1}{2} [1 - \cos x]$$

So, $f(x) = 0 \Rightarrow \cos x = 1 \Rightarrow x = 2n\pi, n \in \mathbb{I}$

The sum of the solution in $(0, 629)$

$$S = 2[\pi + 2\pi + 3\pi + \dots + 100\pi] = 10100\pi$$

$$K = 10100.$$

54. 9.00

Sol. Relation is [(1, 1) (1, 2) (2, 1) (2, 3) (3, 2) (1, 3) (3, 1) (2, 2) (3, 3)]

FIITJEE
ALL INDIA INTEGRATED TEST SERIES
JEE (Advanced)-2024
PART TEST – III
PAPER –2
TEST DATE: 20-10-2023

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 04, 19 – 22, 37 – 40): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A (05 – 10, 23 – 28, 41 – 46): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	–1	In all other cases.

Physics

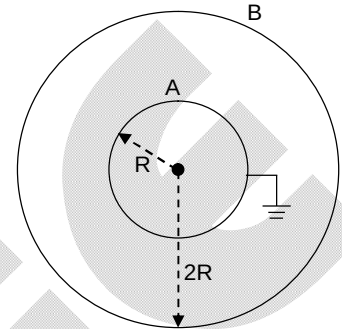
PART – I

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

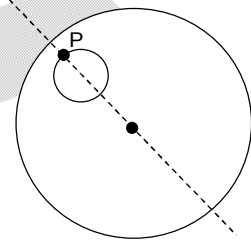
1. Two concentric conducting shells are given. Shell 'A' is earthed whereas shell 'B' is given charge Q_0 . Both shells start expanding at $t = 0$ with speed $v_A = 2v_0$ and $v_B = v_0$ outwards. What will be current through earthing wire at $t = 0$?

- (A) $\frac{3Q_0 v_0}{4R}$
 (B) $\frac{3Q_0 v_0}{2R}$
 (C) $\frac{Q_0 v_0}{4R}$
 (D) None of these



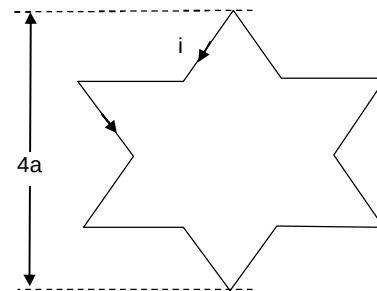
2. Consider an infinite long cylindrical wire of radius a . It has a cylindrical cavity of radius $a/3$ distance between axis of cylindrical wire and cavity is $a/2$. If this wire carries a current of current density 'J', find magnetic field at point P.

- (A) $\frac{13}{36} \mu_0 J a$
 (B) $\frac{\mu_0 J a}{4}$
 (C) $\frac{8}{15} \mu_0 J a$
 (D) None of these



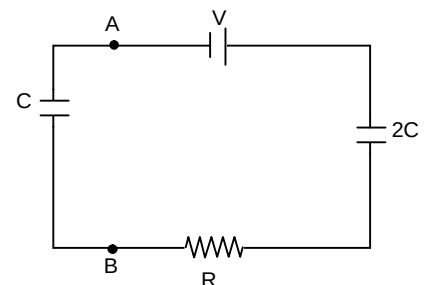
3. What is magnetic dipole moment of the loop?

- (A) $4a^2 i$
 (B) $4\sqrt{3} a^2 i$
 (C) $8\sqrt{3} a^2 i$
 (D) None of these



4. What amount of heat will be generated when the long charged capacitor between A and B is shorted by a conducting wire?

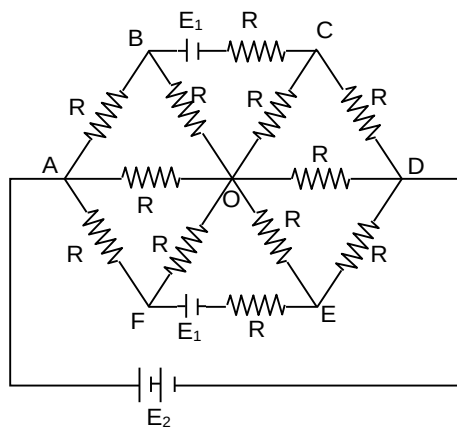
- (A) $\frac{CV^2}{3}$
 (B) $\frac{2CV^2}{3}$
 (C) $\frac{CV^2}{2}$
 (D) None of these



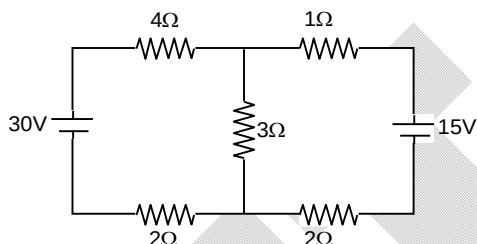
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. All resistors are of resistance 5Ω , $E_1 = 5V$ and $E_2 = 10V$. Internal resistance of cells is negligible.

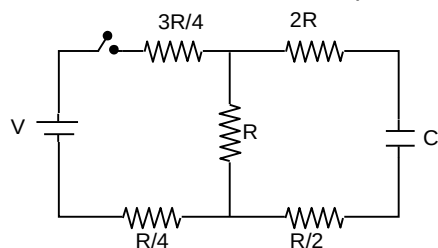


- (A) current through AO is 1 A
(B) current through BC is 0 A
(C) current through E_2 is 2 A
(D) current through OC is 0.5 A
6. For the given circuit choose the correct option(s):

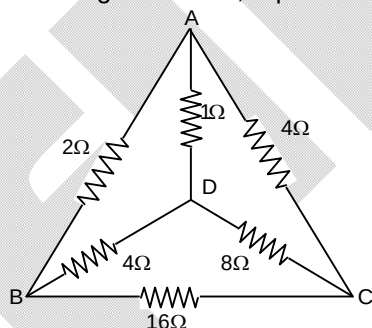


- (A) Power delivered by 30 V battery is 90 W
(B) Power delivered by 15 V battery is 15 W
(C) Current through 3Ω resistor is 3 A
(D) Power loss in 1Ω resistor is 1 W
7. Current density in a cylindrical wire is varying with radial distance r as $J = \frac{J_0}{R^2} r(R-r)$, where R is radius of wire. If B is magnetic field due to this wire, then choose the correct option(s).
- (A) maximum magnetic field will be at a distance $\frac{8R}{9}$
(B) maximum magnetic field will be at a distance R
(C) maximum magnetic field will be $\frac{\mu_0 J_0 R}{12}$
(D) maximum magnetic field will be $\frac{64\mu_0 J_0 R}{729}$

8. Circuit is closed at $t = 0$ and capacitor was uncharged initially. Choose the correct option(s):



- (A) charge on the capacitor is $q = \frac{CV}{2} \left(1 - e^{-\frac{t}{3RC}} \right)$
- (B) charge on the capacitor is $q = \frac{CV}{2} \left(1 - e^{-\frac{2t}{5RC}} \right)$
- (C) current in resistor R is $i = \frac{V}{2R} \left(1 - e^{-\frac{2t}{5RC}} \right)$
- (D) current in resistor R is $i = \frac{V}{2R} \left(1 - e^{-\frac{t}{3RC}} \right)$
9. A parallel plate capacitor of plate Area 'A'. Plate separation 'd' is carrying a charge Q_0 . A dielectric slab, of dielectric constant 'k', completely fills the space between plates. If E_0 , V_0 and C_0 are electric field potential energy and capacitor before insertion of slab and E , V and C are after insertion, Then, chose correct option.
- (A) $E = \frac{E_0}{k}$
- (B) $C = kC_0$
- (C) $V = \frac{V_0}{k}$
- (D) Induced charge on slab face has magnitude of $Q_0 \left(1 - \frac{1}{k} \right)$
10. For the given circuit, equivalent resistance between terminals.

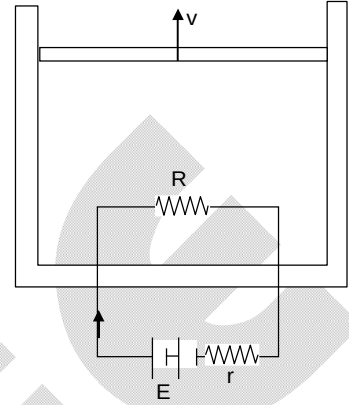


- (A) B and C is $\frac{16}{5} \Omega$
- (B) A and D is $\frac{4}{5} \Omega$
- (C) D and C is $\frac{8}{3} \Omega$
- (D) A and C is $\frac{12}{5} \Omega$

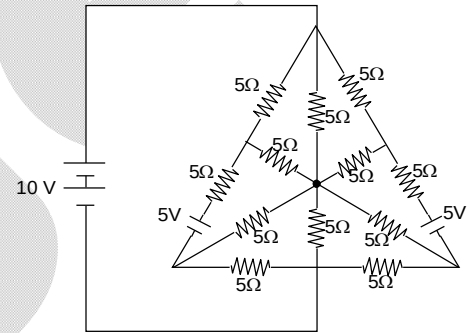
Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

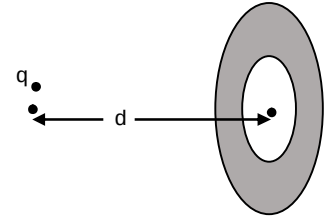
11. A resistance coil with battery is placed inside thermally insulated cylinder fitted with a frictionless piston. Cylinder contains an ideal diatomic gas and mass of piston is 10 kg. Current flowing in the circuit is 100 mA and $R = 1400 \Omega$. If pressure of gas remains constant and atmospheric pressure is 10^5 Pa. Find the speed in cm/sec with which piston is moving out.



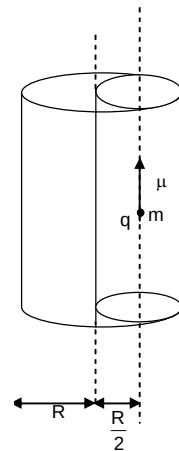
12. Find the current provided by 5 V cell in the given circuit (internal resistance of each cell is zero).



13. A charge q_0 is placed at a distance of $d = 12$ cm from the hollow disc of inner radius 9 cm and outer radius 16 cm. If flux of electric field through the disc area is $\frac{n \cdot q_0}{5 \epsilon_0}$, find n .

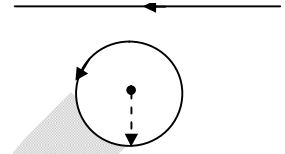


14. Inside a very long cylindrical wire of radius 'R' there is long cylindrical cavity of radius ' $\frac{R}{2}$ '. Axes of cylinder and cylindrical cavity are parallel and separated by distance ' $\frac{R}{2}$ '. A particle with charge q and mass m is thrown from the axis of cavity parallel to it with small speed ' μ '. Time period of motion is given by $N \cdot \frac{\pi m}{q \mu_0 J R}$. Find N .

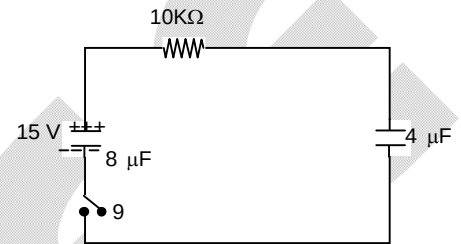


15. Let there be a spherically symmetric charge distribution with charge density varying as $\rho(r) = \rho_0 \left(\frac{5}{4} - \frac{r}{R} \right)$ up to $r = R$. The electric field at a distance $r = \frac{R}{2}$ is given by $E = \frac{x\rho_0 R}{48\epsilon_0}$. Find the value of x .

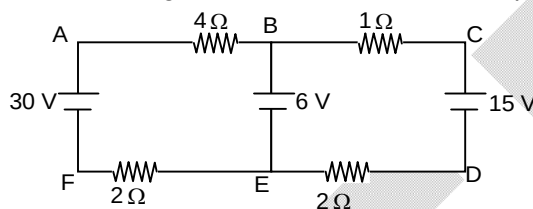
16. Radius of a coil of wire with N turns is 0.22 m and a current of 3.5A flows in clockwise direction in it. Another long straight wire carrying a current 54 A⁰ towards left is located 5 cm from the edge of the coil. Magnetic field at the center of the coil is zero. The number of turns in the coil are $N = ?$



17. A capacitor of $8\mu\text{F}$ is charged at 15 V and connected to circuit as shown in figure. After closing the circuit heat lost in the resistance is x mJ. Find the value of $10x$.



18. Current through Branch BE in the circuit is (in Ampere)



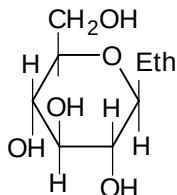
Chemistry

PART – II

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

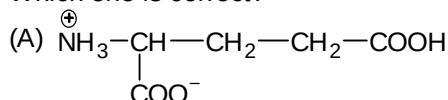
19. Which is correct about



- (A) It is a monomer of cellulose
(C) It will go under mutarotation

- (B) It's a reducing agent
(D) It will react with 4 moles of $(\text{CH}_3\text{CO})_2\text{O}$

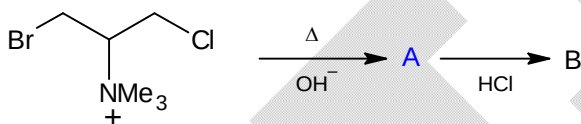
20. Which one is correct?



have 3 acidic group present

- (B) Lysin have two $-\text{COOH}$ group present
(C) During the polymerization of Bakelite C – C bond formation happens
(D) Polyethylene is a step growth polymer

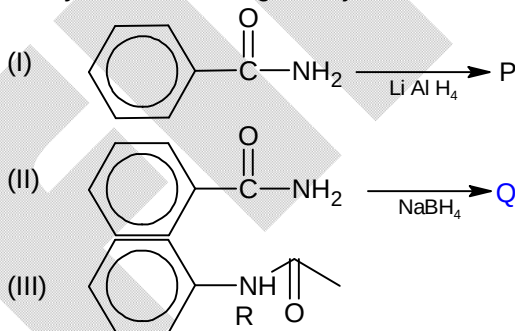
- 21.



Product B has

- (A) One chiral carbon
(C) It gives positive test with Lucas reagent
(B) It's a vicinal dichloride
(D) it's a gem dichloride

22. Carbyl amine test is given by



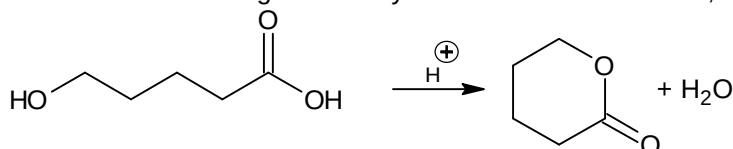
- (A) Only P
(C) P, Q, R

- (B) P and Q
(D) None of these

Section – A (Maximum Marks: 24)

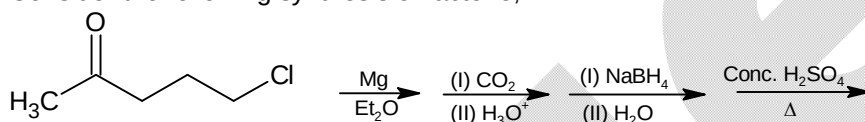
This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

23. Consider the following acid catalyzed esterification reaction,



The correct statement concerning the above reaction is/are

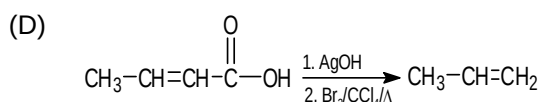
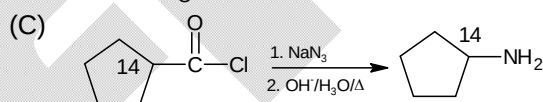
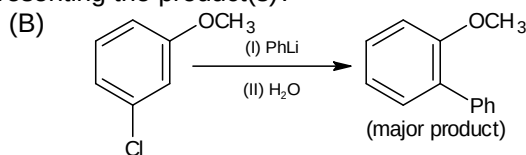
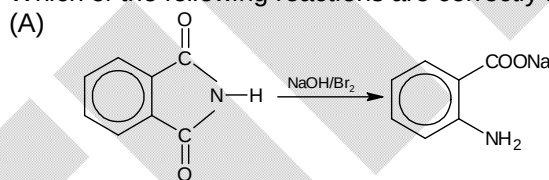
- (A) The above ester is the major product rather than the one formed through intermolecular condensation
 (B) If the hydroxyl (-OH) group of alcohol contain O^{18} , radioactivity will be retained in the ester
 (C) If the sp^2 oxygen of acid is O^{18} , it will essentially be absent in the product ester
 (D) Intramolecular condensation of a hydroxyl acid is always favoured over intermolecular condensation
24. Consider the following synthesis of lactone,



Expected product(s) is/are

- (A)
- (B)
- (C)
- (D)

25. Which of the following reactions are correctly representing the product(s)?

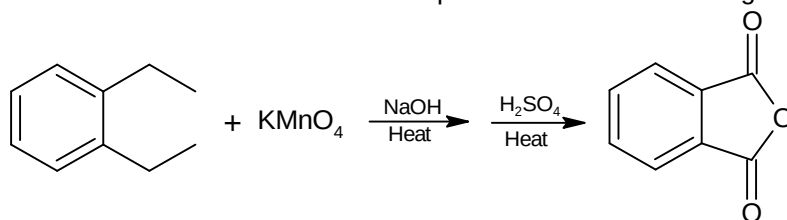


26. Which of the following statements is/are true?

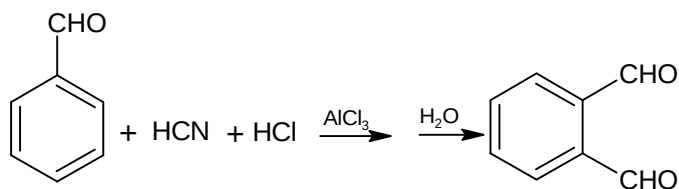
- (A) Major part of energy needed for the heterolysis of C – X bond in S_N1 reaction mechanism is obtained by the solvation of X^- by the polar protic solvent.
 (B) Greater solvation of attacking nucleophile & poor solvation of transition state promotes the reaction by S_N2 mechanism.
 (C) Increase in the number of phenyl ring at C of C – X causes a shift in mechanism from S_N2 to S_N1
 (D) Allyl & benzyl halides undergo S_N2 reactions as they are primary halides which do not give S_N1 reaction

27. Choose the correct match of reactants and products from the following list:

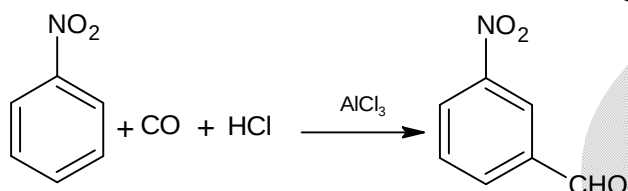
(A)



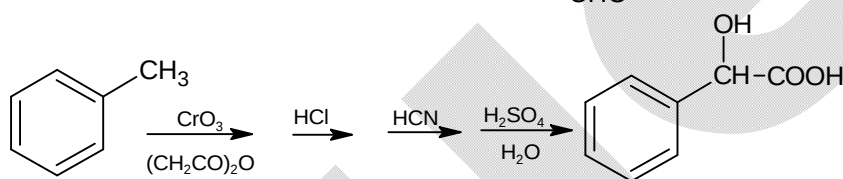
(B)



(C)

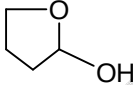


(D)



28. Correct statement (s) among the following:

- (A) Formic acid gives positive test with Tollen's reagent
 (B) Benzaldehyde gives positive test with Fehling's solution

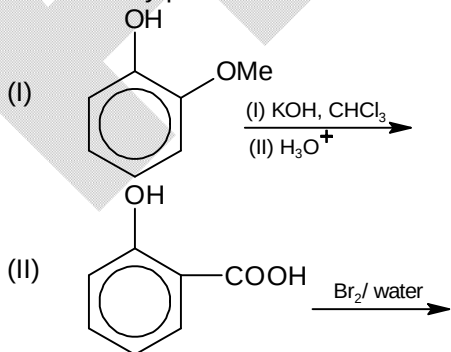
(C)  gives positive test with Benedict's solution

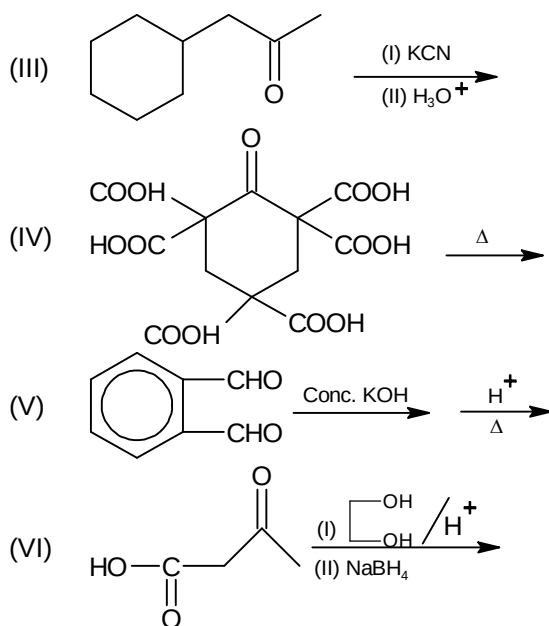
(D) $\text{H}_2\text{N}-\text{C}(=\text{O})-\text{NH}-\text{C}(=\text{O})-\text{NH}_2$ gives Biuret test positive

Section – B (Maximum Marks: 24)

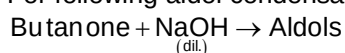
This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

29. In how many products of the reaction will give CO_2 with NaHCO_3 ?



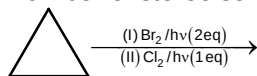


30. For following aldol condensation

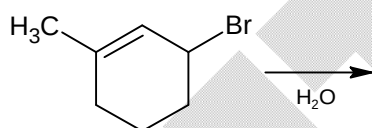


How many different products (including Stereoisomers) are formed above?

31. Number of stereoisomers possible for product in

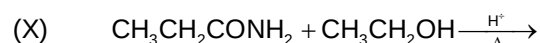
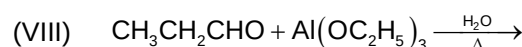
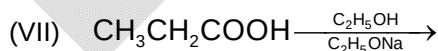
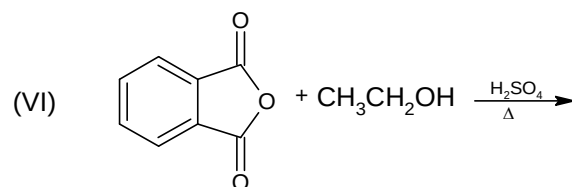
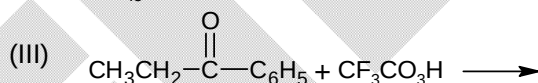
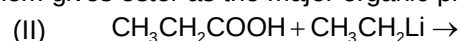
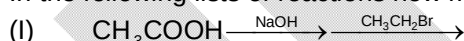


32.



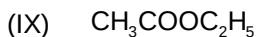
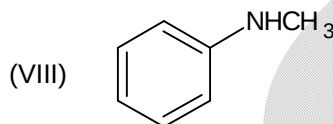
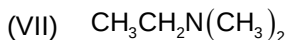
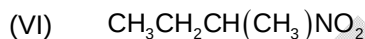
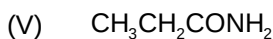
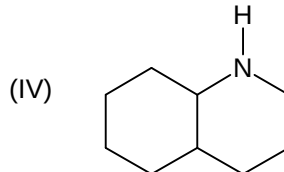
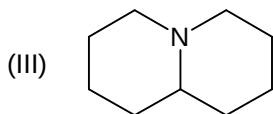
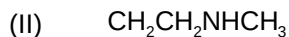
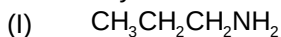
Total number of SN^1 product in the reaction is

33. In the following lists of reactions how many of them gives ester as the major organic product?



34. How many different isomers exist for $\text{C}_3\text{H}_6\text{O}_2$ which reduces Tollen's reagent as well as forms $\text{C}_3\text{H}_6\text{O}_3$ upon treatment with acetic anhydride?

35. How many of the following gives visible change when treated with nitrous acid ($\text{NaNO}_2 + \text{HCl}$)?



36. An organic acid X has molecular formula $\text{C}_8\text{H}_{16}\text{O}_2$. X on heating with NH_3 forms Y. Y on treatment with alkaline Br_2 solution forms Z. Z on treatment with nitrous acid followed by heating with concentrated H_2SO_4 solution gives 2, 4-dimethyl-pentene. How many different acids X can satisfy the above mentioned condition?

Mathematics**PART – III****Section – A (Maximum Marks: 12)**

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

37. $\lim_{x \rightarrow 1} \frac{1 + \sin \pi \left(\frac{3x}{1+x^2} \right)}{1 + \cos \pi x}$ is equal to
 (A) 0 (B) 1
 (C) 2 (D) None of these
38. If $x > 0$, let $f(x) = 5x^2 + Ax^{-5}$, where A is a positive constant. Find the smallest A such that $f(x) \geq 24$ for all $x > 0$.
 (A) $2 \left(\frac{24}{7} \right)^{7/2}$ (B) $2 \left(\frac{24}{7} \right)^{1/2}$
 (C) $2 \left(\frac{24}{7} \right)^{2/7}$ (D) None of these
39. Let $g_a(x) = \frac{a^x}{a^x + \sqrt{a}}$, ($a > 0$) then the value of $g_5(x) + g_5(1-x)$ is
 (A) 1 (B) 5
 (C) 10 (D) None of these
40. Let $f(x)$ be real valued function not identically zero such that $f(x+y^n) = f(x) + \{f(y)\}^n \forall x, y \in \mathbb{R}$. where, n is odd natural number > 1 and $f'(0) \geq 0$ then the function $f(x)$ is
 (A) discontinuous at one point
 (B) continuous everywhere but non differentiable at some point
 (C) discontinuous at three point
 (D) continuous and differentiable everywhere

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

41. An isosceles triangle is inscribed in a circle of radius r . If the angle 2α at the apex is restricted to lie between 0 and $\frac{\pi}{2}$. 'P' denotes the perimeter of triangle then
 (A) $P_{\max} = 3\sqrt{3}r$ (B) $P_{\max} = 4r$
 (C) $P_{\min} = 2r(\sqrt{2} + 1)$ (D) No minimum value of P can be found
42. If $\frac{a_0}{n+1} + \frac{a_1}{n} + \frac{a_2}{n-1} + \dots + \frac{a_{n-1}}{2} + a_n = 0$ then the equation $a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n = 0$ will have
 (A) At least one +ve real root (B) At least one root in (0, 1)
 (C) At least one root in (-1, 0) (D) At least one -ve real root

43. $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=n+1}^{2n} \ln\left(1 + \frac{r}{n}\right)$ equals

(A) $\ln\left(\frac{27}{4e}\right)$

(C) $3\ln 3 - 2\ln 2 - 1$

(B) $\ln\left(\frac{27}{e^2}\right)$

(D) $3\ln 3 - 2$

44. If $I = \int_0^1 \sqrt{1+x^3} dx$, then

(A) $I < 1$

(C) $I < \frac{\sqrt{7}}{2}$

(B) $I \neq \frac{\sqrt{5}}{2}$

(D) $\frac{3\pi}{4}$

45. Let $W = \{1, 2, 3, 4\}$. Consider the following relation of W .

$R_1 = \{(1, 1)(1, 2)\}$

$R_2 = \{(1, 1)(2, 3)(4, 1)\}$

$R_3 = \{(1, 3)(2, 4)\}$

$R_4 = \{(1, 1), (2, 2), (3, 3)\}$

$R_5 = W \times W$

$R_6 = \phi$

(A) R_5 is the only reflexive relation

(B) All relation are transitive.

(C) All relation are either reflexive or symmetric

(D) R_4 , R_5 and R_6 are the only symmetric relation

46. Consider a real valued continuous function f such that $f(x) = \sin x + \int_{-\pi/2}^{\pi/2} (\sin x + tf(t)) dt$. If m and M are maximum and minimum value of the function f , then

(A) $M = 3(\pi + 1)$

(C) $\frac{M}{m} = \frac{3}{2}$

(B) $m = 2(\pi + 1)$

(D) $\frac{M}{m} = 3$

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

47. If $\int_0^1 \tan^{-1}\left(\frac{1}{1-x+x^2}\right) dx = k \int_0^1 \tan^{-1} x dx$, then the number of solution of $k \sin \pi x = \log_k x$ is/are

48. For $a > 0$, the minimum value of the integral $\int_0^{1/a} (a^3 + 4x - a^5 x^2) e^{ax} dx$ is

49. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function satisfying $f\left(\frac{x+y}{3}\right) = \frac{2+f(x)+f(-1)}{3}$ for all real x and y and $f'(2) = 2$, then the number of solution of the equation $x^2 + (f(|x|))^2 = 9$ are
50. $\lim_{a \rightarrow \infty} \int_0^1 e^{x^2} \cos ax \, dx$ is equal to
51. If $\lim_{x \rightarrow a} \frac{\int_a^x f(x) dx - \left(\frac{x-a}{2}\right)(f(x) + f(a))}{(x-a)^3} = 0$, then $f(x)$ is of maximum degree?
52. Inside a semicircle of diameter 1, a right angle triangle is drawn with one of its legs along the diameter. If the maximum area of the triangle is M , then the value of $32\sqrt{3}M$ is:
53. Let f be a differentiable function satisfying $\int_0^{f(x)} f^{-1}(t) dt - \int_0^x (\cos t - f(t)) dt = 0$ and $f\left(\frac{\pi}{2}\right) = \frac{2}{\pi}$ then $\left[\int_0^{\pi/2} f(x) dx\right]$ is Here $[.]$ denotes G.I.F.
54. If $\int (\sin(101x) \sin^{99} x) dx = \frac{\sin(ax)(\sin x)^b}{c} + k$, then the value of $\frac{a+b}{c}$ is equal to

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PART TEST – III
PAPER –2
TEST DATE: 20-10-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. A

Sol. $\frac{kq_1}{r_1} + \frac{kq_2}{r_2} = 0$
$$-\frac{kq_1}{r_1^2} \frac{dr_1}{dt} + \frac{k \cdot dq_1}{r_1^2 dt} - \frac{kq_2}{r_2^2} \cdot \frac{dr_2}{dt} = 0$$

2. B

Sol. $B_{\text{net}} = B - B_{\text{cav}}$
$$= \frac{5\mu_0 J a}{12} - \frac{\mu_0 J a}{6} = \frac{\mu_0 J a}{4}$$

3. B

Sol. $\mu = iA$
$$= i \cdot 12 \times \frac{a}{2} \times \left(\frac{2a}{\sqrt{3}} \right)$$

$$= (4\sqrt{3}a^2 i)$$

4. B

Sol. $Q_1 = \frac{2CV}{3}, Q_2 = 2CV, \Delta Q = \frac{4CV}{3}$
$$V_1 = \frac{1}{2} \cdot \frac{2C}{3} \cdot V^2, V_2 = \frac{1}{2} \cdot 2C \cdot V^2, W_{\text{bat}} = \frac{4CV^2}{3}$$

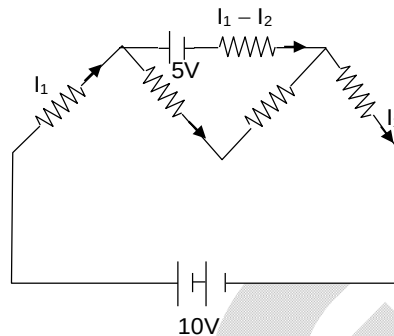
$$\text{Heat} = \frac{2CV^2}{3} - \frac{4CV^2}{3} = -\frac{2CV^2}{3}$$

5. ABCD

Sol. $3I_1 - I_2 = 1$

$3I_2 - I_1 = 1$

$I_1 = 0.5A, \quad I_2 = 0.5A$



6. ABD

Sol. $E_{\text{eff}} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} = 20V$

$R_{\text{eff}} = 2\Omega$

7. BC

Sol. $B2\pi r = \mu_0 \int \frac{J_0}{R^2} r(R-r) \cdot 2\pi r dr$

$$B = \frac{J_0 \cdot r^2}{R^2} \left[\frac{R}{3} - \frac{r}{4} \right]$$

8. A

Sol. $R_{\text{eff}} = 3R$

$Q_{\text{max}} = \frac{V}{2}$

$$q = Q_{\text{max}} \left(1 - e^{-\frac{t}{R_{\text{eff}} \cdot C}} \right)$$

9. ABCD

Sol. Basic concept.

10. ABCD

Sol. Basic concept of Wheatstone bridge.

Section – B

11. 2

Sol. $i^2 R = nC_p \frac{dT}{dt} + mgv$

$PV = nRT$

$PdV = nRdT$

$PAvdT = nRdT$

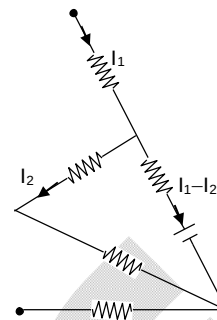
12. 0

Sol. $3I_1 - I_2 = 1$

$3I_2 - I_1 = 1$

$I_1 = 0.5, I_2 = 0.5$

$I_1 - I_2 = 0$



13. 1

Sol. $\phi = \frac{q}{\epsilon_0} (\cos \theta_1 \sim \cos \theta_2)$

$\cos \theta_1 = \frac{3}{5}, \cos \theta = \frac{4}{5}$

14. 8

Sol. B inside cavity is $\frac{\mu_0 JR}{4}$ and is uniform.

$$\begin{aligned} \text{Then time period } T &= \frac{2\pi m}{qB} \\ &= \frac{8\pi m}{q\mu_0 JR} \end{aligned}$$

15. 7

Sol. $E \cdot 4\pi r^2 = \int_0^r \rho \cdot 4\pi r^2 dr$

$E = \frac{\rho_0 r}{4\epsilon_0} \cdot \left(\frac{5}{3} - \frac{r}{R} \right)$

16. 4

Sol. $\frac{\mu_0 N I_1}{2R} = \frac{\mu_0 \cdot I_2}{2\pi d}$

$N = \frac{I_2 R}{\pi d \cdot I_1}$

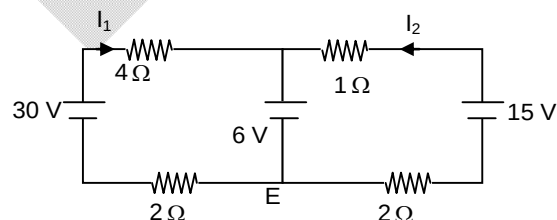
17. 7

$$\begin{aligned} \text{Sol. Heat loss} &= \frac{1}{2} \times 8 \times 15^2 - \left(\frac{1}{2} \times 8 \times 10^2 + \frac{1}{2} \times 4 \times 10^2 \right) \mu\text{J} \\ &= (900 - 200) \mu\text{J} \\ &= 700 \mu\text{J} \end{aligned}$$

18. 7

Sol. $30 - 4I_1 - 6 - 2I_1 = 0$

$15 - I_2 - 6 - 2I_2 = 0$



Chemistry

PART – II

Section – A

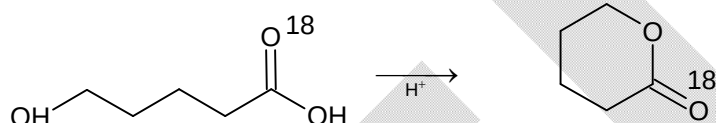
19. D
Sol. Property of carbohydrates

20. C
Sol. Polyethylene is an addition polymer.

21. D
Sol. $A = \text{Br}-\text{CH}_2-\text{CH}=\text{CH}-\text{Cl}$
 $B = \text{Br}-\text{CH}_2-\text{CH}_2-\text{CH}(\text{Cl})-\text{CH}_2-\text{Cl}$

22. A
Sol. Factual.

23. AB
Sol.

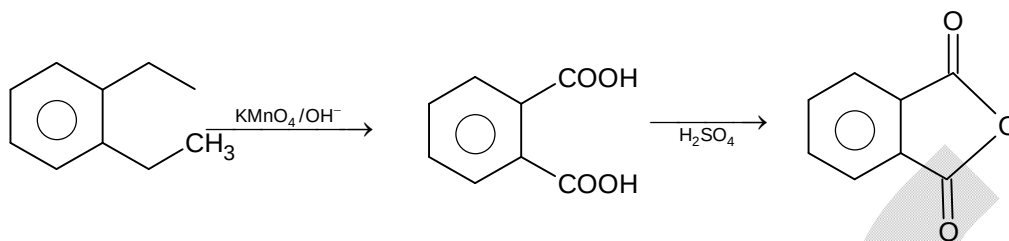


24. CD
Sol. $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_2\text{Cl} \xrightarrow[\text{Et, H}_2\text{O}]{\text{Mg}} \text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_2\text{COOH} \xrightarrow[\text{H}^+]{\text{CO}_2} \text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_2\text{COOH} \xrightarrow[\Delta]{\text{H}^+} \text{cyclic ester}$
 $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_2\text{COOH} \xrightarrow{\text{NaBH}_4} \text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_2\text{COOH} \xrightarrow[\Delta]{\text{H}^+} \text{cyclic ester}$
(d,l)

25. AC
Sol. (A) $\rightarrow$ Hoffmann bromide
(B) $\rightarrow$
(C) Curtius reaction
(D) Hunsdieker reaction

26. AC
Sol. Factual.

27. ABC
Sol.

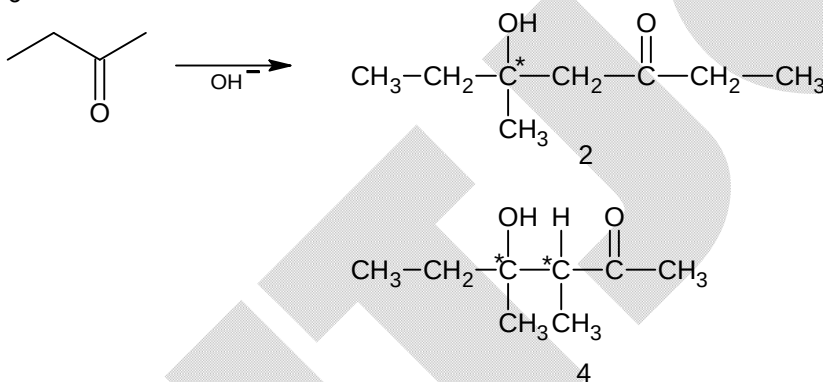


28. ACD
Sol. Hemiacetal gives benedict test.

Section – B

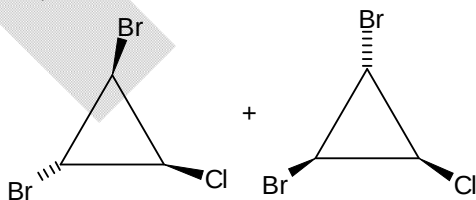
29. 3
Sol. II, III, IV Total answer = 3

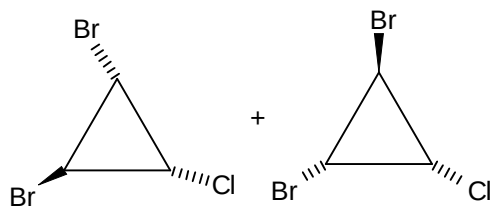
30. 6
Sol.



31. 4
Sol. Product is

It has 2 chiral carbons + 1 pseudo chiral
So,



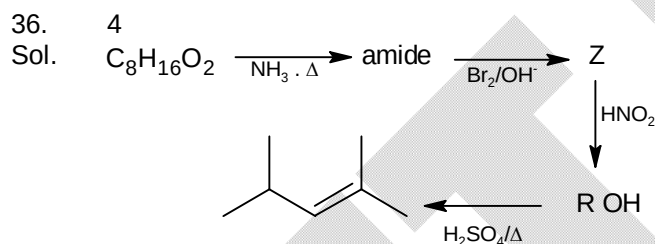
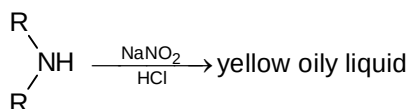
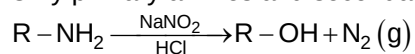


32. 4
Sol. Factual.

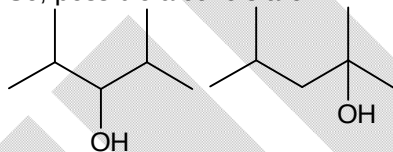
33. 7
Sol. Reaction I, III, IV, V, VI, VIII, IX

34. 4
Sol. Factual.

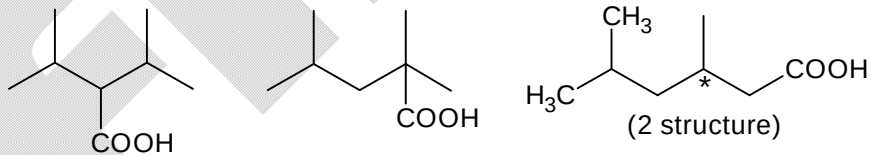
35. 4
Sol. Only primary amines and secondary amines gives visible change with nitrous acid



So, possible alcohols are



So, acids are



Mathematics**PART – III****Section – A**

37. A

Sol. Take $x = 1 + \theta$ then $\theta \rightarrow 0$ and solve

38. A

Sol. $f'(x) = 10x - 5Ax^{-6}$

$$\text{and } f''(x) = 10 + 30Ax^{-6} > 0$$

i.e. $f'(x) = 0$ gives a minima.

$$\Rightarrow x^7 = \frac{A}{2} \Rightarrow x = \left(\frac{A}{2}\right)^{1/7}$$

Since $A > 0$, we get only one minima and no maxima.Hence the smallest value of $f(x)$ will be at $x = \left(\frac{A}{2}\right)^{1/7}$

$$\text{i.e. } f(x)_{\min} = 5 \cdot \left(\frac{A}{2}\right)^{1/7} + A \left(\frac{A}{2}\right)^{-5/7} = 24$$

$$\Rightarrow 5 \left(\frac{A}{2}\right)^{2/7} + 2 \left(\frac{A}{2}\right)^{2/7} = 24 \Rightarrow A = 2 \left(\frac{24}{7}\right)^{7/2}$$

39. A

$$\text{Sol. } \frac{a^x}{a^x + \sqrt{a}} + \frac{a^{1-x}}{a^{1-x} + \sqrt{a}} = 1$$

40. D

Sol. Put $x = y = 0 \Rightarrow f(0) = 0$

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h)}{h} = \lambda (\text{say}) \dots (i)$$

$$\begin{aligned} \text{Again } f'(0) &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f\left(0 + \left(\frac{1}{h^n}\right)^n\right) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(0) + \left\{f\left(\frac{1}{h^n}\right)\right\}^n - f(0)}{h} = \lim_{h \rightarrow 0} \left\{ \frac{f\left(\frac{1}{h^n}\right)}{\frac{1}{h^n}} \right\}^n = \lambda^n \dots (ii) \end{aligned}$$

From (i) and (2)

$$\lambda = \lambda^n \Rightarrow \lambda(\lambda^{n-1} - 1) = 0$$

$$\lambda = 0 \quad \text{or } \lambda = 1$$

$$\Rightarrow f'(x) = 0 \quad f'(x) = 1$$

$$f(x) = c = 0 \quad f(x) = x + k$$

Hence $l < 1$ and $l < \frac{\sqrt{7}}{2}$

45. ABD

Sol. Use definition of reflective, symmetric and transitive relation.

46. AD

Sol. $f(x) = \sin x + \int_{-\pi/2}^{\pi/2} (\sin x + tf(t)) dt$
 $\Rightarrow f(x) = \sin x + \pi \sin x + \int_{-\pi/2}^{\pi/2} tf(t) dt$
 $\Rightarrow f(x) = (\pi + 1) \sin x + A$

Now, $A = \int_{-\pi/2}^{\pi/2} (t(\pi + 1) \sin t + A) dt = 2(\pi + 1) \int_0^{\pi/2} tf(t) dt$

$\therefore A = 2(\pi + 1)$

Hence $f(x) = (\pi + 1) \sin x + 2(\pi + 1)$

$f(x)_{\max} = 3(\pi + 1) = M$

$f(x)_{\min} = (\pi + 1) = m$

$\frac{M}{m} = 3$

Section – B

47.

Sol. $\int_0^1 \tan^{-1} \frac{x - (x-1)}{1 + x(x-1)} dx = \int_0^1 [\tan^{-1} x - \tan^{-1}(x-1)] dx$
 $= \int_0^1 \tan^{-1} x dx - \int_0^1 \tan^{-1}(x-1) dx$
 $= 2 \int_0^1 \tan^{-1} x dx$ Hence $k = 2$

Now $2 \sin \pi x = \log_2 x$

Number of solution is 3.

48.

Sol. $\int_0^{1/a} (a^3 + 4x - a^5 x^2) dx = e^{ax} (Ax^2 + Bx + C) + D.$

Differentiating both sides, $(a^3 + 4x - a^5 x^2) e^{ax} = e^{ax} (2Ax + B) + [Ax^2 + Bx + C] e^{ax} \cdot a$

$\Rightarrow A = -a^4, B = \frac{4 + 2a^4}{a}, C = \frac{-a^4 - 4}{a^2}$

$\therefore I = \left\{ e^{ax} \left[-a^4 x^2 + \frac{4 + 2a^4}{a} x - \frac{a^4 + 4}{a^2} \right] \right\}_0^{1/a}$

$\therefore I = a^2 + \frac{4}{a^2}$

Least value of I is 4.

49. 2

Sol. As $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{2 + f(3x) + f(3h)}{3} - \frac{2 + f(3x) + f(0)}{3}$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(3h) - f(0)}{3h - 0} = f'(0) = f'(2)$$

$$f(x) = 2x + 2 \text{ so } x^2 + (f(|x|))^2 = 9 \Rightarrow x^2 + (2|x| + 2)^2 = 9 \text{ number of } = 2$$

50. 0

Sol. Let $u = e^{x^2}$ and $v = \cos ax$
Using by-part

$$I = \left(\frac{e^{x^2} \sin ax}{a} \right)_0^1 - \int_0^1 \frac{2xe^{x^2} \sin ax}{a} dx$$

$$I = \frac{e \sin a}{a} - \frac{2}{a} \int_0^1 xe^{x^2} \sin ax dx$$

For $x \in [0, 1]$, $-e \leq xe^{x^2} \sin ax \leq e$

$$-e = \int_0^1 -e dx \leq \int_0^1 xe^{x^2} \sin ax dx \leq \int_0^1 e dx = e$$

$$\text{and } \lim_{a \rightarrow \infty} \left(\frac{e \sin a}{a} - \frac{1}{a} \int_0^1 2xe^{x^2} \sin ax dx \right) = 0$$

$$\text{Thus, } \lim_{a \rightarrow \infty} \int_0^1 e^{x^2} \cos ax dx = 0$$

51. 1

Sol. $x = a + h$ then $h \rightarrow 0$

$$\lim_{h \rightarrow 0} \frac{\int_a^{a+h} f(x) dx - \frac{h}{2}(f(a+h) + f(a))}{h^3} = 0$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(a+h) - \frac{1}{2}(f(a+h) + f(a)) - \frac{h}{2}f'(a+h)}{3h^2} \quad (\text{Using L' Hospital's Rule})$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{\frac{1}{2}f(a+h) - \frac{1}{2}f(a) - \frac{h}{2}f'(a+h)}{3h^2} = 0$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{\frac{1}{2}f'(a+h) - 0 - \frac{1}{2}f'(a+h) - \frac{h}{2}f''(a+h)}{6h} = 0 \quad (\text{Using L' Hospital Rule})$$

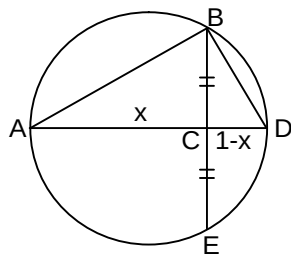
$$\Rightarrow \lim_{h \rightarrow 0} \frac{-f''(a+h)}{12} = 0$$

Or $f''(a) = 0 \quad \forall a \in \mathbb{R}$

Thus $f(x)$ must be of maximum degree 1

52. 9

Sol. $BC \times CE = AC \times CD \Rightarrow BC \times CE = x(1-x)$



Here $BC = CE \Rightarrow BC = \sqrt{x(1-x)}$.

$$\text{Area} = \frac{x\sqrt{x(1-x)}}{2} = \Delta$$

$$\Delta^2 = \frac{x^3 - x^4}{2}, \text{ it is max at } x = \frac{3}{4}$$

$$\Delta_{\max} = \frac{3\sqrt{3}}{32}.$$

53. 1

Sol. $\int_0^{f(x)} f^{-1}(t) dt - \int_0^x (\cos t - f(t)) dt = 0$

Differentiating w.r.t. to x

$$\Rightarrow f^{-1}(f(x)) \cdot f'(x) - (\cos x - f(x)) = 0$$

$$\Rightarrow xf'(x) + f(x) = \cos x$$

$$\Rightarrow (xf(x))' = \cos x$$

$$\Rightarrow xf(x) = \sin x + C$$

As $f\left(\frac{\pi}{2}\right) = \frac{2}{\pi} \Rightarrow C = 0$

So $f(x) = \frac{\sin x}{x}$

Now, $\int_0^{\pi/2} f(x) dx = \int_0^{\pi/2} \frac{\sin x}{x} dx$

As, $\frac{2}{\pi} < \frac{\sin x}{x} < 1$

$$1 < \int_0^{\pi/2} \frac{\sin x}{x} dx < \frac{\pi}{2}$$

So $\left[\int_0^{\pi/2} \frac{\sin x}{x} dx \right] = 1$

54. 2

Sol. $I = \int \sin(101x) \sin^{99} x dx = \int \sin(100+x) \cdot (\sin x)^{99} dx$
 $= \int \underbrace{\sin(100x)}_u \underbrace{\cos x \cdot (\sin x)^{99}}_v dx + \int \cos(100x) (\sin x)^{100} dx$

Use by-part

$$I = \frac{\sin(100x)(\sin x)^{100}}{100} + k$$

Then, $a = 100$, $b = 100$, $c = 100$

$$\frac{a+b}{c} = 2$$

FIITJEE

ALL INDIA INTEGRATED TEST SERIES

PART TEST – III

JEE (Main)-2024

TEST DATE: 15-09-2023

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

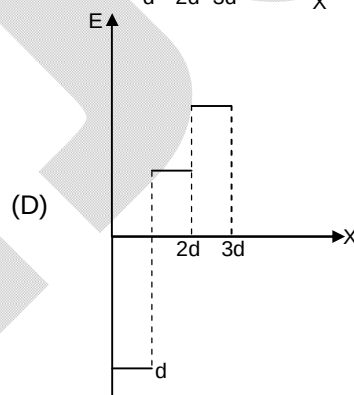
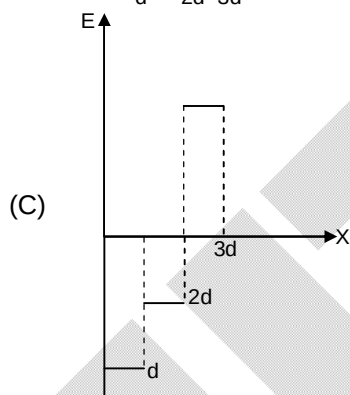
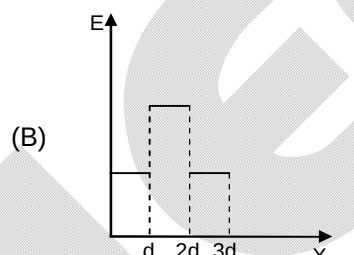
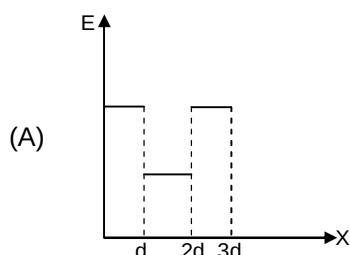
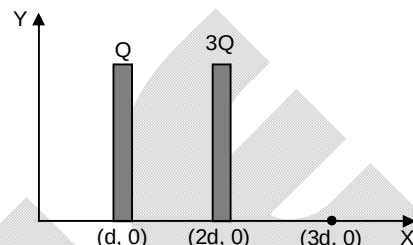
PART – A

SECTION – A

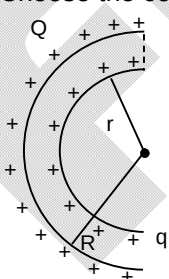
(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

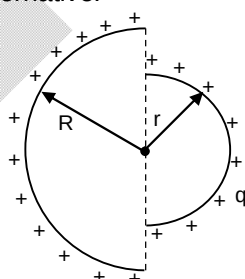
1. Two very large thin conducting plates having same cross-sectional area are placed as shown in figure. They are carrying charges Q and $3Q$ respectively. The variation of electric field as a function of x (for $x = 0$ to $x = 3d$) will be best represented by:



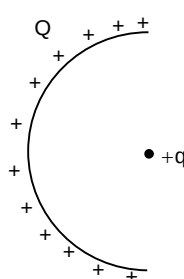
2. Refer figure (a), (b) and (c). All charges are uniformly distributed on respective configurations. Force on charge $+q$ due to hemisphere is F_1 , F_2 and F_3 in figure (a), (b) and (c) respectively. Choose the correct alternative:



(a)



(b)



(c)

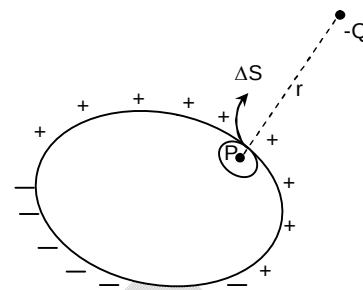
(A) $F_1 = F_3 = F_2$

(B) $F_1 > F_3 > F_2$

(C) $F_1 = F_3 > F_2$

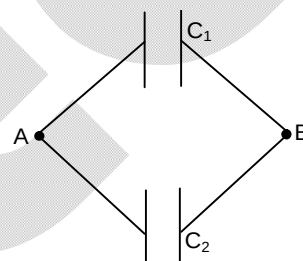
(D) $F_1 > F_3 = F_2$

3. A charge $-Q$ is placed at some distance from a neutral conductor. Charge is induced on its surface. In the neighborhood of a point P on its surface, the charge density is $\sigma \text{ C/m}^2$. Consider a small area ΔS on the surface of the conductor encircling point P. Find the resultant force experienced by the area ΔS due to charge present on the surface elsewhere and the charge $-Q$.



- (A) $\frac{\sigma^2}{\epsilon_0} \Delta S$ (B) $\frac{\sigma^2}{2\epsilon_0} \Delta S$
 (C) $\frac{\sigma^2}{\epsilon_0} \Delta S + \frac{\sigma Q}{4\pi \epsilon_0 r^2} \Delta S$ (D) $\left(\frac{\sigma^2}{2\epsilon_0} + \frac{\sigma Q}{4\pi \epsilon_0 r^2} \right) \Delta S$

4. Two identical capacitors are connected in parallel and a charge Q is given to this combination. At instant $t = 0$, plates of capacitor C_1 are moved apart with speed v and of capacitor C_2 are moved towards each other with speed v . Find the intensity of current which flows from A to B.

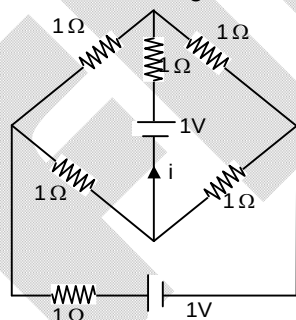


Take, $|Q| = d_0 = \text{initial distance between plates of capacitors and}$
 $v = 1 \text{ m s}^{-1}$.

- (A) 1 A (B) 2.5 A
 (C) 0.5 A (D) 0 A
5. A charge q_1 is at a distance r from another charge q_2 in vacuum. Force experienced by charge $q_1 = F$. In another arrangement charge q_2 is placed in a dielectric medium of dielectric constant K , whereas charge q_1 remains in vacuum. The distance between them remains unchanged. Now force experienced by, charge $q_1 = F_1$ and charge $q_2 = F_2$. Mark the correct option.

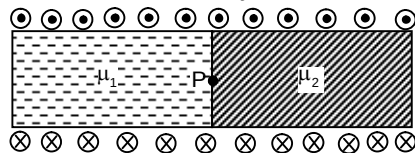
- (A) $F_1 = F_2 = F$ (B) $F_1 = F, F_2 = \frac{F}{K}$
 (C) $F_1 = F_2 = \frac{F}{K}$ (D) $F_1 = \frac{F}{K}, F_2 = F$

6. Current i in the given circuit is

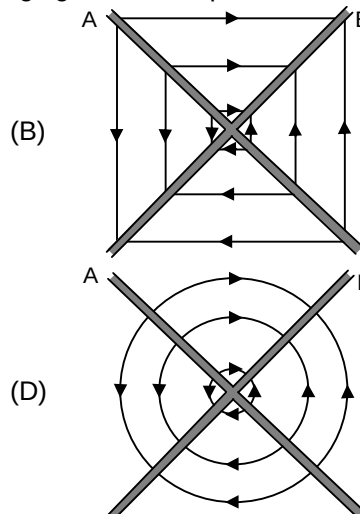
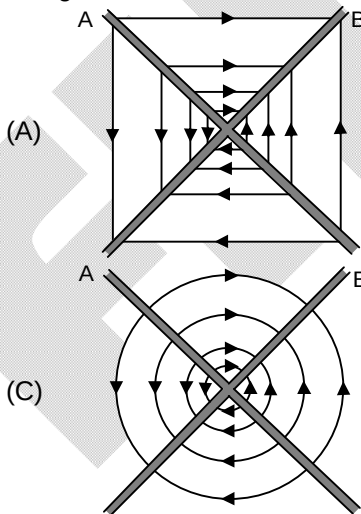


- (A) 0 amp (B) 1 amp
 (C) 2 amp (D) $\frac{1}{2}$ amp

7. A long narrow solenoid is half filled with material of relative permeability μ_1 and half filled with another material of relative permeability μ_2 . Calculate the magnetic field (B) on the axis of the solenoid at boundary of the two material (i.e. at point P). The current in solenoid coil is I.

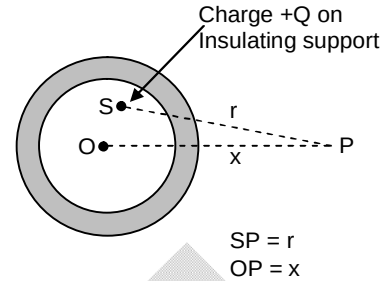


- (A) $(\mu_1 + \mu_2)\mu_0 nI$
- (B) $\frac{1}{2}(\mu_1 + \mu_2)\mu_0 nI$
- (C) $\frac{2\mu_1\mu_2}{\mu_1 + \mu_2}\mu_0 nI$
- (D) $\frac{\mu_1 + \mu_2}{\mu_1 + \mu_2}\mu_0 nI$
8. Line integral of magnetic field B along curve AB, due to infinite conductor carrying current I, as shown in figure, is
- (A) $\frac{2\theta\mu_0 I}{2\pi}$
- (B) $\frac{\theta\mu_0 I}{\pi}$
- (C) $\frac{\theta\mu_0 I}{2\pi}$
- (D) $\frac{\theta\mu_0 I}{4\pi}$
9. Several α -particles of different speeds enter a uniform magnetic field confined into a cylindrical region. If all the α -particles enter the field radially, what can you say about time intervals spent by them in the magnetic field?
- (A) Faster is the particle, lesser is the time
- (B) Slower is the particle, lesser is the time
- (C) The time is same for all the particles
- (D) Insufficient information
10. Two infinitely large planes A and B intersect each other at right angles and carry uniform surface charge densities $+\sigma$ and $-\sigma$. Which of the following figures best represents electric field lines?



11. The diagram shows a charge $+Q$ held on an insulating support S and enclosed by a hollow spherical conductor. O is the center of the spherical conductor and P is a point such that $OP = x$ and $SP = r$. The electric field at point P will be given as:

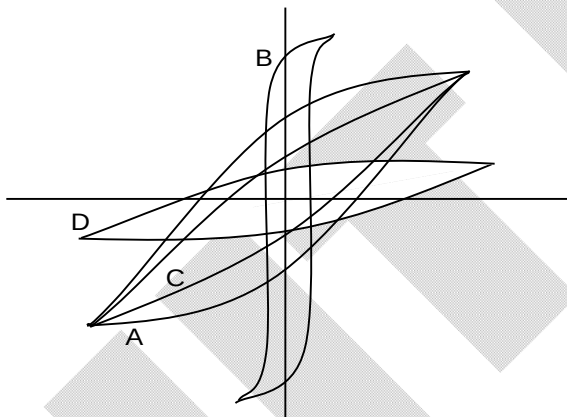
- (A) $\frac{Q}{4\pi\epsilon_0 x^2}$
 (B) $\frac{Q}{4\pi\epsilon_0 r^2}$
 (C) 0
 (D) None of these



12. A small conductor spherical shell of radius r is connected to earth through battery of potential difference V . A charge $+q$ is given a velocity towards shell such that it stops at a distance l from shell where $l \gg a$. The kinetic energy given to the charge $+q$ is _____.

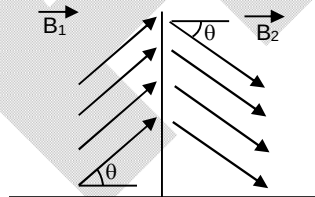
- (A) $\frac{qVa}{l}$
 (B) $\frac{kq^2a}{2l^2}$
 (C) $\frac{qVa}{l} - \frac{kq^2a}{2l^2}$
 (D) $\frac{qVa}{l} + \frac{kq^2a}{2l^2}$

13. Hysteresis curves for four different materials is as shown in figure. Which material should be selected for electromagnet?



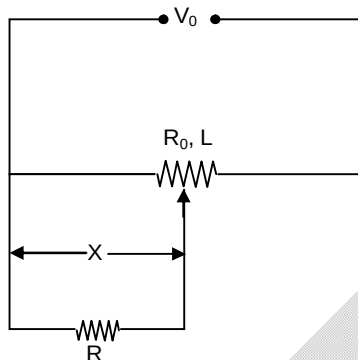
- (A) A
 (B) B
 (C) C
 (D) D

14. A uniform current carrying plane is placed in a uniform magnetic field such that the magnetic field in the region is as shown. Current per unit length in the plane is given by



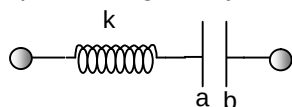
- (A) $\frac{|B_1 - B_2|}{\mu_0}$, outwards
 (B) $\frac{|B_1 + B_2|}{\mu_0}$, outwards
 (C) $\frac{|B_1 - B_2|}{\mu_0}$, inwards
 (D) $\frac{|B_1 + B_2|}{\mu_0}$, inwards

15. A circular loop of N turns is made from wire of length L . Another circular loop is made from identical wire but number of turns are half. Ratio of torque on second coil to that on first coil when placed in a uniform magnetic field in same orientation is given by ____.
- (A) 2 : 1 (B) 1 : 2
(C) 4 : 1 (D) 1 : 1
16. A particular galvanometer serves as a 2 V full scale voltmeter when a 2500Ω resistor is connected in series with it. It serves as a 0.5 A full scale ammeter when a 0.22Ω resistor is connected in parallel with it. Determine the internal resistance of the galvanometer and the current required to produce full scale deflection.
- (A) 45Ω (B) 95Ω
(C) 145Ω (D) 195Ω
17. Figure illustrates potentiometer circuit by means of which we can vary a voltage V applied to a certain device possessing a resistance R . The potentiometer wire has a length L and a resistance R_0 . A voltage V_0 is applied to the terminals of wire. Find the voltage V fed to the device as a function of distance x .



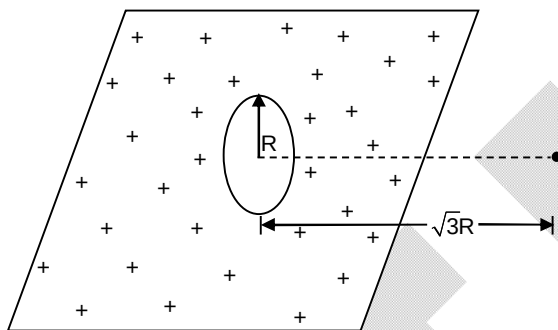
- (A) $\frac{V_0 R x}{R_L - R_0 x \left(1 - \frac{x}{L}\right)}$ (B) $\frac{V_0 R x}{R_L - R_0 x \left(1 + \frac{x}{L}\right)}$
(C) $\frac{V_0 R x}{R_0 x \left(1 - \frac{x}{L}\right) - R_L}$ (D) $\frac{V_0 R x}{R_L + R_0 x \left(1 + \frac{x}{L}\right)}$
18. An oil drop carrying a charge of two electrons has a mass of 3.2×10^{-17} kg. It is falling freely in air with terminal speed. The electric field required to make the drop move upwards with the same speed is (neglect buoyant force)
- (A) $2 \times 10^3 \text{ Vm}^{-1}$ (B) $4 \times 10^3 \text{ Vm}^{-1}$
(C) $3 \times 10^3 \text{ Vm}^{-1}$ (D) $8 \times 10^3 \text{ Vm}^{-1}$

19. Consider an air filled parallel plate capacitor with one plate connected to a spring having a force constant k , and another plate held fixed. The system rests on a frictionless table top. If the charges placed on plates a and b are $+Q$ and $-Q$ respectively, the spring expand will have an expansion x , in equilibrium, given by



Capacitor connected to a spring

- (A) $\frac{Q^2}{kA\epsilon_0}$ (B) $\frac{2Q^2}{kA\epsilon_0}$
 (C) $\frac{Q^2}{2kA\epsilon_0}$ (D) None of these
20. An infinite dielectric sheet having charge density σ has a hole of radius R in it. An electron is released on the axis of the hole at a distance $\sqrt{3}R$ from the center. The speed with which it crosses the center of the hole is



- (A) $\sqrt{\frac{\sigma e R}{2m\epsilon_0}}$ (B) $\sqrt{\frac{\sigma e R}{m\epsilon_0}}$
 (C) $\sqrt{\frac{2\sigma e R}{m\epsilon_0}}$ (D) None of these

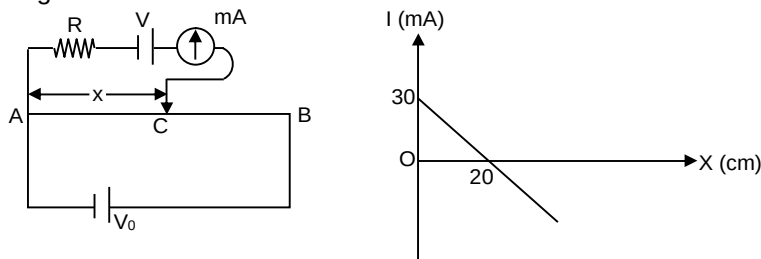
SECTION – B

(Numerical Answer Type)

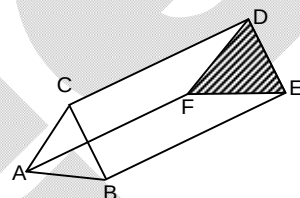
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. In a conducting material, voltage across it is proportional to $\sqrt{l}$ such that $V = K\sqrt{l}$ where K is the proportionality constant. Two elements of this material is with value of $R = 1.66$ and 1.25 volt amp $^{-\frac{1}{2}}$ are connected in parallel. The value of equivalent R of element of same conducting material is _____.

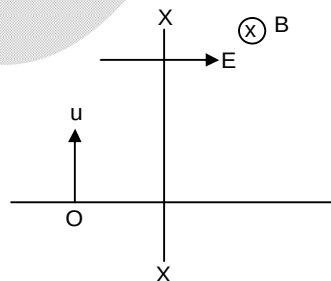
22. AB is a uniform wire of length $L = 100$ cm. A cell of emf $V_0 = 12$ volt is connected across AB. A resistance R , cell of emf V and a milliammeter (which can show deflection in both directions) is connected to the circuit as shown. Contact C can be slid on the wire AB. Distance $AC = x$. The current (I) through the milliammeter is taken positive when the cell of emf V is discharging. A graph of I vs x has been shown. If resistance $R = 10n$, the value of n is _____. Neglect internal resistance of the cells.



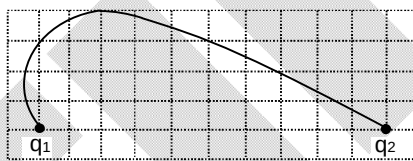
23. A prism shaped network of resistors has been shown in figure. Each arm (like AB, AC, CD, DF ...) has resistance R . Equivalent resistance between C and D is $\frac{mR}{n}$, where m and n doesn't have any common factor determine $m + n$.



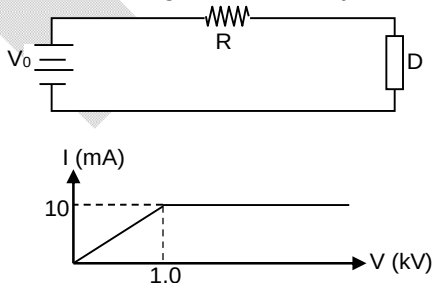
24. In the figure shown XX represents a vertical plane perpendicular to the plane of the figure. To the right of this plane there is a uniform horizontal magnetic field B directed into the plane of the figure. A uniform electric field E exists horizontally perpendicular to the magnetic field in entire space. A charge particle having charge q and mass m is projected vertically upward from point O. It crosses the plane XX after time T . If the speed of projection of particle is K gT, it was observed to move uniformly after time T . $qE = mg$. The value of K is _____.



25. In the figure, a line of electric field created by two point charges q_1 and q_2 is shown. If it is known that $q_1 = 1 \mu\text{C}$, the magnitude of charge q_2 is

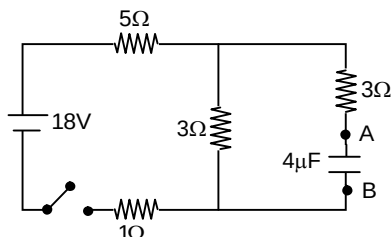


26. Volt-ampere (V-I) characteristics of an unknown device D connected in a circuit in series with a resistance and a battery is shown in the figure. If value of the resistance is $R = 300 \text{ k}\Omega$ and terminal voltage of the battery is $V_0 = 5 \text{ kV}$, find potential drop (in kV) across the device.



27. Two metal pieces having a potential difference of 800 V are 0.02m apart horizontally. A particle of mass 1.96×10^{-15} kg is suspended in equilibrium between the plates. If e is the elementary charge, then charge on the particle is 'me'. Find m .

28. In the circuit shown in figure, the charge on capacitor $q_c = a \left(1 - e^{-\frac{t}{b}} \right) \mu\text{C}$ as a function of time t (μsec). At $t = 0$, switch is closed. LCM of a and b is ____.



29. A magnetic needle performs 20 oscillations per minute in a horizontal plane and 15.2 oscillations per minute in vertical east west plane. The angle of dip (in degree) at the location is δ . The value of $\frac{\delta}{5} = \underline{\hspace{2cm}}$.
30. A semicircular ring has a total charge $+Q$ distributed uniformly over it. The magnetic field at its center when it moves with velocity v along its axis is given by $\frac{X\mu_0 QV}{\pi^y R^z}$. The value of $xyz = \underline{\hspace{2cm}}$.

Chemistry

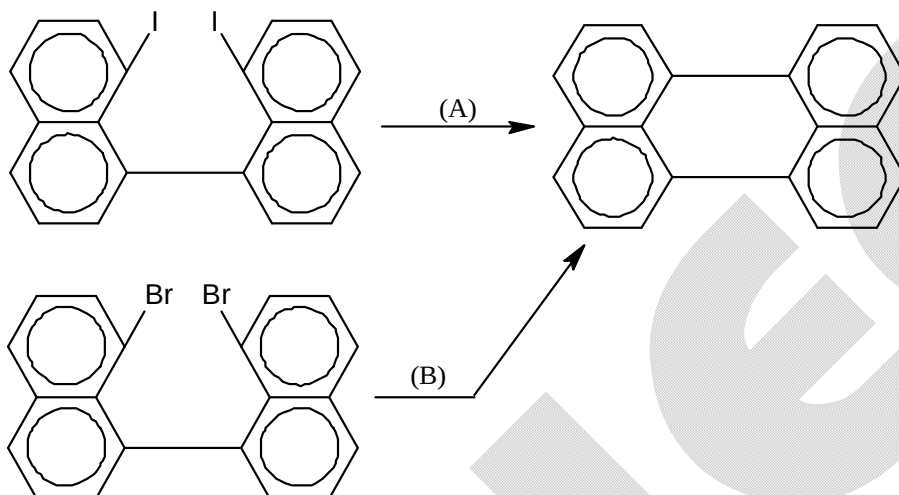
PART – B

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

31.



(A) and (B) are:

(A) Both are Cu / Δ

(C) $\text{A} = \text{Zn} + \text{Et}_2\text{O}$, $\text{B} = \text{Cu} / \Delta$

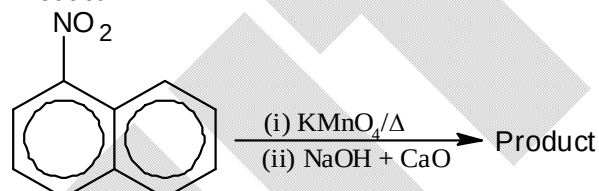
(B) $\text{A} = \text{Cu} / \Delta$, $\text{B} = \text{Zn} + \text{Et}_2\text{O}$

(D) Both are Zn / Δ

32.

Identify the product of the following reaction.

Product



(A) Nitrobenzene

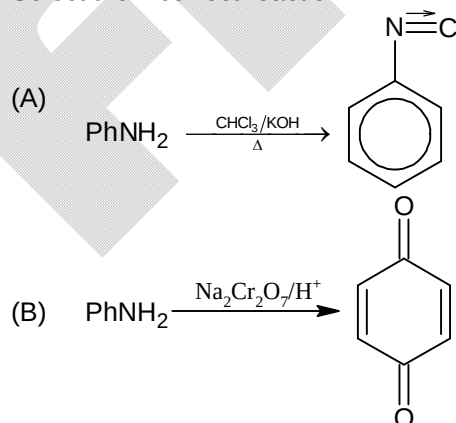
(C) $\text{CO}_2 + \text{H}_2\text{O} + \text{NO}_2$

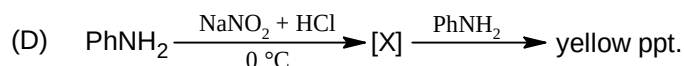
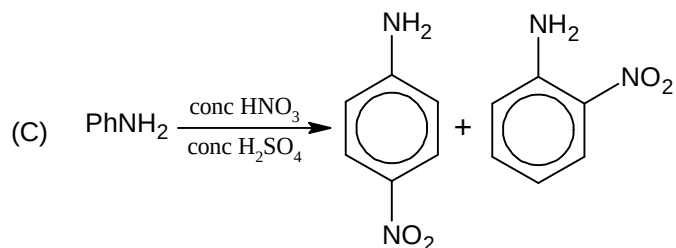
(B) Benzene

(D) O – nitro benzoic acid

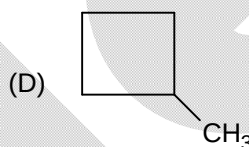
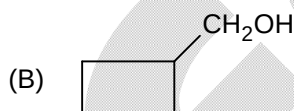
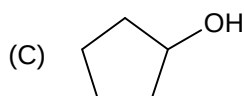
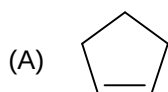
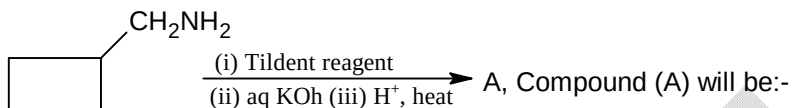
33.

Select the incorrect reaction.





34.



35.

OH group of ROH is a good leaving group in:

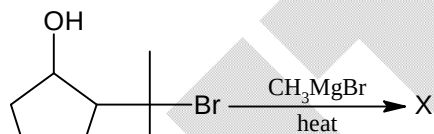
(A) Basic medium

(B) Acidic medium

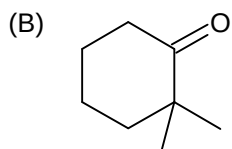
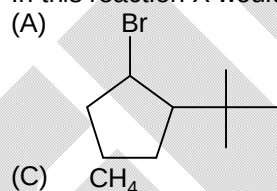
(C) Neutral medium

(D) Presence of sun light

36.



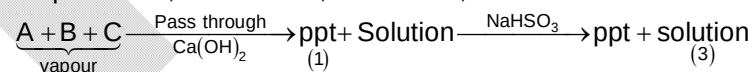
In this reaction X would be:



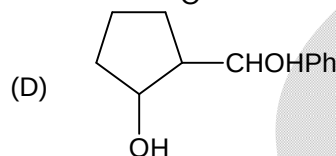
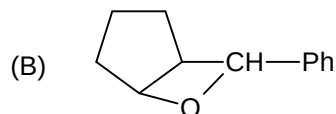
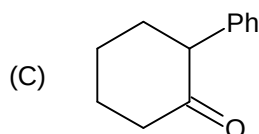
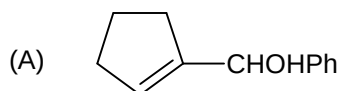
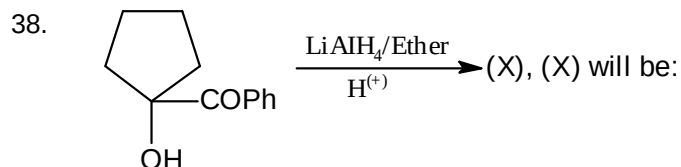
(D) Both (B) & (C)

37.

Choose the answer that has the following compounds located correctly in the separation scheme. Propanone-A, Methanol-B, AcOH-C,



	ppt (1)	ppt (2)	solution (3)
(A)	A	B	C
(B)	B	C	A
(C)	C	B	A
(D)	C	A	B



39. Cyclo Pentanone can be converted in to 5 – Hydroxy pentanoic acid by:

(A) hot KMnO_4

(B) $\text{SeO}_2, \text{H}_3\text{O}^+$

(C) $\text{NH}_2\text{OH} / \text{H}_3\text{O}^+$

(D) $\text{CH}_3\text{CO}_3\text{H} \& \text{H}_3\text{O}^+$

40. γ – hydroxyl butanal can be converted in to γ – lactones by:

(A) $\text{H}^+ / \text{Cr}_2\text{O}_7^{2-}, \text{H}_3\text{O}^+$

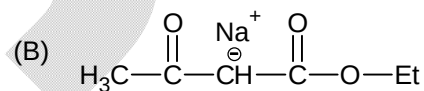
(B) $\text{KMnO}_4 / \text{H}^+, \text{H}_3\text{O}^+$

(C) Alkaline $\text{CuSO}_4, \text{H}^+ / \text{heat}$

(D) All of these

41. $\text{CH}_3\text{COOC}_2\text{H}_5 \xrightarrow{\text{EtONa}}$ Product,
Product of this reaction would be:

(A) $\text{CH}_3\text{COCH}_2\text{COOEt}$



(C) $\text{CH}_3\text{COCH}_2 - \text{COH}(\text{OEt})_2$

(D) $\text{CH}_2(\text{COOEt})_2$

42. Ethyl acetate can be converted into acetone by:

(A) $\text{CH}_3\text{MgBr}(\text{Excess}), \text{H}_3\text{O}^+$

(B) $\text{EtONa} / \text{H}^+, \text{H}_3\text{O}^+ / \text{heat}$

(C) $\text{OH}^- / \text{H}_2\text{O}, \text{CaO} / \text{NaOH}, \text{heat}$

(D) both (A) & (B)

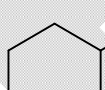
43. Formaldehyde and acetaldehyde can be differentiated by:

(A) Phenol/ dil NaOH

(B) $[\text{Ag}(\text{NH}_3)_2]\text{OH}$

(C) NH_2NH_2 in glycol/ NaOH

(D) All of these

44.  &  can be differentiated by:

(A) KOBBr

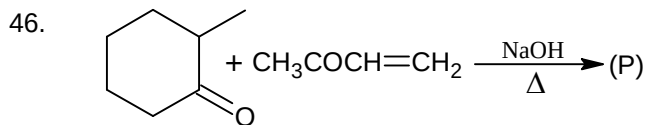
(B) $\text{NaNO}_2 + \text{H}_2\text{SO}_4$

(C) Na

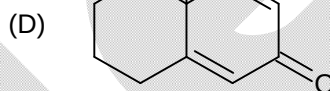
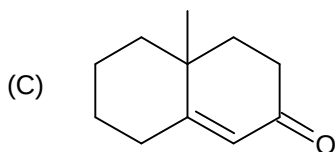
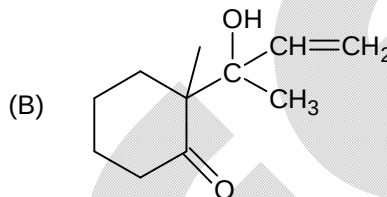
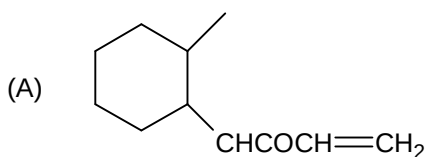
(D) $\text{Cu}^{+2} / \text{OH}^{(-)}$

45. Which is wrong about $(\text{Me}_3\text{C})_2\text{CO}$?

- (A) It will not show haloform test.
 (B) It will give $(\text{Me}_3\text{C})_2\text{CH}_2$ when treated with $\text{N}_2\text{H}_4 / \text{NaOH}$,
 (C) It will not show ketol condensation
 (D) All are correct statements



Here P is:



47. Which among the following will not show cross aldol with formaldehyde?

- (A) Glyoxal (B) nitro methane
 (C) cyclopentadiene (D) 3-oxo butanal

48. $\text{HO}_2\text{CCH}_2\text{CHNH}_2\text{COOH}$ is called aspartic acid

If pK_a of $\alpha\text{-COOH}$ is 1.88, pK_a of $\alpha\text{-NH}_3^+$ is 9.60 & pK_a of side chain is 3.65, then isoelectric point of aspartic acid will be:

- (A) 5.74 (B) 2.77
 (C) 6.13 (D) 3.65

49. Which is incorrect about fructose?

- (A) It gives Benedict's test
 (B) It gives CO_2 with HIO_4
 (C) It forms Glucosazone when treated with Phenyl hydrazine
 (D) It reacts with bromine water

50. (A) $+ \text{Na} \xrightarrow[\text{EtOH}]{\text{NH}_3}$ (B) $\xrightarrow[\text{Me}_2\text{S}]{\text{O}_3}$ $\text{CH}_3\text{COCH}_2\text{CHO}$, compound (A) is

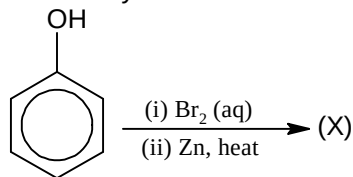
- (A) o-xylene (B) m-xylene
 (C) p-xylene (D) Mesitylene

SECTION – B

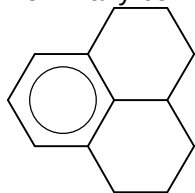
(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

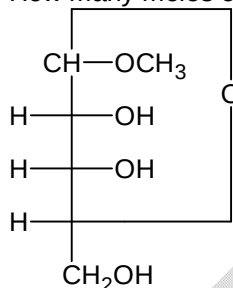
51. How many mono nitro derivatives will be obtained by the nitration of [X]?



52. How many benzylic hydrogens are present in the hydrocarbon shown?



53. How many moles of HIO_4 is required to break down the given molecule here?



54. $\xrightarrow[\text{(ii) (b) moles of NaNH}_2]{\text{(i) (a) moles of PCl}_5}$ $\text{CH}_2\text{—C}\equiv\text{C—CH}_3$

Sum of $(a + b = ?)$

55. How many compound give positive Tollen's test?

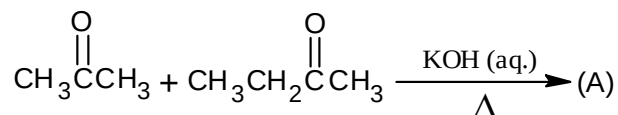
a. D—Glucose, b. D—Fructose,

c. $\text{CH}_3\text{—CHO}$, d. PHCH_2OH ,

e. CH_3CH_3 , f.

g.

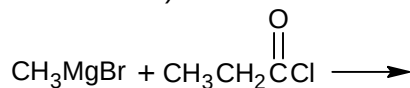
56.



(A) = number of aldol condensation product (including stereoisomer) = _____

57.

Total number of enol possible for the compound formed during given reaction will be (including stereoisomer):



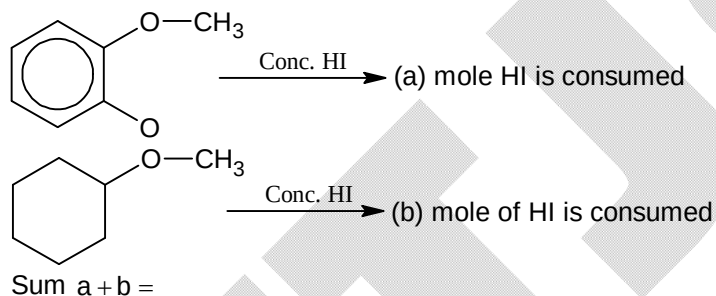
58.

In Zeisel's method for the determination of methoxyl groups, a sample of 2.68 gm of a compound (A) gave 14.08 gm of AgI. If the molecular weight of compound (A) is 134, the number of ($-\text{OCH}_3$) group(s) in the compound (A) is:

59.

If (dl) or ($\pm$) 2-methyl butanoic acid were esterified by reaction with (dl) or ($\pm$) 2-butanol, how many optically active compounds would be present in the final equilibrium reaction mixture?

60.



Mathematics**PART – C****SECTION – A****(Single Choice Answer Type)**

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

61. Exhaustive set of values of x satisfying $\log_{|x|}(x^2 + x + 1) \geq 0$ is
 (A) $(-1, 0)$ (B) $(-\infty, -1) \cup (1, \infty)$
 (C) $(-\infty, \infty) \sim (-1, 0, 1)$ (D) $(-\infty, -1) \cup (-1, 0) \cup (1, \infty)$
62. Complete solution set of the equation $|x^2 - 1 + \cos x| = |x^2 - 1| + |\cos x|$ belonging to $(-2\pi, \pi)$, is
 (A) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - (-1, 1)$ (B) $\left[-\frac{3\pi}{2}, -\frac{\pi}{2}\right] \cup [-1, 1] \cup \left[\frac{\pi}{2}, \pi\right]$
 (C) $\left[-\frac{3\pi}{2}, -\frac{\pi}{2}\right] \cup \left[-\frac{\pi}{2}, \pi\right)$ (D) $\left(-2\pi, -\frac{3\pi}{2}\right] \cup \left[-\frac{\pi}{2}, -1\right] \cup \left[1, \frac{\pi}{2}\right]$
63. Domain of $f(x) = \sqrt{2\{x\}^2 - 3(x) + 1}$, where $\{.\}$ denotes the fractional part, in $[-1, 1]$, is
 (A) $[-1, 1] \sim \left(\frac{1}{2}, 1\right)$ (B) $\left[-1, -\frac{1}{2}\right] \cup \left[0, \frac{1}{2}\right] \cup \{1\}$
 (C) $\left[-1, -\frac{1}{2}\right]$ (D) $\left[-\frac{1}{2}, 1\right]$
64. Range of $f(x) = \cos^{-1}\left(\frac{1+x^2}{2x}\right) + \sqrt{2-x^2}$ is
 (A) $\left\{0, 1 + \frac{\pi}{2}\right\}$ (B) $\{0, 1 + \pi\}$
 (C) $\left\{1, 1 + \frac{\pi}{2}\right\}$ (D) $\{1, 1 + \pi\}$
65. If $|\sin^{-1} x| > [\cos^{-1} x]$, where $[.]$ denotes the greater integer function, then complete set of values of x is
 (A) $[\cos 1, 1]$ (B) $[\sin 1, 1]$
 (C) $[\cos 1, \sin 1]$ (D) $[0, 1]$
66. If $\lim_{x \rightarrow 0} \frac{x^a \sin^b x}{\sin(x^c)}$, where $a, b, c \in \mathbb{R} \sim \{0\}$, exists and has zero value, then
 (A) $a + c = b$ (B) $b + c = a$
 (C) $a + b = c$ (D) None of these
67. $\lim_{x \rightarrow 0} \frac{\sin(6x^2)}{\ln \cos(2x^2 - x)}$ is equal to
 (A) 12 (B) -12
 (C) 6 (D) -6

68. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying $|f(x)| \leq x^2 \forall x \in \mathbb{R}$, then
 (A) 'f' is continuous but not-differentiable at $x = 0$
 (B) 'f' is discontinuous at $x = 0$
 (C) 'f' is differentiable at $x = 0$
 (D) None of these
69. Let $f(x) = \cos x$,
 $g(x) = \begin{cases} \text{minimum}\{f(t) : 0 \leq t \leq x\}, & x \in [0, \pi] \\ \sin x - 1, & x > \pi \end{cases}$, Then
 (A) $g(x)$ is discontinuous at $x = \pi$
 (B) $g(x)$ is continuous for $x \in [0, \infty)$
 (C) $g(x)$ is differentiable at $x = \pi$
 (D) $g(x)$ is differentiable for $x \in [0, \infty)$
70. The curves $x^2 - 4y^2 + c = 0$ and $y^2 = 4x$ will intersect orthogonally for
 (A) $c \in (0, 16)$
 (B) $c \in (-3, 4)$
 (C) $c \in (3, 4)$
 (D) None of these
71. l_1 and l_2 are the side lengths of two variable squares S_1 and S_2 respectively. If $l_1 = l_2 + l_2^3 + 6$, then rate of change of the area of S_2 with respect to rate of change of the area of S_1 and $l_2 = 1$ is
 (A) $\frac{3}{4}$
 (B) $\frac{4}{3}$
 (C) $\frac{3}{2}$
 (D) $\frac{1}{32}$
72. $f(x)$ be a differentiable function such that $f'(x) = \frac{2}{(\log_3(\log_{1/4}(\cos x + a)))}$.
 If $f(x)$ is increasing for all values of x , then
 (A) $a \in (5, \infty)$
 (B) $a \in \left(1, \frac{5}{4}\right)$
 (C) $a \in \left(\frac{5}{4}, 5\right)$
 (D) None of these
73. $f(x) = \int_0^x (e^t - 1)(t - 1)(\sin t - \cos t) \sin t dt \forall x \in \left(-\frac{\pi}{2}, 2\pi\right]$, then $f(x)$ is decreasing in
 (A) $\left(-\frac{\pi}{2}, 0\right) \cup \left(\frac{\pi}{4}, 1\right) \cup \left(\pi, \frac{\pi}{4}\right)$
 (B) $\left(-\frac{\pi}{2}, \frac{\pi}{4}\right) \cup (1, \pi) \cup \left(\frac{5\pi}{4}, 2\pi\right)$
 (C) $\left(\frac{\pi}{4}, 1\right) \cup \left(\pi, \frac{5\pi}{4}\right)$
 (D) $\left(0, \frac{\pi}{4}\right) \cup (1, \pi) \cup \left(\frac{5\pi}{4}, 2\pi\right)$
74. A straight line is drawn through the point $P(3, 4)$ meeting the positive direction of co-ordinate axes at the points A and B. If 'O' is the origin, then minimum area of triangle OAB is equal to
 (A) 12 sq. units
 (B) 6 sq. units
 (C) 24 sq. units
 (D) 48 sq. units

75. $\int e^x \left(\frac{1-x}{1+x^2} \right)^2 dx$ is equal to

(A) $\frac{e^x}{1+x^2} + C$

(B) $-\frac{e^x}{1+x^2} + C$

(C) $\frac{e^x}{(1+x^2)^2} + C$

(D) $-\frac{e^x}{(1+x^2)^2} + C$

76. $\int \frac{dx}{x^2 \sqrt{2+x^2}}$ is equal to

(A) $x\sqrt{1+x^2} + C$

(B) $-\frac{1}{x}\sqrt{1+x^2} + C$

(C) $-\frac{1}{x}\sqrt{1-x^2} + C$

(D) $\frac{1}{x}\sqrt{1+x^2} + C$

77. $\int \frac{dx}{(1+\sqrt{x}\sqrt{x-x^2})}$ is equal to

(A) $\frac{2(\sqrt{x}-1)}{\sqrt{1-x}} + C$

(B) $\frac{2(1+\sqrt{x})}{\sqrt{1-x}} + C$

(C) $\frac{2(\sqrt{x}-1)}{\sqrt{x-1}} + C$

(D) $\frac{2(1+\sqrt{x})}{\sqrt{x-1}} + C$

78. If $\int x \cdot \frac{\ln(x + \sqrt{1+x^2})}{\sqrt{1+x^2}} dx$
 $= a\sqrt{1+x^2} \ln(x + \sqrt{1+x^2}) + bx + c$, then

(A) $a = 1, b = -1$

(B) $a = 1, b = 1$

(C) $a = -1, b = 1$

(D) $a = -1, b = -1$

79. $f(x)$ and $g(x)$ are continuous functions in $[0, 1]$ satisfying, $f(x) = f(a-x)$ and $g(x) + g(a-x) = a$,
 then $\int_0^a f(x) \cdot g(x) dx$ is equal to

(A) $\frac{a}{2}$

(B) $\frac{a}{2} \int_0^a f(x) dx$

(C) $\int_0^a f(x) dx$

(D) $a \int_0^a f(x) dx$

80. If $f(x)$ is a continuous function for all real values of x and satisfies $\int_n^{n+1} f(x) dx = \frac{n^2}{2} \forall x \in I$ then,

$\int_{-3}^5 f(|x|) dx$ is equal to

(A) $\frac{19}{20}$

(B) $\frac{35}{2}$

(C) $\frac{17}{2}$

(D) None of these

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Total number of positive real values of x satisfying $2[x] = x + \{x\}$, where $[.]$ and $\{.\}$ denote the greatest integer function and fractional part respectively, is equal to
82. Total number of solutions of $2^{\cos x} = |\sin x|$ in $[-2\pi, 5\pi]$ is equal to
83. Total number of solutions of the equation $\sin \pi x = |\ln|x||$ are
84. Total number of points of discontinuity of $f(x) = [3 + 4 \sin x]$, where $[.]$ denotes the greatest integer function, in $[\pi, 2\pi]$ is equal to
85. Total number of critical points of $f(x) = \max\{\sin x, \cos x\} \forall x \in (-2\pi, 2\pi)$ is equal to
86. If $\int \tan^4 x dx = a \tan^3 x + b \tan x + cx + d$, then $15(a + b + c)$ is
87. If $\int \frac{1-x^7}{x(1+x^7)} dx = a \ln|x| + b \ln|x^7 + 1| + C$, then $14(a + b)$ is
88. If $x = \int_0^y \frac{dt}{\sqrt{1+9t^2}}$ and $\frac{d^2y}{dx^2} = ay$, then 'a' is equal to
89. The value of $\int_{1/e}^{\tan x} \frac{t dt}{1+t^2} + \int_{1/e}^{\cot x} \frac{dt}{t(1+t^2)}$, where $x \in \left(\frac{\pi}{6}, \frac{\pi}{3}\right)$, is equal to
90. $\frac{16}{\pi} \cdot \lim_{n \rightarrow \infty} \left(\frac{n}{n^2} + \frac{n}{n^2+1} + \frac{n}{n^2+2^2} + \dots + \frac{n}{2n^2-2n+1} \right)$ is equal to

FIITJEE

ALL INDIA INTEGRATED TEST SERIES

PART TEST – III

JEE (Main)-2024

TEST DATE: 15-09-2023

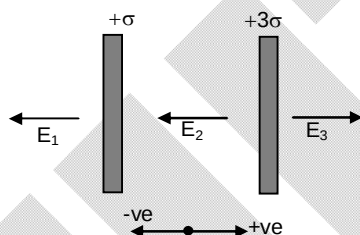
ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. C
Sol. The direction of electric fields at different points is shown in figure below.

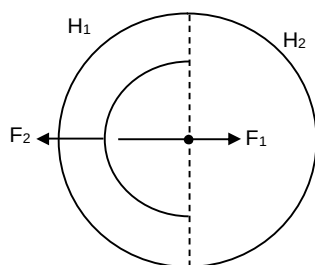


$$E_1 = \frac{\sigma}{2\epsilon_0} + \frac{3\sigma}{2\epsilon_0} = \frac{2\sigma}{\epsilon_0} = E_3$$

$$E_2 = \frac{3\sigma}{2\epsilon_0} - \frac{\sigma}{2\epsilon_0} = \frac{\sigma}{\epsilon_0}$$

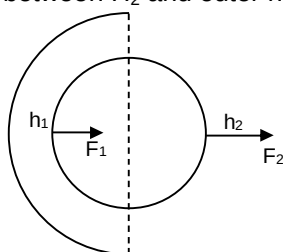
E_1 and E_2 are in the negative direction and E_3 in positive direction.

2. A
Sol. Complete the outer sphere.
Now this sphere will exert no force on the inner hemisphere.
This means the two hemispheres in the outer sphere exert equal and opposite force on the inner hemisphere.
Call the forces on inner due to two halves H_1 and H_2 of outer sphere as F_1 and F_2 .



$$F_1 = F_2 = F(\text{say})$$

Now complete the inner sphere with outer sphere as hemispherical
Now, force between one half of inner sphere (H_1) and the outer hemisphere is very much like that between H_2 and outer hemisphere is like F_2 in previous drawing.



$\therefore$ Resultant force on inner ball due to outer hemisphere is $F_1 = F_2 = 2F(\rightarrow)$

3. B

Sol. EF due to all charges other than charges on ΔS at the concerned surface = $\frac{\sigma}{2\epsilon_0}$

4. C

Sol. At some instant t , $q_1 + q_2 = Q$ and $\frac{q_1}{C_1} = \frac{q_2}{C_2}$

$$\text{Also, } C_1 = \frac{\epsilon_0 A}{(d_0 + vt)} \text{ and } C_2 = \frac{\epsilon_0 A}{(d_0 - vt)}$$

Hence, from there, we get

$$q_1 = Q \left(\frac{d_0 - vt}{2d_0} \right)$$

$$\text{And } q_2 = Q \left(\frac{d_0 + vt}{2d_0} \right)$$

Current in circuit = Reduction in charge of first capacitor = Charge increase in second capacitor.

$$\begin{aligned} \therefore I &= \frac{-dq_1}{dt} = \frac{dq_2}{dt} \\ &= \frac{Qv}{2d_0} = \frac{Q \times 1}{2Q} = 0.50A \end{aligned}$$

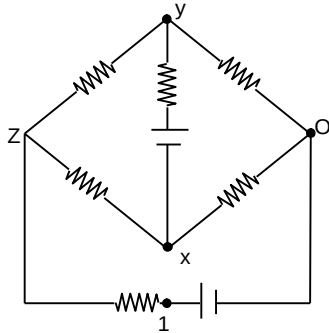
5. B

Sol. Electric field in dielectric is $\frac{q_1}{4\pi k \epsilon_0}$

Electric field in vacuum is $\frac{q_2}{4\pi \epsilon_0}$

6. D

Sol. KCL at node x $3x - y - 2 = -1$
 at node y $3y - x - 2 = 1$
 at node z $3z - y - x = 1$



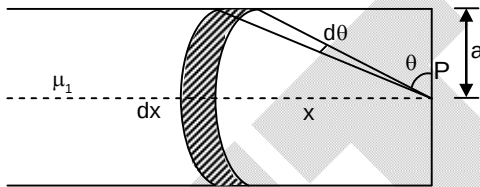
$$x = 0, y = \frac{1}{2}, z = \frac{1}{2}$$

$$i = \frac{x + 1 - y}{1} = \frac{1}{2} \text{ amp}$$

7. B

Sol. Let's calculate the field due to one half by way of integration. (This is just to demonstrate; otherwise the result is obviously half the field due to a solenoid extending to large length on both sides.)

Consider a rig element of angular width $d\theta$ as shown in the figure.



$$x = a \tan \theta$$

$$\therefore dx = a \sec^2 \theta d\theta$$

Number of turns in width dx is ndx

Field due to ring at P is along the axis given by

$$dB = \frac{\mu_0 \mu_1 n dx a^2}{2(a^2 + x^2)^{3/2}} = \frac{\mu_0 \mu_1 n a^2}{2} \frac{(a \sec^2 \theta d\theta)}{(a \sec \theta)^3} \left[\because \sqrt{a^2 + x^2} = a \sec \theta \right]$$

$$dB = \frac{\mu_0 \mu_1 n}{2} \cos \theta d\theta$$

$$B_1 = \frac{\mu_0 \mu_1 n}{2} \int_0^{\pi/2} \cos \theta d\theta = \frac{1}{2} \mu_0 \mu_1 n$$

Similarly, field due to second half will be

$$B_2 = \frac{1}{2} \mu_0 \mu_2 n$$

Resultant field at P is $B = B_1 + B_2 = \frac{1}{2} (\mu_1 + \mu_2) \mu_0 n$

By solving the equation

$$\frac{1}{2}mV^2 = q \frac{Va}{l} - \frac{kq^2a}{2l^2}$$

13. C

Sol. It should possess high susceptibility and poor retentivity.

14. C

Sol. $\vec{B}_0 + \vec{B} = \vec{B}_1$

$$\vec{B}_0 - \vec{B} = \vec{B}_2$$

$$\vec{B} = \frac{\vec{B}_1 - \vec{B}_2}{2} = \frac{1}{2} \mu_0 i$$

15. A

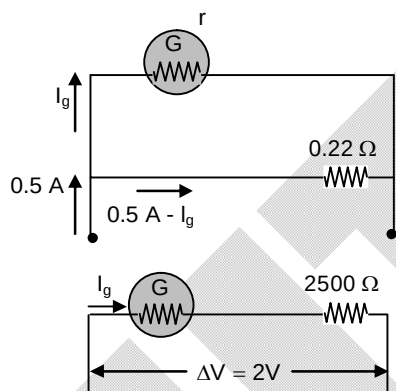
Sol. $N_1 R_1 = N_2 R_2 \Rightarrow R_2 = 2R_1$

$$\Rightarrow \frac{\tau_2}{\tau_1} = \frac{M_2}{M_1} = \frac{N_2 R_2^2}{N_1 R_1^2} = 2 : 1$$

16. C

Sol. For ammeter, we have $I_g r = (0.5A - I_g)(0.22\Omega) \dots (1)$

$$\Rightarrow I_g (r + 0.22\Omega) = 0.11V$$



For voltmeter, we have $2v = I_g (r + 2500\Omega) \dots (2)$

Solve (1) and (2) simultaneously, we get

$$I_g = 0.756 \text{ mA and } r = 145 \Omega$$

17. D

Sol. Resistance of x meter length of wire,

$$R_1 = \frac{R_0}{L} x$$

Resistance R and R_1 are in parallel, so their effective resistance,

$$R' = \frac{RR_1}{R + R_1}$$

This is in series with length $(L - x)$ of wire. Resistance of length $(L - x)$ of wire is

$$R_2 = \frac{R_0}{L} (L - x)$$

Hence, total resistance is given by

$$R_{\text{eff}} = R' + R_2 = \frac{RR_1}{R + R_1} + R_2$$

So, current in the main circuit is given by

$$I = \frac{V_0}{R_{\text{eff}}} = \frac{V_0}{\left(\frac{RR_1}{R + R_1} + R_2 \right)}$$

Potential difference across the length x is

$$V = IR' = I \frac{RR_1}{R + R_1}$$

$$\Rightarrow V = \frac{V_0}{\left(\frac{RR_1}{R + R_1} + R_2 \right)} \left(\frac{RR_1}{R + R_1} \right)$$

$$\Rightarrow V = \frac{V_0 RR_1}{RR_1 + RR_2 + R_1 R_2}$$

$$\Rightarrow V = \frac{V_0 RR_1}{R(R_1 + R_2) + R_1 R_2} = \frac{V_0 RR_1}{R_0 R + R_1 R_2}$$

$$\Rightarrow V = \frac{V_0 R \left(\frac{R_0}{L} x \right)}{R_0 R + \left(\frac{R_0}{L} x \right) \left(\frac{R_0}{L} (L - x) \right)}$$

$$\Rightarrow V = \frac{V_0 R x}{RL + R_0 x \left(1 - \frac{x}{L} \right)}$$

18.

A

Sol.

For downward motion with constant velocity, we have

$$q = 2e \text{ and } m = 3.2 \times 10^{-17} \text{ kg}$$

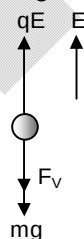
Since motion is downwards, so viscous force is upwards



$$6\pi\eta rv = mg \quad \dots (i)$$

For upwards motion with same constant speed, we have

$$6\pi\eta rv + mg = qE \quad \dots (ii)$$



Substitute (i) in (ii), we get

$$qE = 2mg$$

$$\Rightarrow E = \frac{2mg}{q} = \frac{2 \times 3.2 \times 10^{-17} \times 10}{2 \times 1.6 \times 10^{-19}} = 2 \times 10^3 \text{ Vm}^{-1}$$

19. C

Sol. The spring force $\vec{F}_s$ acting on plate a is given by

$$\vec{F}_s = -kx\hat{i}$$

Similarly, the electrostatic force $\vec{F}_e$ due to the electric field created by plate b is

$$\vec{F}_e = QE\hat{i} = Q\left(\frac{\sigma}{2\epsilon_0}\right)\hat{i} = \frac{Q^2}{2A\epsilon_0}\hat{i}$$

Where A is the area of the plate. Notice that charges on plate a cannot exert a force on itself, as required by Newton's Third Law. Thus, only the electric field due to plate b is considered. At equilibrium the two forces cancel and we have

$$kx = Q\left(\frac{Q}{2A\epsilon_0}\right)$$

Which gives

$$x = \frac{Q^2}{2kA\epsilon_0}$$

Hence, the correct answer is C.

20. B

Sol. Since the potential function is not defined for an infinite conducting sheet, hence to solve this problem we either calculate potential difference or use force equations. Electric field due to an infinite dielectric sheet

$$E_1 = \frac{\sigma}{2\epsilon_0}$$

Electric field at the axis of a disc of radius R is

$$E_2 = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{x}{\sqrt{x^2 + R^2}} \right]$$

The resultant electric field is given by

$$E = E_1 - E_2 = \frac{\sigma}{2\epsilon_0} \frac{x}{\sqrt{x^2 + R^2}}$$

Force on an electron is

$$F = -eE \Rightarrow F = -\frac{\sigma ex}{2\epsilon_0 \sqrt{x^2 + R^2}}$$

$$\Rightarrow mv \frac{dv}{dx} = -\frac{\sigma ex}{2\epsilon_0 \sqrt{x^2 + R^2}}$$

$$\Rightarrow m \int_0^v v dv = -\frac{\sigma e}{2\epsilon_0} \int_{\sqrt{3}R}^0 \frac{x}{\sqrt{x^2 + R^2}} dx$$

$$\Rightarrow \frac{mv^2}{2} = -\frac{\sigma e}{2\epsilon_0} \left(\sqrt{x^2 + R^2} \right) \Big|_{\sqrt{3}R}^0$$

$$\Rightarrow v = \sqrt{\frac{\sigma e R}{m\epsilon_0}}$$

Hence, the correct answer is B.

SECTION – B

21. 1

Sol.
$$\frac{1}{R_{eq}} = \sqrt{\frac{1}{R_1^2} + \frac{1}{R_2^2}} = \sqrt{\frac{1}{1.66^2} + \frac{1}{1.25^2}} = 1$$

22. 8

Sol. When $I = 0$

$$x = 20 \text{ cm}$$

It means
$$V_{AC} = \frac{V_0 x}{L} = \frac{12 \times 20}{100} = 2.4 \text{ volt}$$

Since there is no current through the milliammeter

Hence
$$V_{AC} = V = 2.4 \text{ volt}$$

When $x = 0$; $I = 30 \text{ mA}$

$\therefore 2.4 = I R$

$$2.4 = 30 \times 10^{-3} R$$

$$R = 80 \Omega$$

23. 8

Sol.
$$R_{eq} = \frac{\frac{3R}{q} \times R}{\frac{3R}{q} + R} = \frac{3R}{5}$$

$$m + n = 3 + 5 = 8.$$

24. 2

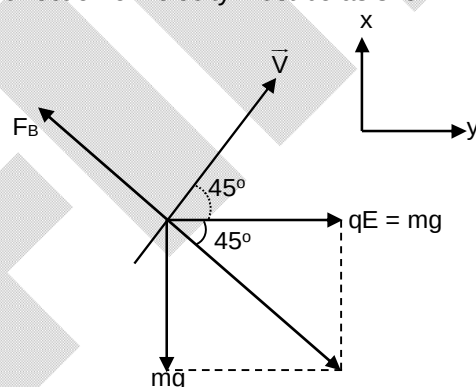
Sol. After the crossing the XX plane, the forces on the particle are

(1) $mg(\downarrow)$

(2) $qE(\rightarrow)$

(3) Magnetic force perpendicular to velocity ($F_B = qVB$)

For particle to move uniformly, the magnetic force must balance the resultant of mg and qE . Hence, direction of velocity must be as shown in figure.



Component of velocity in X and Y direction must be same.

$$V_x = V_y$$

Since, acceleration in y direction before entering magnetic field is $a_y = \frac{qE}{m} = \frac{mg}{m} = g$;

We have

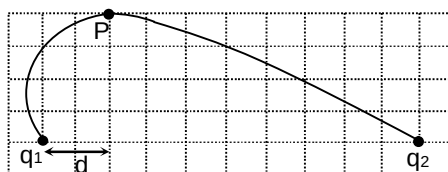
$$u - gT = gT$$

$$u = 2gT$$

25.

8

Sol.



At point P

$$E_{1y} = \frac{kq_1}{4d^2 + d^2} \times \frac{2}{\sqrt{5}}$$

$$E_{2y} = \frac{-kq_2}{4d^2 + 16d^2} \times \frac{1}{\sqrt{5}}$$

$$E_{1y} + E_{2y} = 0$$

$$\Rightarrow \frac{2q_1}{5\sqrt{5}} = \frac{q_2}{20\sqrt{5}}$$

$$\Rightarrow q_2 = -8 \mu\text{C}$$

26.

2

Sol.

$$R = \frac{V}{I} = \frac{1000}{10 \times 10^{-3}} = 100 \text{ k}\Omega$$

$$i = \frac{5 \times 10^3}{400 \times 10^3} = 12.5 \text{ mA}$$

I cannot be greater than 10 mA

$\Rightarrow$ Potential drop across device D = 2kV.

27.

3

Sol.

The electric field strength between the plates is given as

$$E = \frac{V}{d} = \frac{800}{0.02} = 40000 \frac{\text{V}}{\text{m}}$$

For the equilibrium of the particle, we use

$$Mg = Eq$$

$$Mg = E(ne)$$

$$\Rightarrow n = \frac{mg}{Ee} = \frac{1.96 \times 10^{-15} \times 9.8}{40000 \times 1.6 \times 10^{-19}}$$

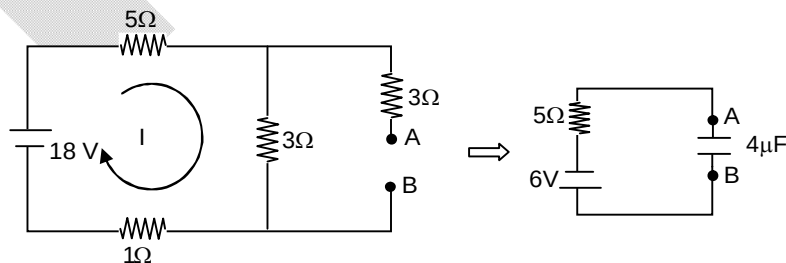
$$\Rightarrow n = 3$$

28.

4

Sol.

To do the transient analysis of the circuit we replace the circuit across the capacitor by an equivalent battery using Thevenin's analysis for which the circuit is redrawn as shown in figure with internal resistance across A and B are given by replacing batteries of circuit by straight wires of zero resistance.



$$r_{AB} = \frac{6 \times 3}{6 + 3} + 3\Omega$$

$$R_{AB} = 2\Omega + 3\Omega = 5\Omega$$

The current in left loop of circuit is given as

$$I = 2A$$

Open circuit potential difference across terminals A and B is given by writing equation of potential drop from A to B as

$$V_A - 3 \times 2 = V_B$$

$$\Rightarrow V_A - V_B = 6V$$

Now charge as a function of time on capacitor plates can be calculated by using equation-(110) which is given as

$$q_C = CV \left(1 - e^{-\frac{t}{RC}} \right)$$

$$\Rightarrow q_C = 24 \left(1 - e^{-\frac{t}{20}} \right) \mu C$$

29. 6

Sol. In horizontal plane, the magnetic needle oscillates due to horizontal component of earth's magnetic field B_H for which the time period of oscillation is given as

$$T = 2\pi \sqrt{\frac{I}{MB_H}}$$

Above result of time period is already discussed earlier in an illustrative example. Students are advised to remember this result for oscillation of a magnetic in magnetic field.

In the vertical north-south plane in magnetic meridian the needle oscillates in the total earth's magnetic field B_E and in vertical east-west plane which is a plane perpendicular to the magnetic meridian it oscillates only in earth's vertical component B_V . If the time period be T_1 and T_2 , then we have

$$T_1 = 2\pi \sqrt{\frac{I}{MB_E}} \text{ and } T_2 = 2\pi \sqrt{\frac{I}{MB_V}}$$

From above equations, we have

$$\frac{T_1^2}{T^2} = \frac{B_H}{B_E}$$

If n is the initial frequency of oscillation in horizontal plane then we have

$$\frac{n_1^2}{n^2} = \frac{B_E}{B_H}$$

And
$$\frac{n_2^2}{n^2} = \frac{B_V}{B_H}$$

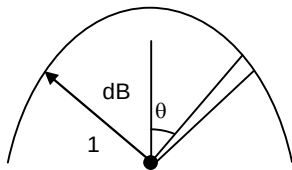
We also have
$$\frac{B_E}{B_H} = \sec \theta$$

By solving above equation,

$$\theta = 30^\circ$$

30. 2

Sol.
$$dB \cos \theta = \frac{\mu_0}{4\pi} \frac{Q}{\pi R^2} V \cos \theta d\theta$$



$$B = \int_{-\pi/2}^{\pi/2} dB \cos \theta = \frac{\mu_0 Q V}{2\pi^2 R^2} \Rightarrow x \times y \times z = 2$$

Chemistry**PART – B****SECTION – A**

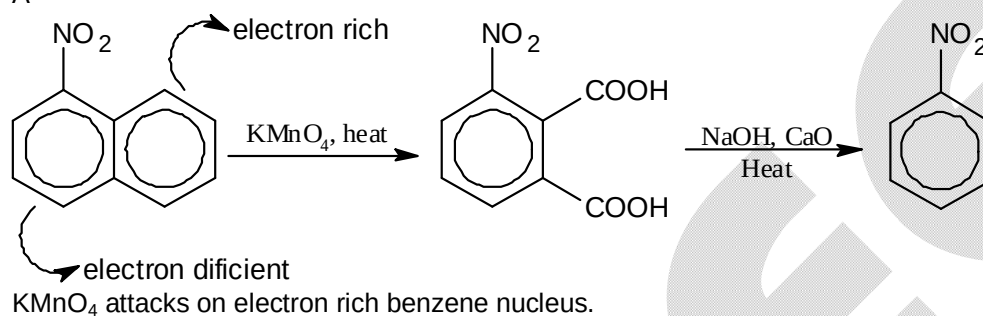
31. B

Sol. A = Cu (Ullman reaction)

B = Zn + Et₂O (Fitting reaction)

32. A

Sol.

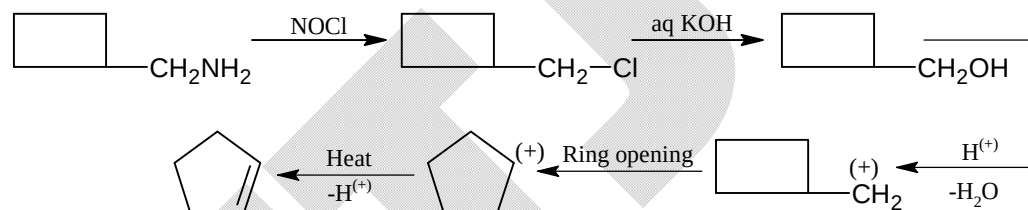


33. C

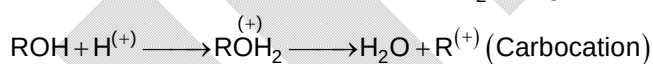
Sol. Along with ortho and para products meta derivative also forms with 47%.

34. A

Sol.

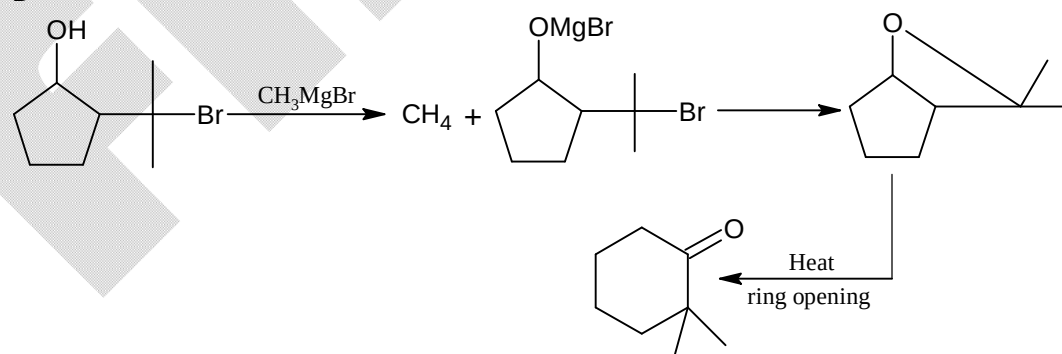


35. B

Sol. Because in acidic medium it removes H₂O & generates carbocation

36. D

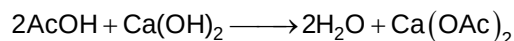
Sol.



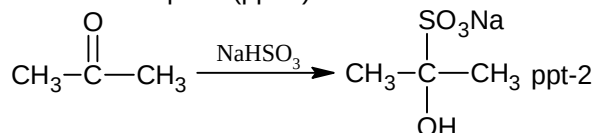
37.

D

Sol. In ppt(1) acetic acid separates as cal. Acetate



After separation solution contains MeOH & propanone. Propanone separates as propanone sodium bi sulphite (ppt-2)

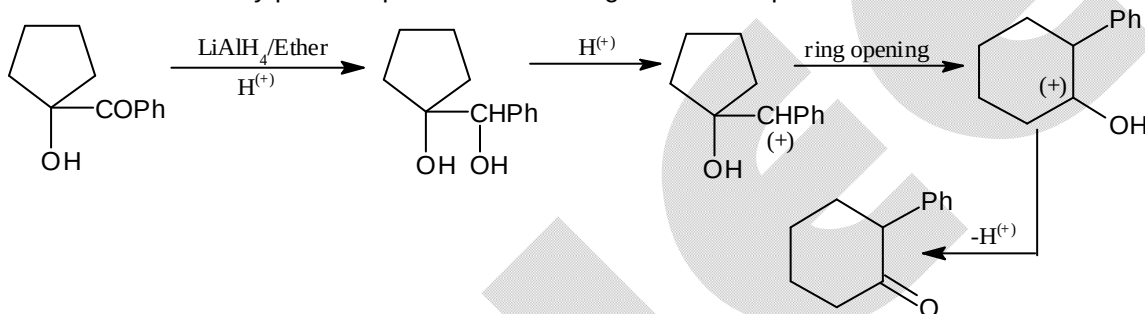


Thus, solution contains methyl alcohol.

38.

C

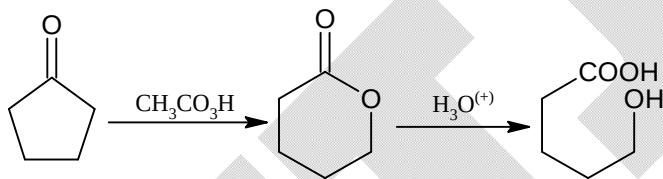
Sol. Reduction followed by pinacole pinacolone rearrangement takes place.



39.

D

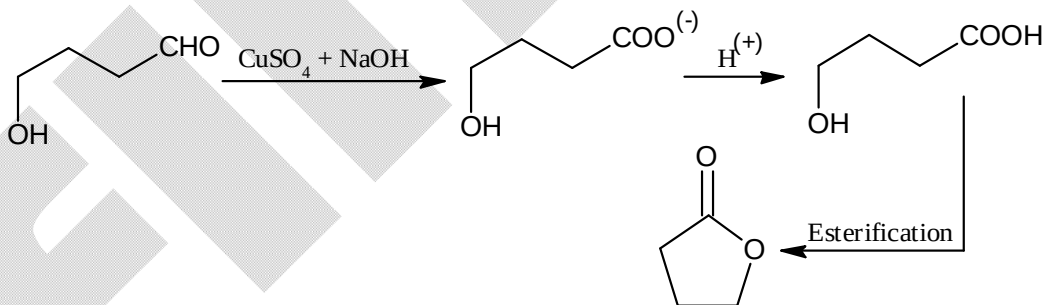
Sol.



40.

C

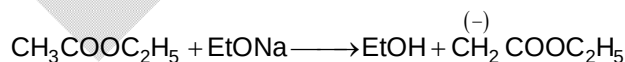
Sol.

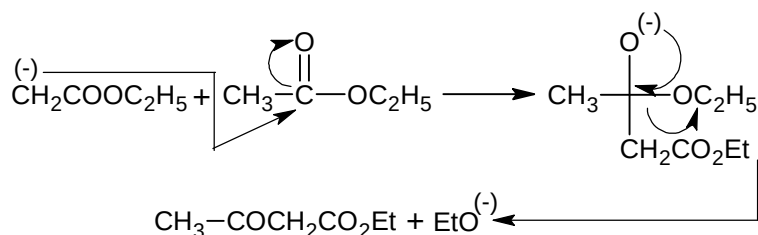


41.

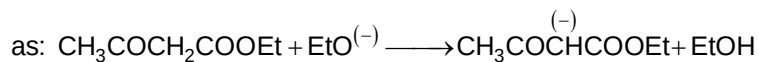
A

Sol. This is claisen condensation:



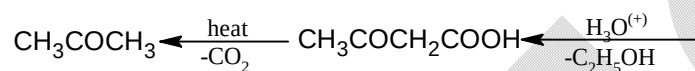
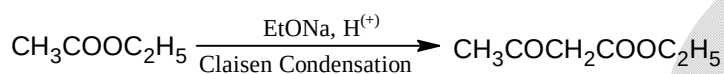


If we do not add $\text{H}^{(+)}$ at the last of the reaction then $\text{EtO}^{(-)}$ will remove $\text{H}^{(+)}$ from $\text{CH}_3\text{COCH}_2\text{CO}_2\text{Et}$



42. B

Sol.



43. A

Sol. Formaldehyde form a very hard plastic (Bakelite) with Phenol & $\text{H}^{(+)}$ ion.

44. B

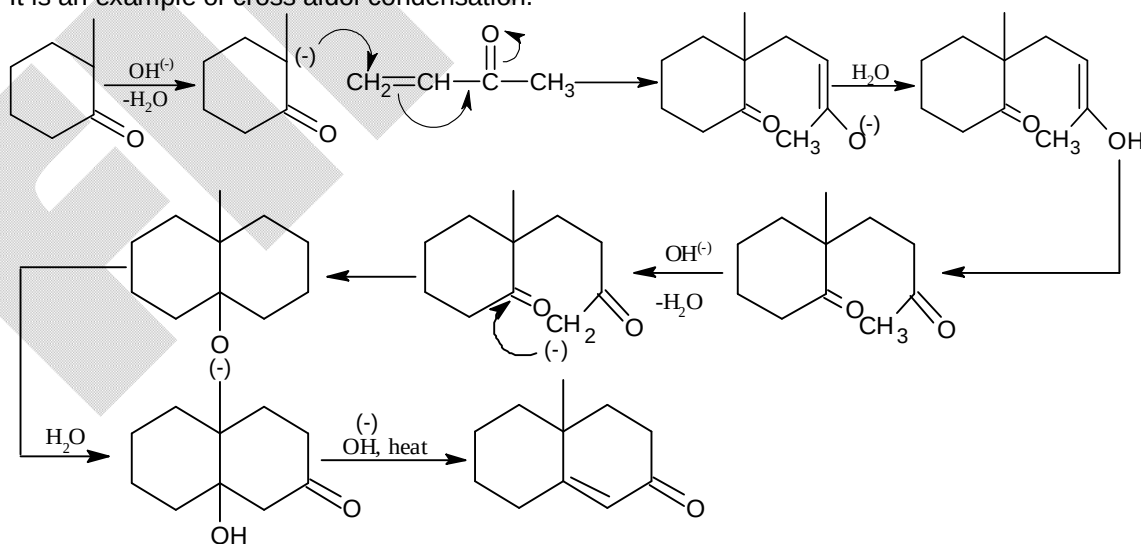
Sol. Phenol shows Liberman's nitroso test with $\text{NaNO}_2 + \text{H}_2\text{SO}_4$.

45. B

Sol. Due to steric hindrance CO group is not reduced by ($\text{N}_2\text{H}_4, \text{NaOH}, \text{heat}$)

46. C

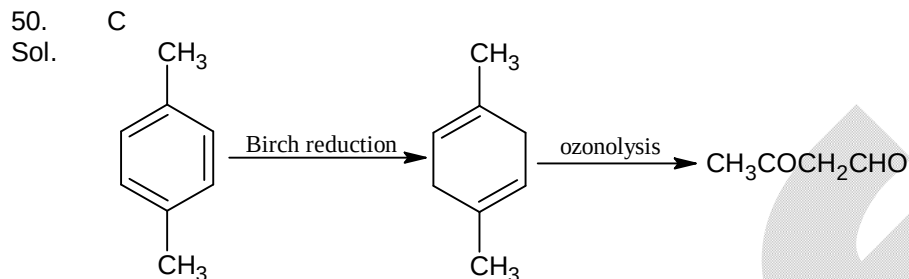
Sol. It is an example of cross aldol condensation.



47. A
Sol. Glyoxal % formaldehyde both are independent from alpha hydrogen atom thus, these two will show cross Cannizzaro reaction

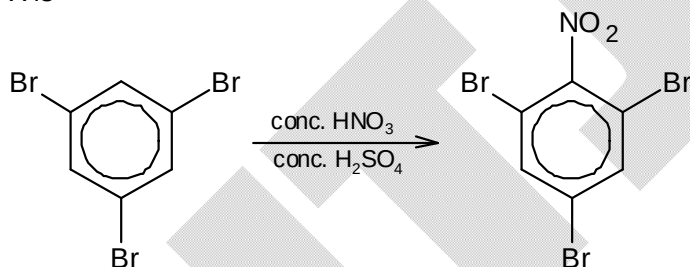
48. B
Sol. Iso electric point is the average of p^{K_a} of similar groups.
Iso electric point = $(1.88 + 3.65) / 2 = 2.77$

49. D
Sol. Keto group cannot be oxidized by bromine water.



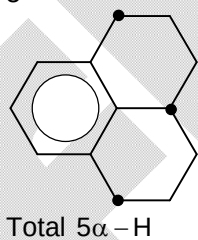
SECTION - B

51. 1
Sol. X is



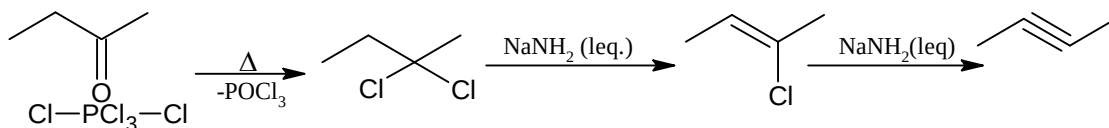
Here all position are identical hence, only one product will be obtained after mono nitration.

52. 5
Sol.



53. 1
Sol. Theoretical

54. 3
Sol.

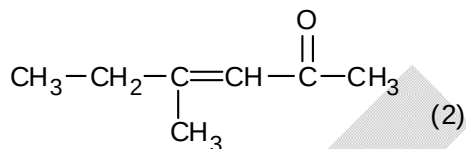
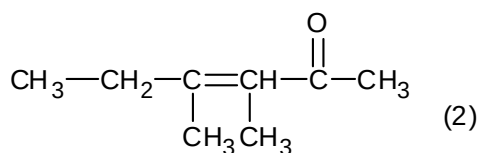
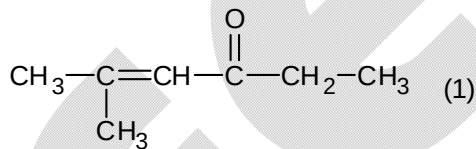
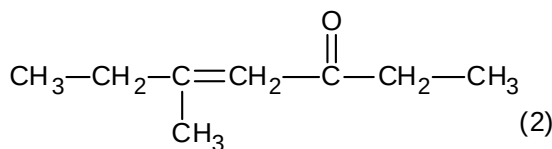
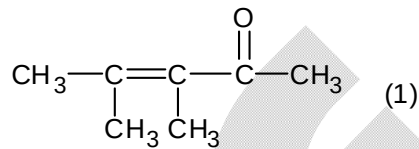
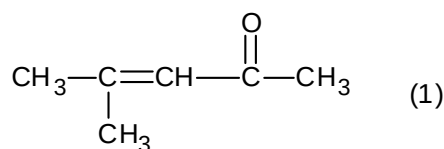


55. 4

 Sol. a. D-Glucose, b. D-Fructose (α -hydroxy ketone)

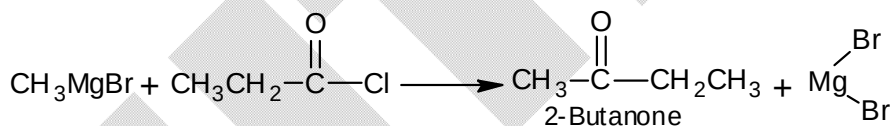

56. 9

Sol.

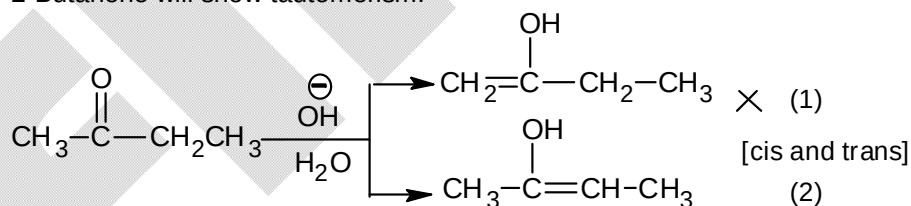

 Total products = $1 + 2 + 2 + 2 + 1 + 1 = 9$

57. 3

Sol.



2-Butanone will show tautomerism.


 Total enol products = $1 + 2 = 3$.

58. 3

Sol. 2.68 gm of (A) gives 14.08 of AgI

$$134 \text{ gm of (A) gives } \frac{14.08 \times 134}{2.68}$$

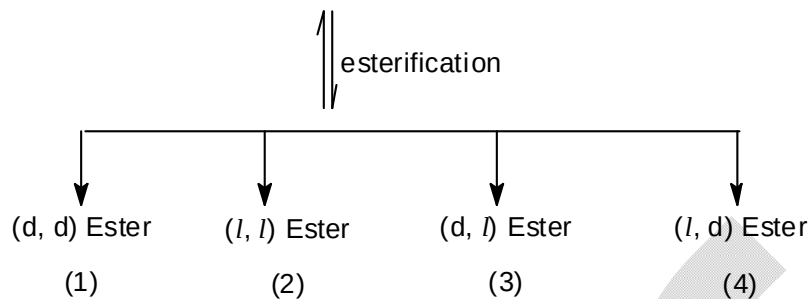
= 704 gm of AgI

$$= \frac{704}{235} = \text{mol of AgI}$$

$$= 3 \text{ (OMe) groups}$$

59. 4

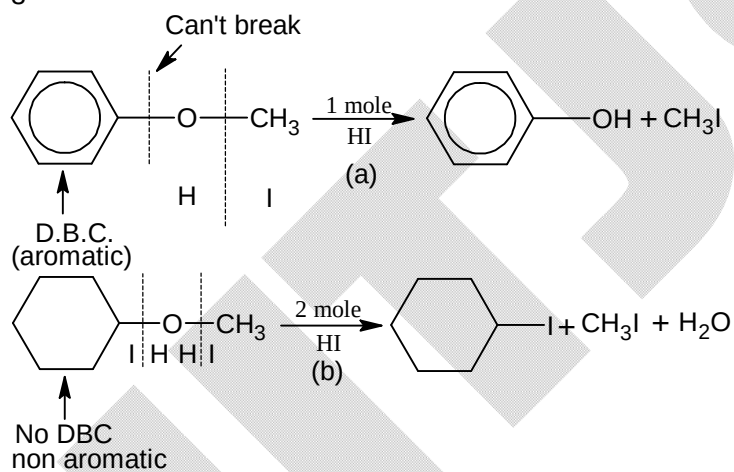
Sol. (dl) 2-methyl butanoic acid and (dl) 2-butanol



Ester consists of two stereocentres. Chiral center during whole reaction are not effected, that's why all ester are optical active.

60. 3

Sol.



Mathematics

PART – C

SECTION – A

61. D

Sol. In this case base is variable. Thus we must take two separate cases:

 (i) $|x| \in (0, 1)$. In this case we have to ensure that $0 < x^2 + x + 1 \leq 1 \Rightarrow x \in [-1, 0]$

 Common part of $|x| \in (0, 1)$ and $x \in [-1, 0]$ is $x \in (-1, 0)$

 (ii) $|x| > 1$. In this case we have must have

$$\begin{aligned} x^2 + x + 1 &\geq 1 \\ \Rightarrow x &\in (-\infty, -1] \cup [0, \infty) \end{aligned}$$

 Common part of $|x| > 1$ and $x \in (-\infty, -1] \cup [0, \infty)$ is $(-\infty, -1) \cup (1, \infty)$

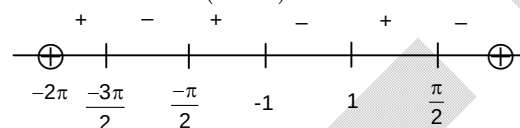
Thus, the final solution is

$$x \in (-\infty, -1) \cup (-1, 0) \cup (1, \infty)$$

62. D

$$\text{Sol. } |x^2 - 1 + \cos x| = |x^2 - 1| + |\cos x|$$

 It implies that $(x^2 - 1) \cdot \cos x \geq 0$ because $|x + y| = |x| + |y|$ if $xy \geq 0$.

 Sign scheme of $(x^2 - 1)\cos x$ is


Thus solution is

$$\left[-\frac{\pi}{2}, -1\right] \cup \left[1, \frac{\pi}{2}\right] \cup \left(-2\pi, \frac{-3\pi}{2}\right]$$

63. B

Sol. We must have

$$\begin{aligned} 2\{x\}^2 - 3x + 1 &\geq 0 \\ \Rightarrow \{x\} &\geq \{x\} \text{ or } \{x\} \leq \frac{1}{2} \end{aligned}$$

 Thus we have $0 \leq \{x\} \leq \frac{1}{2}$

$$\Rightarrow x \in \left[n, n + \frac{1}{2}\right], n \in \mathbb{I}$$

64. D

$$\text{Sol. } \cos^{-1}\left(\frac{1+x^2}{2x}\right) \text{ is defined if } \left|\frac{1+x^2}{2x}\right| \leq 1$$

$$\Rightarrow x \neq 0$$

$$\Rightarrow 1 + x^2 - 2|x| \leq 0$$

$$\Rightarrow (|x| - 1)^2 \leq 0$$

$$\Rightarrow x = 1, -1$$

Thus domain of $f(x)$ is $\{1, -1\}$.

Hence range is $\{1, 1 + \pi\}$

65. B

Sol. $[\sin^{-1} x] = 0, \forall x \in [0, \sin 1]$

$$= 1, \forall x \in [\sin 1, 1]$$

$$\text{and } [\cos^{-1} x] = \begin{cases} 1, & \forall x \in [0, \cos 1] \\ 0, & \forall x \in [\cos 1, 1] \end{cases}$$

Thus $[\sin^{-1} x] > [\cos^{-1} x]$
 $x \in [\sin 1, 1]$

66. C

Sol. $\lim_{x \rightarrow 0} \frac{x^a \cdot \sin^b x}{\sin x^c}$

$$= \lim_{x \rightarrow 0} x^a \cdot \left(\frac{\sin x}{x}\right)^b \cdot \frac{x^c}{\sin x^c} \cdot x^{b-c}.$$

$$= \lim_{x \rightarrow 0} x^{a+b-c} \left(\frac{\sin x}{x}\right)^b \frac{x^c}{\sin x^c}$$

This limit will have non-zero value if $a + b = c$.

67. B

Sol. $\lim_{x \rightarrow 0} \frac{\sin 6x^2}{\ln \cos(2x^2 - x)}$

$$= \lim_{x \rightarrow 0} \frac{(\sin 6x^2) 6x^2}{6x^2 \cdot \left(1 - 2\sin^2\left(\frac{2x^2 - x}{2}\right)\right)}$$

$$= \lim_{x \rightarrow 0} \frac{\sin 6x^2}{6x^2} \cdot \frac{6x^2}{-2\sin^2\left(\frac{2x^2 - x}{2}\right)} \cdot \frac{-2\sin^2\left(\frac{2x^2 - x}{2}\right)}{\ln\left(1 - 2\sin^2\frac{2x^2 - x}{2}\right)}$$

$$= -3 \cdot \lim_{x \rightarrow 0} \frac{x^2}{\sin^2\left(\frac{2x^2 - x}{2}\right)}$$

$$= -3 \cdot \lim_{x \rightarrow 0} \frac{x^2 \cdot \left(\frac{2x^2 - x}{2}\right)^2}{\left(\frac{2x^2 - x}{2}\right)^2 \cdot \sin^2\left(\frac{2x^2 - x}{2}\right)} = -3 \times \frac{1}{\frac{1}{4}} = -12$$

68. C

Sol. Putting $x = 0$, we get

$$|f(x)| \leq 0$$

$$\Rightarrow f(0) = 0$$

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h - 0}$$

$$= \lim_{h \rightarrow 0} \frac{f(h)}{h}$$

$$\text{Now } \left| \frac{f(h)}{h} \right| \leq |h|$$

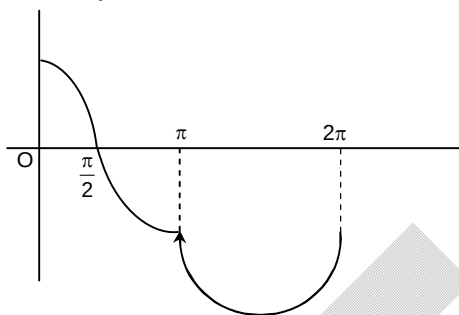
$$\Rightarrow -|h| \leq \frac{f(h)}{h} \leq |h|$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(h)}{h} = 0.$$

Thus $f(x)$ is differentiable at $x = 0$.

69. B

$$\text{Sol. } g(x) = \begin{cases} \cos x, & 0 \leq x \leq \pi \\ \sin x - 1, & x > \pi \end{cases}$$



Adjacent figure represents the graph of $g(x)$. Clearly, $g(x)$ is continuous but non-differentiable at $x = \pi$.

70. D

Sol. Curves will intersect if $x^2 - 16x + c = 0$ has real roots.

Thus, $c \leq 64$.

$$\text{For } x^2 - 4y^2 + c = 0, \quad \frac{dy}{dx} = \frac{x}{4y}$$

$$\text{For } y^2 = 4x, \quad \frac{dy}{dx} = \frac{2}{y}$$

If curves intersect orthogonally, then

$$\frac{x}{4y} \cdot \frac{2}{y} = -1$$

$$\Rightarrow -2y^2 = -8x$$

$$\Rightarrow x = 0$$

$$\Rightarrow y = 0.$$

But if $y = 0$, slope of both curves are undefined.

71. D

Sol. Let Δ_1 and Δ_2 be area of squares S_1 and S_2 . Then

$$\Delta_1 = l_1^2, \Delta_2 = l_2^2$$

$$\Rightarrow \frac{d\Delta_1}{dl_1} = 2l_1, \frac{d\Delta_2}{dl_2} = 2l_2$$

$$\Rightarrow \frac{d\Delta_2}{d\Delta_1} = \frac{l_2}{l_1} \cdot \frac{dl_2}{dl_1} = \frac{l_2}{l_1(1+3l_2)}$$

When $l_2 = 1, l_1 = 8$, then

$$\Rightarrow \frac{d\Delta_2}{d\Delta_1} = \frac{1}{32}$$

72. D

Sol. For $f(x)$ to be increasing $\forall x \in \mathbb{R}$, we must have

$$\log_3(\log_{1/4}(\cos x + a)) > 0, \forall x \in \mathbb{R}$$

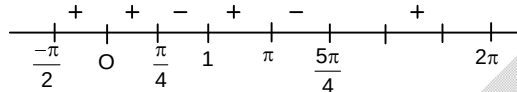
$$\Rightarrow 0 < \cos x + a < \frac{1}{4} \quad \forall x \in \mathbb{R}$$

$$\Rightarrow -\cos x < a < \frac{1}{4} - \cos x, \forall x \in \mathbb{R}$$

73. C

Sol. $f'(x) = (e^x - 1)(x - 1)(\sin x - \cos x)\sin x$

Sign scheme of $f'(x)$ is



Clearly $f(x)$ is increasing in

$$\left(-\frac{\pi}{2}, \frac{\pi}{4}\right) \cup (1, \pi) \cup \left(\frac{5\pi}{4}, 2\pi\right) \text{ and decreasing in } \left(\frac{\pi}{4}, 1\right) \cup \left(\pi, \frac{5\pi}{4}\right)$$

74. C

Sol. Let the equation of drawn line be $\frac{x}{a} + \frac{y}{b} = 1$.

Where $a > 3, b > 4$.

As the line passes through $(3, 4)$ and meets the positive direction of co-ordinate axis, we have

$$\frac{3}{a} + \frac{4}{b} = 1.$$

$$\Rightarrow b = \frac{4a}{(a-3)}$$

Now area of triangle OAB,

$$\Delta = \frac{1}{2}ab = \frac{2a^2}{(a-3)} \cdot \frac{d\Delta}{da} = \frac{2a(a-6)}{(a-3)^2}$$

Clearly, $a = 6$ is the point of minima for Δ .

$$\text{Thus, } \Delta_{\min} = \frac{2 \times 36}{3} = 24 \text{ sq. units.}$$

75. A

$$\text{Sol. } I = \int e^x \left(\frac{1-x}{1+x^2} \right)^2 dx$$

$$= \int e^x \frac{(1-x^2-2x)}{(1+x^2)^2} dx$$

$$\begin{aligned}
 &= \int e^x \left(\frac{1}{1+x^2} - \frac{2x}{(1+x^2)^2} \right) dx \\
 &= \frac{1}{1+x^2} \cdot e^x + \int \frac{2x \cdot e^x}{(1+x^2)^2} dx \\
 &\quad - \int \frac{2x \cdot e^x}{(1+x^2)} dx \\
 &= \frac{e^x}{1+x^2} + C
 \end{aligned}$$

76. D

Sol.
$$\int \frac{dx}{x^2 \sqrt{1+x^2}} = \int \frac{dx}{x^3 \sqrt{1+\frac{1}{x^2}}}$$

Let $t = \sqrt{1+\frac{1}{x^2}}$

$$\Rightarrow \frac{dt}{dx} = \frac{1 - \sqrt{\frac{2}{x^3}}}{2\sqrt{1+\frac{1}{x^2}}}$$

$$\Rightarrow \frac{dx}{x^3} = -tdt$$

$$\Rightarrow I = -\int \frac{tdt}{t} = -t + C$$

$$= -\sqrt{1+\frac{1}{x^2}} + C$$

$$= \frac{1}{x} \sqrt{1+x^2} + C$$

77. A

Sol.
$$I = \int \frac{dx}{(1+\sqrt{x})\sqrt{x-x^2}}$$

$$= \int \frac{(1-\sqrt{x})}{(1-x)\sqrt{x}(1-x)^{1/2}} dx$$

$$= \int \frac{(1-\sqrt{x})dx}{\sqrt{x}(1-x)^{3/2}}$$

$$= \int \frac{dx}{x^2 \left(\frac{1}{x} - 1 \right)^{3/2}} - \int \frac{dx}{(1-x)^{3/2}}$$

$$\begin{aligned}
 &= -\int \frac{d\left(\frac{1}{x}-1\right)}{\left(\frac{1}{x}-1\right)^{3/2}} - \int \frac{dx}{(1-x)^{3/2}} \\
 &= \frac{2}{\left(\frac{1}{x}-1\right)^{1/2}} - \frac{2}{(1-x)^{1/2}} + C \\
 &= \frac{2\sqrt{x}}{\sqrt{1-x}} - \frac{2}{\sqrt{1-x}} + C \\
 &= \frac{2(\sqrt{x}-1)}{\sqrt{1-x}} + C
 \end{aligned}$$

78. A

Sol. $I = -\int x \cdot \frac{\ln(x + \sqrt{x^2 + 1})}{\sqrt{x^2 + 1}} dx$

Let $t = \sqrt{x^2 + 1}$

$$\Rightarrow \frac{dt}{dx} = \frac{x}{\sqrt{x^2 + 1}}$$

$$\Rightarrow I = \int \ln(t + \sqrt{t^2 - 1}) dt$$

$$= \ln(t + \sqrt{t^2 - 1}) \cdot t - \int \frac{1 + \frac{1}{\sqrt{t^2 - 1}}}{t + \sqrt{t^2 - 1}} \cdot t dt$$

$$= t \cdot \ln(t + \sqrt{t^2 - 1}) - \frac{1}{2} \int \frac{2t}{\sqrt{t^2 - 1}} dt$$

$$= \sqrt{1+x^2} \cdot \ln(x + \sqrt{1+x^2}) - x + C$$

$$\Rightarrow a = 1, b = -1$$

79. B

Sol. $I = \int_0^a f(x) \cdot g(x) dx$

$$= \int_0^a f(a-x) \cdot g(a-x) dx$$

$$= \int_0^a f(x) \cdot (a - g(x)) dx$$

$$= a \int_0^a f(x) dx - \int_0^a f(x) \cdot g(x) dx$$

$$\Rightarrow 2I = a \int_0^a f(x) dx$$

$$\Rightarrow I = \frac{a}{2} \int_0^a f(x) dx$$

80. B

$$\begin{aligned}
 \text{Sol. } \int_{-3}^5 f(|x|) dx &= \int_{-3}^3 f(|x|) dx + \int_3^5 f(|x|) dx \\
 &= 2 \int_0^3 f(x) dx + \int_3^5 f(x) dx \\
 &= 2 \left(\int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_2^3 f(x) dx \right) + 2 \left(\int_0^1 f(x) dx + \int_4^5 f(x) dx \right) \\
 &= 2 \left(0 + \frac{1}{2} + \frac{2^2}{2} \right) + \left(\frac{9}{2} + \frac{16}{2} \right) = \frac{35}{2}
 \end{aligned}$$

SECTION - B

81. 1

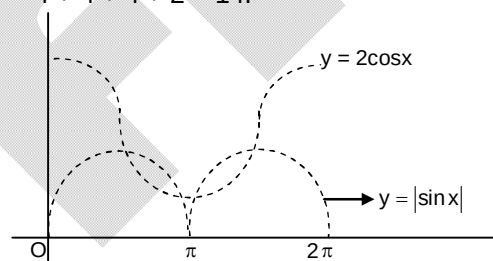
$$\begin{aligned}
 \text{Sol. } 2[x] &= x + \{x\} \\
 \Rightarrow [2x] &= [x] + 2(x) \\
 \Rightarrow \{x\} &= \frac{[x]}{2} \\
 \Rightarrow 0 &\leq \frac{[x]}{2} < 1 \\
 \Rightarrow [x] &= 0, 1 \\
 \text{For } [x] = 0, \text{ we get } \{x\} &= 0 \Rightarrow x = 0 \\
 \Rightarrow x &= 0 \\
 \text{For } [x] = 1, \text{ we get} \\
 \{x\} &= \frac{1}{2} \\
 \Rightarrow x &= \frac{3}{2} \\
 \therefore \text{Number of positive real } x &= 1.
 \end{aligned}$$

82. 14

Sol. Graphs of $y = 2^{\cos x}$ and $y = |\sin x|$ meet four times in $[0, 2\pi]$.

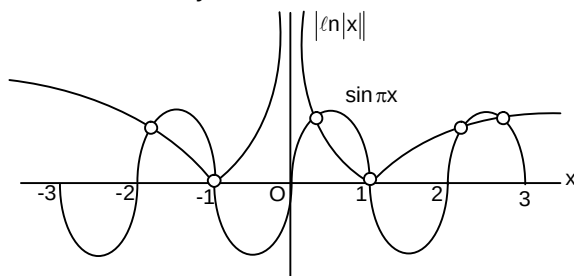
Thus, total number of solutions

$$= 4 + 4 + 4 + 2 = 14.$$

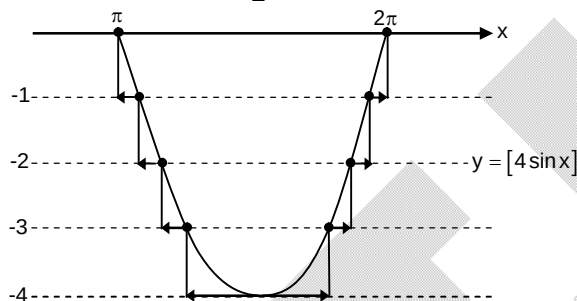


83. 6

Sol. There are exactly six solutions.



84. 8

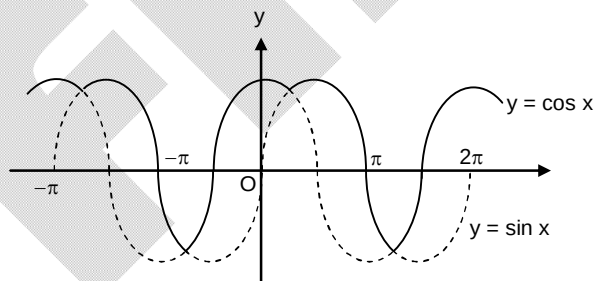
Sol. $f(x) = 3 + [4\sin x]$ Clearly, $f(x)$ is non-differentiable at all these points where $f(x) = [4\sin x]$ is non-differentiable.In $[\pi, 2\pi]$ $g(x)$ is clearly non differentiable at eight points. It should be noted that $g(x)$ is differentiable at $x = \frac{3\pi}{2}$.

85. 4

Sol. Bold line represents the graph of $y = f(x)$. Clearly $f(x)$ is non-differentiable at

$$x = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{3\pi}{4}, \frac{7\pi}{4},$$

$$\text{Also, } f'(x) = 0 \text{ at } x = -\frac{3\pi}{4}, 0, \frac{\pi}{2}.$$

Thus there are 7 critical points in $(-2\pi, 2\pi)$.

86. 5

Sol. $\int \tan^4 x dx = a \tan^3 x + b \tan x + cx + d$

Differentiating both sides, we get

$$\tan^4 x = 3a \tan^2 x \cdot \sec^2 x + b \sec^2 x + c$$

$$\begin{aligned}\sin^4 x &= 3a \sin^2 x + b \cos^2 x + c \cos^4 x \\ (1 - \cos^2 x)^2 &= 3a \sin^2 x + b \cos^2 x + c \cos^4 x \\ \Rightarrow 1 + \cos^4 x - 2 \cos^2 x &= 3a + (b - 3a) \cos^2 x + c \cos^4 x \\ \Rightarrow 3a &= 1, c = 1, b - 3a = -2 \\ \Rightarrow a &= \frac{1}{3}, b = -1, c = 1 \\ \therefore 15(a + b + c) &= 5\end{aligned}$$

87. 10

Sol. $I = \int \frac{1-x^7}{x(1+x^7)} dx$

$$= a \ln|x| + b \ln|1+x^7| + C$$

Differentiating both sides, we get

$$\frac{1-x^7}{x(1+x^7)} = \frac{a}{x} + b \cdot \frac{7x^6}{1+x^7}$$

$$\Rightarrow a = 1, a + 7b = -1$$

$$\Rightarrow b = -\frac{2}{7}$$

$$\therefore 14(a+b) = 10$$

88. 9

Sol. $x = \int_0^y \frac{dt}{\sqrt{1+9t^2}}$

$$\Rightarrow \frac{dx}{dy} = \frac{1}{\sqrt{1+9y^2}}$$

$$\Rightarrow \frac{dx}{dy} = \frac{1}{\sqrt{1+9y^2}}$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{(18y)}{2\sqrt{1+9y^2}} \cdot \frac{dy}{dx} = 9y$$

$$\Rightarrow a = 9$$

89. 0

Sol. $f(x) = \int_{1/e}^{\tan x} \frac{tdt}{(1+t^2)} + \int_{1/e}^{\cot x} \frac{dt}{t(1+t^2)}$

$$\Rightarrow f'(x) = \frac{\tan x}{1+\tan^2 x} \cdot \sec^2 x + \frac{1}{\cot x(1+\cot^2 x)} (-\operatorname{cosec}^2 x)$$

$$\Rightarrow f'(x) = \frac{\tan x}{1+\tan^2 x} \cdot \sec^2 x + \frac{1}{\cot x(1+\cot^2 x)} - \operatorname{cosec}^2 x$$

$$= \tan x - \tan x = 0$$

90. 4

Sol. Let $L = \frac{n}{n^2} + \frac{n}{n^2 + 2^2} + \dots + \frac{n}{n^2 + (n-1)^2}$

$$= \sum_{r=0}^{n-1} \frac{n}{n^2 + r^2}$$

$$= \frac{1}{n} \sum_{r=0}^{n-1} \frac{1}{1 + \frac{r^2}{n^2}}$$

Thus the given limit is equal to

$$\int_0^1 \frac{dx}{1+x^2} = \tan^{-1} x \Big|_0^1 = \frac{\pi}{4}$$

$$\therefore \frac{16}{\pi} \cdot L = 4$$

FIITJEE
ALL INDIA INTEGRATED TEST SERIES
JEE (Advanced)-2024
PART TEST – IV
PAPER –1
TEST DATE: 20-10-2023

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 06, 19 – 24, 37 – 42): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A (07 – 10, 25 – 28, 43 – 46): This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

Physics

PART – I

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

- A nucleus X of mass M, initially at rest, undergoes alpha decay according to the equation ${}^A_Z X \rightarrow {}^{A-4}_{Z-2} Y + {}^4_2 \text{He}$. The alpha particle emitted in the above process is found to move in a circular track of radius r in a uniform magnetic field B. Then (mass and charge of α -particle are m and q respectively)

(A) The ratio of kinetic energies of the alpha particle and the daughter nucleus Y approximately equals $\frac{M-m}{m}$

(B) The ratio of kinetic energies of the alpha particle and the daughter nucleus Y approximately equals $\frac{m}{M}$

(C) The energy released in the process approximately equals $\frac{M}{M-m} \left(\frac{r^2 q^2 B^2}{2m} \right)$

(D) The energy released in the process approximately equals $\frac{m}{M-m} \left(\frac{r^2 q^2 B^2}{2m} \right)$
- A diverging lens of focal length f_1 is placed in front of and coaxially with a concave mirror of focal length f_2 . Their separation is d. A parallel beam of light incident on the lens returns as a parallel beam from the arrangement. Then,

(A) The beam diameters of the incident and reflected beams must be the same

(B) $d = 2|f_2| - |f_1|$

(C) $d = |f_2| - |f_1|$

(D) If the entire arrangement is immersed in water, the conditions will remain unaltered
- Let λ_α , λ_β and λ'_α denote the wavelengths of the X-rays of the K_α , K_β , and L_α lines in the characteristic X-rays for a metal. Then,

(A) $\lambda'_\alpha > \lambda_\alpha > \lambda_\beta$

(B) $\lambda'_\alpha > \lambda_\beta > \lambda_\alpha$

(C) $\frac{1}{\lambda_\beta} = \frac{1}{\lambda_\alpha} + \frac{1}{\lambda'_\alpha}$

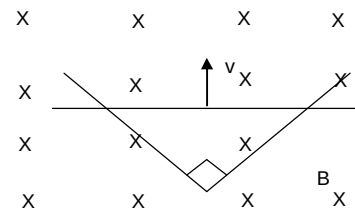
(D) $\frac{1}{\lambda_\alpha} = \frac{1}{\lambda_\beta} + \frac{1}{\lambda'_\alpha}$
- Two straight conducting rails form a right angle where their ends are joined. A conducting bar in contact with the rails starts at the vertex at time $t = 0$ and moves with constant velocity v along them as shown in figure. A magnetic field $\vec{B}$ is directed into the page. The induced emf in the circuit at any time t is proportional to

(A) t^2

(B) t

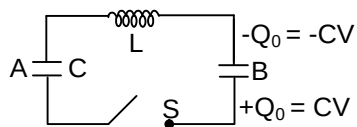
(C) v

(D) v^2

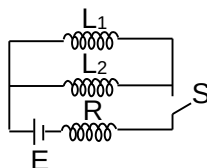


5. An inductor and two capacitors are connected in the circuit as shown in figure. Initially capacitor A has no charge and capacitor B has CV charge. Assume that the circuit has no resistance at all. At $t = 0$, switch S is closed, then

[Given $LC = \frac{2}{\pi^2 \times 10^4} \text{ s}^2$ and $CV = 100 \text{ mC}$]



- (A) When current in the circuit is maximum, charge on each capacitor is same
 (B) When current in the circuit is maximum, charge on capacitor A is twice the charge on capacitor B.
 (C) $q = 50(1 + \cos 100\pi t) \text{ mC}$, where q is the charge on capacitor B at time t
 (D) $q = 50(1 - \cos 100\pi t) \text{ mC}$, where q is the charge on capacitor B at time t
6. In the circuit, a battery of emf E , a resistance R and inductance coil L_1 and L_2 and switch S are connected as shown in figure. Initially the switch is open.



- (A) The time constant of the circuit is $\frac{1}{R} \left(\frac{L_1 L_2}{L_1 + L_2} \right)$
 (B) Steady state current in the inductor L_1 is $\frac{EL_2}{R(L_1 + L_2)}$
 (C) Steady state, current in the inductor L_2 is $\frac{EL_1}{R(L_1 + L_2)}$
 (D) In steady state the total energy stored in the inductor coils is $\frac{1}{2} \left(\frac{L_1 L_2}{L_1 + L_2} \right) \frac{E^2}{R^2}$

Section – A (Maximum Marks: 12)

This section contains **FOUR (04) Matching List Type Questions**. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

7. Two transparent media of refractive indices μ_1 and μ_3 have a solid lens shaped transparent material of refractive index μ_2 between them as shown in figures in Column II. A ray traversing these media is also shown in the figures. In List I different relationships between μ_1, μ_2 and μ_3 are given. Match them to the ray diagram shown in List II.

List - I		List - II	
(I)	$\mu_1 < \mu_2$	(P)	

(II)	$\mu_1 > \mu_2$	(Q)	
(III)	$\mu_2 = \mu_3$	(R)	
(IV)	$\mu_2 > \mu_3$	(S)	
		(T)	

Codes:

	I	II	III	IV
(A)	PR	QST	PRT	QS
(B)	RP	ST	TRP	QST
(C)	ST	RPT	QST	PR
(D)	PRT	QST	PR	QS

8.

Match List-I with List-II:

List - I		List - II	
(I)	Radius of orbit depends on principal quantum number as	(P)	Increase
(II)	Due to orbital motion of electron, magnetic field arises at the center of nucleus is proportional to principal quantum	(Q)	Decrease
(III)	If electron is going from lower energy level to higher energy level, then velocity of electron will	(R)	Proportional $\frac{1}{n^2}$
(IV)	If electron is going from lower energy level to energy level, then total energy of electron will	(S)	Proportional to n^2
		(T)	Proportional to $\frac{1}{n^5}$

Codes:

	I	II	III	IV
(A)	S	T	Q	P
(B)	T	S	P	Q
(C)	S	T	P	Q
(D)	P	Q	R	S

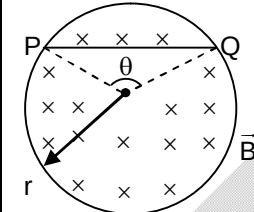
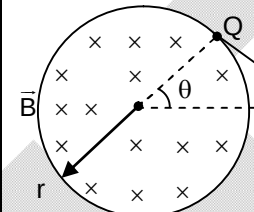
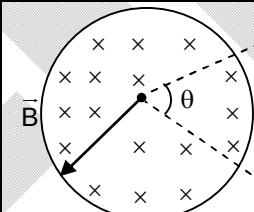
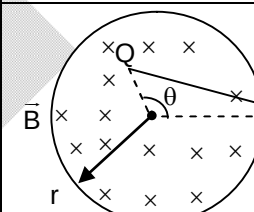
9. Match List-I with List-II:

List - I		List - II	
(I)	Nuclear fusion	(P)	Satisfies $E = mc^2$
(II)	Nuclear fission	(Q)	Generally possible for nuclei with low atomic number
(III)	β - decay	(R)	Generally possible for nuclei with higher atomic number and unstable
(IV)	Exothermic nuclear reaction	(S)	Essentially proceeds by weak nuclear forces
		(T)	Significant moment conservation

Codes:

	I	II	III	IV
(A)	PQT	PR	SQ	Q
(B)	PQT	PR	S	QS
(C)	PR	PQT	PQ	QS
(D)	QT	PQ	ST	PR

10. List I shows the cylindrical region of radius r where a downward magnetic field $\vec{B}$ exists, where $\vec{B}$ is increasing at the rate of dB/dt . A rod PQ is placed in different citation as shown. Match the List I with the correct statement in List II regarding the induced emf in rod.

List - I		List - II	
(I)		(P)	Induced emf in rod PQ is $\frac{1}{2}r^2\theta \frac{dB}{dt}$.
(II)		(Q)	Induced emf in rod PQ is less than $\frac{1}{2}r^2\theta \frac{dB}{dt}$.
(III)		(R)	End P is positive with respect to point Q.
(IV)		(S)	End Q is positive with respect to point P.
		(T)	None of these

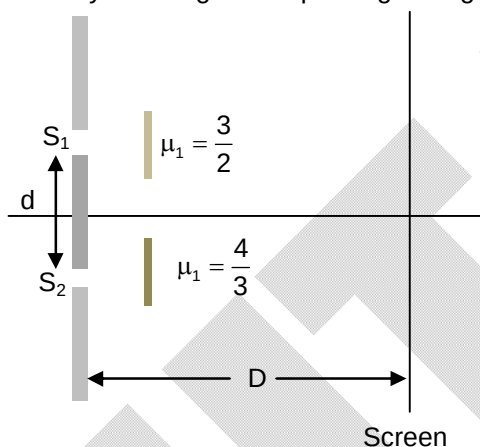
Codes:

	I	II	III	IV
(A)	QR	PS	PR	QS
(B)	RQ	SP	QS	PR
(C)	PQ	RS	S	T
(D)	QS	PR	T	QS

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

11. A parallel beam of light is incident on a transparent sphere of refractive index 'n'. If the beam finally gets focused at a point situated at a distance = $2 \times$ (radius of sphere) from the center of the sphere, then find n.
12. A monochromatic beam of light of $6000 \text{ \AA}$ is used in YDSE set-up. The two slits are covered with two thin films of equal thickness t but of different refractive indices as shown in figure. Considering the intensity of the incident beam on the slits to be I_0 , find the point on the screen at which intensity is I_0 and is just above the central maxima. (Assume that there is no change in intensity of the light after passing through the films.)



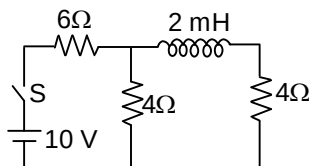
Consider $t = 6 \mu\text{m}$, $d = 1 \text{ mm}$, and $D = 1 \text{ m}$, where d and D have their usual meaning. Give your answer in mm.

13. A cylindrical rod of some laser material $5 \times 10^{-2} \text{ m}$ long and 10^{-2} m in diameter contains 2×10^{25} ions per m^3 . If on excitation all the ions are in the upper energy level and de-excite simultaneously emitting photons in the same direction, calculate the maximum energy contained in a pulse of radiation of wavelength $6.6 \times 10^{-7} \text{ m}$. If the pulse lasts for 10^{-7} s , calculate the average power (in W) of the laser during the pulse.
14. The peak emission from a black body at a certain temperature occurs at a wavelength of $9000 \text{ \AA}$. On increasing its temperature, the total radiation emitted is increased 81 times. At the initial temperature when the peak radiation from the black body is incident on a metal surface, it does not cause any photoemission from the surface. After the increase of temperature, the peak radiation from the black body caused photoemission. To bring these photoelectrons to rest, a potential equivalent to the excitation energy between $n = 2$ and $n = 3$ Bohr levels of hydrogen atoms is required. Find the work function of the metal (in eV).

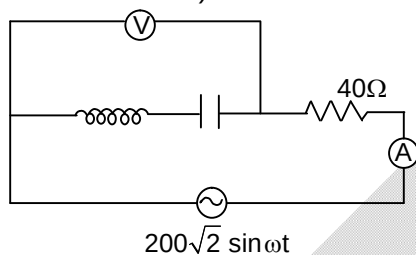
15. A neutron is scattered through ($\equiv$ deviation from its old direction) θ degree in an elastic collision with an initially stationary deuteron. If the neutron loses $\frac{2}{3}$ of its initial KE to the deuteron then find the value of θ (in degree) (In atomic mass unit, the mass of a neutron is 1 u and mass of a deuteron is 2 u).

16. Two concentric coplanar circular loops made of wire, resistance per unit length $10^{-4} \Omega \text{m}^{-1}$, have diameters 0.2 m and 2 m. A time-varying potential difference $(4 + 2.5t)$ volt is applied to the larger loop. Calculate the current in the smaller loop (in A).

17. In the given circuit, what is the current I (in A) drawn from battery at time $t = 0$?



18. In the given LCR series circuit find the reading (in A) of the hot wire ammeter. (Here all hot wire meters are ideal)



Chemistry**PART – II****Section – A (Maximum Marks: 24)**

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

19. Which of the following statements is/are correct for both fluorite and antifluorite structures?
(A) Cation is present in alternate tetrahedral voids.
(B) Anion constitutes lattice
(C) Number of formula unit in one unit cell is 4.
(D) 100% tetrahedral voids are occupied.
20. Which pair(s) of liquids on mixing is /are expected to show no net volume change and no heat effect?
(A) Acetone and ethanol
(B) Chlorobenzene and bromobenzene
(C) Chloroform and benzene
(D) n-Butyl chloride and n-butyl bromide
21. Which of the following is/are correctly matched?
(A) Butter – gel
(B) Milk – emulsion
(C) Fog – aerosol
(D) Dust – solid sol
22. Oxygen has –1 oxidation state in the compound
(A) Caro's acid
(B) Marshall's acid
(C) BaO₂
(D) K₂O
23. Iodine reacts with hypo to give
(A) NaI
(B) Na₂SO₃
(C) Na₂S₄O₆
(D) Na₂SO₄
24. The complex $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$ is formed in the brown ring test for nitrates when freshly prepared FeSO₄ solution is added to aqueous solution of NO₃[–] followed by addition of conc. H₂SO₄. Select correct statements about this complex:
(A) Colour change is due to charge transfer.
(B) It has iron in +1 oxidation state and nitrosyl as NO⁺.
(C) It has magnetic moment of 3.87 BM confirming three unpaired electrons in Fe.
(D) In complex Fe has d²sp³ hybridization

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

25. p_A = partial pressure of component A in liquid mixture, P_A° = vapour pressure of A, χ_A = mole fraction of A in liquid mixture.

List - I		List - II	
(I)	$C_2H_5OH + H_2O$	(P)	Azeotropic mixture
(II)	$C_2H_5Br + C_2H_5I$	(Q)	Obeys Raoult's law
(III)	$p_A = \chi_A P_A^\circ$	(R)	Non-ideal solution with positive deviation
(IV)	$H_2O + H_2SO_4$	(S)	Non-ideal solution with negative deviation
		(T)	Ideal solution

Codes:

	I	II	III	IV
(A)	PR	QT	QT	PS
(B)	PS	Q	PR	P
(C)	QT	PS	T	S
(D)	P	Q	T	P

26. Match the anionic species given in List I that are present in the ore(s) given in List II.

List - I		List - II	
(I)	Carbonate	(P)	Siderite
(II)	Sulphide	(Q)	Malachite
(III)	Hydroxide	(R)	Bauxite
(IV)	Oxide	(S)	Calamine
		(T)	Argentite

Codes:

	I	II	III	IV
(A)	PQR	R	Q	PQ
(B)	QS	QR	T	R
(C)	PQS	T	QR	R
(D)	PQ	PQR	PT	QS

27. Match each of the reactions given in List- I with the corresponding product(s) given in List-II:

List - I		List - II	
(I)	$Cu + \text{dil. } HNO_3$	(P)	NO
(II)	$Cu + \text{conc. } HNO_3$	(Q)	NO_2
(III)	$Zn + \text{dil. } HNO_3$	(R)	N_2O
(IV)	$Zn + \text{conc. } HNO_3$	(S)	$Cu(NO_3)_2$
		(T)	$Zn(NO_3)_2$

Codes:

	I	II	III	IV
(A)	PQ	QR	RT	PT
(B)	QS	PS	PQ	QS
(C)	PS	QT	PQ	QS
(D)	PS	QS	RT	QT

28. All the compounds listed in List-I react with water, Match the result of the respective reactions with the appropriate options listed in List-II.

List - I		List - II	
(I)	$(\text{CH}_3)_2\text{SiCl}_2$	(P)	Hydrogen halide formation
(II)	XeF_4	(Q)	Redox reaction
(III)	Cl_2	(R)	Reacts with glass
(IV)	VCl_5	(S)	Polymerisation
		(T)	O_2 formation

Codes:

	I	II	III	IV
(A)	P	QR	PT	PS
(B)	PS	PQRT	PQT	P
(C)	PQR	QT	PS	RS
(D)	QR	QS	PQ	PT

Section - B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

29. Insulin forms crystals of orthorhombic type with unit cell dimensions of $12.5 \times 8.0 \times 3.0 \text{ nm}^3$. If the density of the crystal is $1.5 \times 10^3 \text{ kg/m}^3$ and there are six insulin molecules per unit cell, the molar mass of insulin (in kg/mol) is: ($N_A = 6 \times 10^{23}$)
30. An element (atomic mass = 125) crystallize in simple cubic structure. If the diameter of the largest sphere which can be placed in the crystal without disturbing the crystal is 366 pm and the density of crystal is 'd' g/cm³, then the value of '300 d' is: ($N_A = 6 \times 10^{23}$)
31. A quantity of 2.0 g of benzoic acid dissolved in 26.0 g of benzene shows a depression in freezing point equal to 1.60 K. Molal depression constant of benzene is $4.88 \text{ K}\cdot\text{kg mol}^{-1}$. The percentage association of the acid is
32. A current of dry air was passed through a series of bulbs containing 1.25 g of a solute A_2B (molar mass = 90 g/mol) in 49 g of water and then through pure water. The loss in weight of the former series of bulbs was 0.98 g and in the later series 0.01 g. The percentage dissociation of solute is
33. It was found that 0.10 mg of an adsorbate covered 0.06 m^2 of a solid surface. If the molar mass of the adsorbate is 0.25 kg/mol, then the effective area of each adsorbate molecule (in 10^{-20} m^2) is: ($N_A = 6.0 \times 10^{23}$). (Consider monolayer)
34. A volume of 585 ml of 1% (w/w) NaCl solution of density 1.2 g/ml is required for complete coagulation of 200 ml of a gold sol in two hours. The coagulation value of NaCl (in millimole/L) is
35. Sodium iodate is treated with calculated amount of sodium bisulphite to prepare iodine. How many moles of sodium bisulphite are required to prepare one mole of iodine from sodium iodate?
36. How many metals are commercially reduced by Goldschmidt's aluminothermic process from the given metals?
Na, Pb, Al, Mg, Cr, Fe

Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

37. In an experimental performance of a single throw of pair of unbiased normal dice (six faced), three events A_1, A_2, A_3 are getting prime number on each dice, getting same number on each dice, getting total on two dice equal to 4, then
- (A) Probabilities $P(A_1), P(A_2)$ & $P(A_3)$ are in A.P. with common difference $\frac{1}{6}$
- (B) Probabilities of $P(A_1), P(A_2)$ & $P(A_3)$ are in A.P. with common difference $\frac{1}{12}$
- (C) A_1 & A_2 are not independent
- (D) $P(A_3/A_1) = \frac{1}{9}$
38. If $A = \begin{bmatrix} x & y & 1 \\ 2 & 1 & 3 \\ 1 & z & 2 \end{bmatrix}$ and $A^{-1} = (5A - A^2)$, then
- (A) $|A| = -3$ (B) $\text{tr}(A) = 5$
- (C) $\text{tr}(A) = 2y + z$ (D) $\text{tr}(A) = x + y + z$
39. If Q is foot of perpendicular drawn from the point P (1, 2, 3) on the plane $x + 2y + z = 14$, a point R on the plane such that $\angle PRQ = \theta$, then
- (A) Greatest value of $\left(QR \sin \theta + \frac{\sqrt{18}}{PR} \right)$ is 3
- (B) If Δ is area of ΔPQR , then least value of $\left(\frac{\Delta^2 + 1}{\Delta} \right)$ is 6
- (C) $(QR + PR) \tan \frac{\theta}{2} = \sqrt{6}$
- (D) If Q is (a, b, c), then $\frac{(a+b+c-1)}{3} = \text{Area of } \Delta PQR$ if $\theta = \frac{\pi}{4}$
40. Given set $S = \{1, 2, 3, \dots, 2022\}$. If a number n is chosen randomly from set S such that $\text{HCF}(n, 2022) = 1$, then probability $p = \frac{m}{n}$, where n is a prime number. Then,
- (A) $\sqrt{|m - n|} = 15$
- (B) $\sqrt{\frac{m}{7}} = 4$
- (C) Number of divisors of m are 10
- (D) Sum of digits of $(m + n)$ is also a prime number

41. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are three vectors such that $\vec{a} \times \vec{b} = \vec{c}$ and $\vec{b} \times \vec{c} = \vec{a}$, then $[\vec{a} \vec{b} \vec{c}]$ is equal to
- (A) $\vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c}$ (B) $\vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} + \vec{c} \cdot \vec{c}$
- (C) $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$ (D) $\frac{1}{2}(|\vec{a} + \vec{b} + \vec{c}|^2 - 1)$
42. If $y = f(x)$ satisfying the differential equation $y\left(\frac{dy}{dx}\right)^2 + (x - y)\left(\frac{dy}{dx}\right) - x = 0$ (given $f(3) = 4$), then
- (A) Greatest value of $(f(x) + x)$ is 5 if $f(x)$ is not linear
- (B) Greatest value of $(f(x) + x)$ is $5\sqrt{2}$ if $f(x)$ is not linear
- (C) $\lim_{x \rightarrow \infty} (f(x) - x) = 1$
- (D) $\lim_{x \rightarrow 2023} \frac{f(x) - f(2023)}{(x - 2023)} = 1$

Section – A (Maximum Marks: 12)

This section contains **FOUR (04) Matching List Type Questions**. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

43. Given three vectors $\vec{a} = \hat{i} + \hat{j} - \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + 2\hat{j} - \hat{k}$.

List - I		List - II	
(I)	A vector perpendicular to the vector $\vec{a}$ and coplanar with $\vec{a}$ and $\vec{b}$	(P)	$(1/\sqrt{2})(-\hat{i} + 2\hat{j})$
(II)	A vector perpendicular to $\vec{a}$ and the vector in (a)	(Q)	$(1/\sqrt{2})(\hat{i} + \hat{k})$
(III)	A vector perpendicular to $\vec{b}$ and $\vec{c}$	(R)	$\frac{1}{\sqrt{6}}(2\hat{i} - \hat{j} + \hat{k})$
(IV)	A vector perpendicular to $\vec{a}$ and $\vec{a} + \vec{c}$	(S)	$(1/\sqrt{2})(\hat{j} + \hat{k})$
		(T)	$3(2\hat{i} - \hat{j} + \hat{k})$

Codes:

	I	II	III	IV
(A)	P	Q	T	S
(B)	R	P	Q	T
(C)	RT	S	P	Q
(D)	S	Q	P	R

44. Match List-I with List-II:

List - I		List - II	
(I)	$\frac{dy}{dx} = \frac{e^{x/y} + 1}{((x/y) - 1)e^{x/y}}$	(P)	$y^3 + 2x = 2x^2y + c$
(II)	$\frac{dy}{dx} = \frac{4xy - 2}{3y^2 - 2x^2}$	(Q)	$x - ke^{\sin^{-1}\frac{y}{x}} = 0$
(III)	$x dy - y dx = \sqrt{x^2 - y^2} dx$	(R)	$\tan^{-1}xy + \frac{1}{3}\left(\frac{y}{x}\right)^3 = 3$
(IV)	$x^4(x dy + y dx) + y^2(1 + x^2y^2)(x dy - y dx) = 0$	(S)	$x + ye^{x/y} = c$
		(T)	$x^2 + ke^{\sin^{-1}\left(\frac{y}{x}\right)} = 0$

Codes:

	I	II	III	IV
(A)	S	P	Q	R
(B)	P	R	S	T
(C)	R	T	Q	S
(D)	S	Q	P	T

45. Let A, B and I (Identify matrix) be the matrices of order 4 such that $A = [a_{ij}]_{4 \times 4}$, where $a_{ij} = 1$ and $A = B + I$.

List - I		List - II	
(I)	$ B =$	(P)	1
(II)	$ B^2 - 4B + I =$	(Q)	0
(III)	$ B^3 + B^2 - 8B =$	(R)	-1
(IV)	$ A^{-1} + I $ is greater than	(S)	-4
		(T)	1/4

Codes:

	I	II	III	IV
(A)	P	Q	T	QR
(B)	P	T	Q	S
(C)	T	Q	P	QRS
(D)	Q	P	Q	RS

46. Match the probabilities of these events:

List - I (Probability)		List - II (Events)	
(I)	$\frac{4}{63}$	(P)	E-1: The number chosen is even
(II)	$\frac{7}{9}$	(Q)	E-2: The first, fifth and ninth digits of the chosen number will be odd
(III)	$\frac{4}{9}$	(R)	E-3: The number will be greater than $3 \cdot 10^8$
(IV)	$\frac{5}{42}$	(S)	E-4: The sum of digits used in the number on first two places from the left, equals the digit used in the last place
		(T)	E-5: The number chosen is divisible by 6

Codes:

	I	II	III	IV
(A)	P	T	Q	R
(B)	S	PT	R	Q
(C)	Q	R	P	S
(D)	S	R	PT	Q

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

47. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ & $P = \begin{bmatrix} 7 & 0 & 2 \\ 0 & 1 & 0 \\ 2 & 0 & 5 \end{bmatrix}$. Then $\det(X^2 + X)$ where $X = P^{-1}AP$ is equal to
48. If $y = x \sin(\ln x)$ is the solution of $x^2 \left(\frac{dy}{dx} \right)^2 - (\lambda - 2)xy \left(\frac{dy}{dx} \right) + 2y^2 - x^2 = 0$, then the value of (λ) is:
49. Area bounded by $y = f(x)$; $x = \frac{1}{\sqrt{2}}$; $x = \frac{\sqrt{3}}{2}$ and x-axis is $\frac{m\pi^2}{n}$ then $(m + n) = ?$
Given $f(x) = x + \frac{2}{3}x^3 + \frac{2}{3} \cdot \frac{4}{5}x^5 + \frac{2 \cdot 4 \cdot 6}{3 \cdot 5 \cdot 7}x^7 + \dots \infty$ for $|x| < 1$.
50. If area bounded by the curve $y = |\cos^{-1}(\sin x)| - |\sin^{-1}(\cos x)|$ & x-axis from $\frac{3\pi}{2} \leq x \leq 2\pi$ is equal to $\frac{\pi^2}{k}$, where $k \in \mathbb{N}$, then $\frac{29}{k}$ is equal to
51. If vectors $\vec{a} = \left(2\sin\theta, 2(\sqrt{3}-1)\sin\frac{\theta}{2}, -3\sqrt{\sin\frac{\theta}{2}} \right)$ and $\vec{b} = \left(-2\sin\theta, 2\cos\frac{\theta}{2}, \frac{1}{\sqrt{3\sin\frac{\theta}{2}}} \right)$ are orthogonal and the vector $\vec{c} = (1, 2, \cos 2\theta)$ makes an obtuse angle with z-axis, then $[\tan^2\theta + \cot^2\theta] = \underline{\hspace{2cm}}$ (where $[.]$ represents greatest integer function).
52. If $T_r = \begin{vmatrix} r-1 & r-2 & r-6 \\ 2r-4 & 2r-6 & 2r-11 \\ 3r-9 & 3r-12 & 3r-18 \end{vmatrix}$, then $\sum_{r=0}^5 T_r =$
53. Let $y = f(x)$ be a curve ' C_1 ' passing through $(2, 2)$ and $\left(8, \frac{1}{2}\right)$ and satisfying a differential equation $y \left(\frac{d^2y}{dx^2} \right) = 2 \left(\frac{dy}{dx} \right)^2$. Curve ' C_2 ' is the director circle of the circle $x^2 + y^2 = 2$. If the shortest distance between the curve C_1 and C_2 is $\sqrt{p} - q$ where $p, q \in \mathbb{N}$, then find the value of $(p^2 - q)$.
54. If $|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$ and $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = \cos\theta$, then maximum value of θ is

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ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. AC

Sol. $\frac{K_{\alpha}}{K_{\gamma}} = \frac{m_{\gamma}}{m_{\alpha}} = \frac{M-m}{m}$

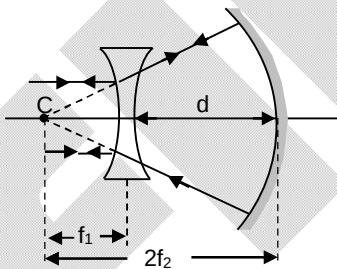
$$r = \frac{\sqrt{2mK_{\alpha}}}{qB} \Rightarrow K_{\alpha} = \frac{q^2 B^2 r^2}{2m}$$

$$K_{\gamma} = \frac{m}{(M-m)} K_{\alpha} = \frac{q^2 B^2 r^2}{2(M-m)}$$

$$\text{Energy released} = K_{\gamma} + K_{\alpha} = \frac{q^2 B^2 r^2}{2m} \left(\frac{M}{M-m} \right)$$

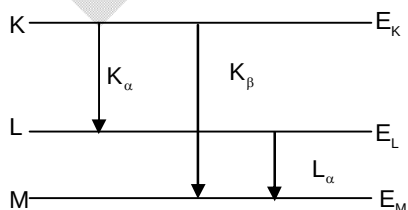
2. AB

Sol.



3. AC

Sol. $E_K - E_L = \frac{hc}{\lambda_{\alpha}} \quad \dots(i)$



$$E_K - E_M = \frac{hc}{\lambda_\beta} \quad \dots(ii)$$

$$(ii) - (i) \Rightarrow E_L - E_M = \frac{hc}{\lambda'_\alpha} = \frac{hc}{\lambda_\beta} - \frac{hc}{\lambda_\alpha}$$

$$\frac{1}{\lambda_\beta} = \frac{1}{\lambda_\alpha} + \frac{1}{\lambda'_\alpha}$$

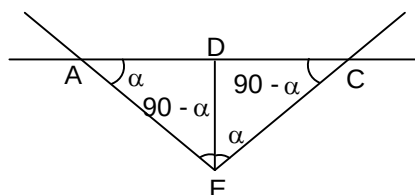
$$\text{Also, } (E_K - E_M) > (E_K - E_L) > (E_L - E_M)$$

$$\frac{hc}{\lambda_\beta} > \frac{hc}{\lambda_\alpha} > \frac{hc}{\lambda'_\alpha}$$

4.

BD

Sol.

 Let $\angle CAE = \alpha = \text{constant}$

 At time t , $DE = vt$

$$AD = DE \cot \alpha$$

 And $DC = DE \tan \alpha$

$$AC = DE (\tan \alpha + \cot \alpha)$$

$$= vt (\tan \alpha + \cot \alpha)$$

 Induced emf $e = BvI$

$$e = Bv^2 t (\tan \alpha + \cot \alpha)$$

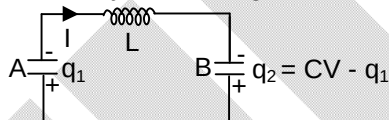
$$e \propto v^2 \text{ and } e \propto t$$

5.

AC

Sol.

Let at any time, charge and current in the circuit are as shown.



$$I = \frac{dq_1}{dt}$$

Applying Kirchhoff's law:

$$\frac{CV - q_1}{C} - \frac{q_1}{C} - L \frac{dI}{dt} = 0 \Rightarrow \frac{CV - 2q_1}{LC} = L \frac{d^2 q_1}{dt^2}$$

$$\Rightarrow \frac{d^2 q_1}{dt^2} = -\frac{2}{LC} \left(q_1 - \frac{CV}{2} \right) \Rightarrow q_1 - \frac{CV}{2} = q$$

$$\Rightarrow q_1 - \frac{CV}{2} = A \sin(\omega t + \delta) \quad \dots(i)$$

$$\text{Where } \omega = \sqrt{\frac{2}{LC}} = 100\pi \text{ rad/s}$$

$$\text{At } t = 0, q_1 = 0 \Rightarrow 0 - \frac{CV}{2} = A \sin \delta \quad \dots(ii)$$

Differentiating (i), $\frac{dq_1}{dt} = A\omega \cos(\omega t + \delta)$

$$\Rightarrow I = A\omega \cos(\omega t + \delta)$$

$$\text{At } t = 0, I = 0 \Rightarrow 0 = A\omega \cos \delta \Rightarrow \delta = \frac{\pi}{2} \quad \dots(\text{iii})$$

$$\text{From (ii), } A = -\frac{CV}{2}$$

$$\text{From (i), } q_1 = \frac{CV}{2} - \frac{CV}{2} \sin\left(\omega t + \frac{\pi}{2}\right)$$

$$\Rightarrow q_1 = \frac{CV}{2}(1 - \cos \omega t) \quad \dots(\text{iv})$$

$$\Rightarrow q_1 = 50(1 - \cos 100\pi t) \text{ mc}$$

$$\text{Now, } q_2 = CV - q_1 = \frac{CV}{2}(1 + \cos \omega t) \quad \dots(\text{v})$$

$$= 50(1 + \cos 100\pi t)$$

$$\text{From (ii), } I = -A\omega \sin \omega t$$

Current in the circuit is maximum for $\omega t = \frac{\pi}{2}$.

And for $\omega t = \frac{\pi}{2}$, we see from (iv) and (v) that $q_1 = q_2 = \frac{CV}{2}$.

6. ABCD

Sol. At any time: $L_1 \frac{di_1}{dt} = L_2 \frac{di_2}{dt}$

$$\Rightarrow \int_0^{i_1} L_1 di_1 = \int_0^{i_2} L_2 di_2 \Rightarrow L_1 i_1 = L_2 i_2 \Rightarrow \frac{i_1}{i_2} = \frac{L_2}{L_1} \quad \dots(\text{i})$$

Total steady state current

$$i = \frac{E}{R} = i_1 + i_2$$

From equation (i) and (ii)

$$i_1 = \frac{EL_2}{R(L_1 + L_2)}, i_2 = \frac{EL_1}{R(L_1 + L_2)}$$

$$U = \frac{1}{2} L_{eq} i^2 = \frac{1}{2} \left(\frac{L_1 L_2}{L_1 + L_2} \right) \frac{E^2}{R^2}$$

7. A

- Sol. (A) $\mu_2 = \mu_3$, as there is no deviation
 $\mu_1 < \mu_2$, as ray bends towards normal
- (B) $\mu_1 > \mu_2$, as ray bends away from normal
 $\mu_2 > \mu_3$, as ray bends away from normal
- (C) $\mu_2 = \mu_3$, as there is no deviation
 $\mu_1 < \mu_2$, as ray bends
- (D) $\mu_1 > \mu_2$, as ray bends away from normal
 $\mu_2 > \mu_3$, as ray bends away from normal

- (E) $\mu_1 > \mu_2$, as ray bends away from normal
 $\mu_2 = \mu_3$, as there is no deviation

8. A

Sol. $r_n = \frac{0.529n^2}{Z} A$

So, (I) $\rightarrow$ (S)

Magnetic field, $B = \frac{12.5Z^3}{n^5} T$

So, (II) $\rightarrow$ (T)

$$B \propto \frac{1}{n^5}$$

$$V_n = \frac{2.2 \times 10^6 Z}{n}$$

$$V_n \propto \frac{1}{n}$$

$n \uparrow V_n \uparrow$

So, (III) $\rightarrow$ (Q)

Total energy, $E_n = \frac{-13.6Z^2}{n^2} \text{ eV}$

$n \uparrow E_n \uparrow$

So, (IV) $\rightarrow$ (P)

9. B

Sol. In nuclear fusion, two lighter nuclei fuse and make big nuclei. In this, mass defect is converted into energy according to $E = mc^2$.

In nuclear fission, heavy nuclei split into two or more than two smaller nuclei. In this process, mass is converted into energy according to $E = mc^2$

In β -decay, neutron proton ratio decreases, so nucleus becomes more stable.

Both nuclear fission and nuclear fusion are exothermic reactions.

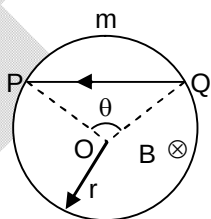
10. A

Sol. (I) Area OPMQO = $\frac{1}{2}r^2\theta$

Flux in this area, $\phi_1 = \frac{1}{2}r^2\theta B$

Induced emf in this area,

$$e_1 = \frac{d\phi_1}{dt} = \frac{1}{2}r^2\theta \frac{dB}{dt}$$



$$\text{Area OPQ} = r \sin\left(\frac{\theta}{2}\right) r \cos\left(\frac{\theta}{2}\right) = \frac{1}{2}r^2 \sin\theta$$

Induced emf in OPQ

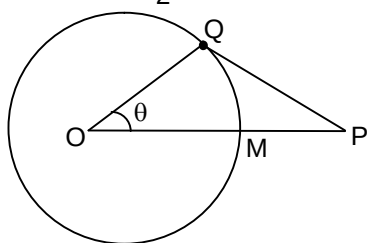
$$e_2 = \frac{d\phi_2}{dt} = \frac{1}{2}r^2 \sin\theta \frac{dB}{dt}$$

e_2 will be only in part PQ, because in OQ and OP induced emf will be zero. Clearly, $e_2 < e_1$, because area OPQ < area OPMQO.

Since B is increasing, so e.m.f. will be in anticlockwise direction. Hence end P will be positive w.r.t. Q.

(II) Here emf in OPQ will be due to flux changing in area OMQ.

This area is $\frac{1}{2}r^2\theta$. The entire emf will be in part PQ. End Q will be positive.

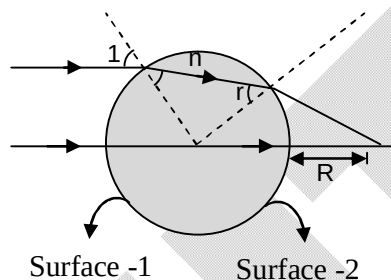


(III) Induced emf = $\frac{1}{2}r^2\theta \frac{dB}{dt}$. End P will be positive.

(IV) Area in which flux is changing is less than $\frac{1}{2}r^2\theta$. End Q will be positive.

Section – B

11. 1.33
Sol.



From surface -1: $\frac{n}{v} - \frac{1}{-\infty} = \frac{n-1}{R}$; $v = \frac{nR}{n-1}$

From surface -2: $\frac{1}{R} - \frac{n}{v-2R} = \frac{1-n}{-R}$; $\frac{2}{R} = \frac{n}{R} + \frac{n}{v-2R}$

$$\Rightarrow \frac{2}{R} = \frac{n(v-2R+R)}{R(v-2R)} \Rightarrow 2v-4R = nv-nR$$

On putting the value of v

$$2\frac{nR}{n-1} - 4R = \frac{n \times nR}{(n-1)} = -nR$$

$$2nR - 4nR + 4R = n^2R - n^2R + nR, n = \frac{4}{3}$$

12. 30.00

Sol. $I_0 = 4I_0 \cos^2 \frac{\phi}{2} \Rightarrow \cos^2 \frac{\phi}{2} = \frac{1}{2} \Rightarrow \phi = \frac{2\pi}{3}$

Therefore, path difference of the two beams producing intensity I_0 on the screen is $\frac{\lambda}{3}$.

Optical path difference between two beams meeting on the screen is

$$\Delta x = \left\{ \left(S_2P - t + \frac{4}{3}t \right) - \left(S_1P - t + \frac{3}{2}t \right) \right\}$$

$$= (S_2P - S_1P) + t \left(\frac{4}{3} - \frac{3}{2} \right)$$

$$\Rightarrow \Delta x = d \sin \theta - \frac{t}{6}$$

$$\therefore \frac{dy}{D} - \frac{t}{6} = \frac{\lambda}{3} \Rightarrow \frac{dy}{D} = \frac{\lambda}{3} + \frac{t}{6}$$

$$\Rightarrow y = \frac{D}{d} \left[\frac{\lambda}{3} + \frac{t}{6} \right] = \frac{1}{10^{-3}} \left(\frac{6000 \times 10^{-10}}{3} + \frac{0.6 \times 10^{-6}}{6} \right)$$

$$= \frac{1}{10^{-3}} \times 3 \times 10^{-7} = 3 \times 10^{-4} \text{ m}$$

$$\therefore y = 30 \text{ mm}$$

13. 23.55

Sol. Total number of ions in the rod
 = number of ions per unit volume $\times$ volume of rod
 $= (2 \times 10^{25}) \times \{ 3.14 \times (0.005)^2 \times (5 \times 10^{-2}) \}$

The number of photons excited in one direction is equal to the total number of ions because all ions are excited. Now, excited energy = number of excited photons $\times$ energy of photon

$$= (7.85 \times 10^{19}) \times \frac{hc}{\lambda}$$

$$= (7.85 \times 10^{19}) \times \frac{(6.6 \times 10^{-34}) \times (3 \times 10^8)}{6.6 \times 10^{-7}}$$

$$= 23.55 \text{ W}$$

14. 2.25

Sol. Let T_1 be the initial temperature and T_2 be the increased temperature of the black body. According to Stefan's law,

$$\left(\frac{T_2}{T_1} \right)^4 = 81 = (3)^4$$

$$\text{Also, } \lambda_1 T_1 = \lambda_2 T_2 \quad \dots (i)$$

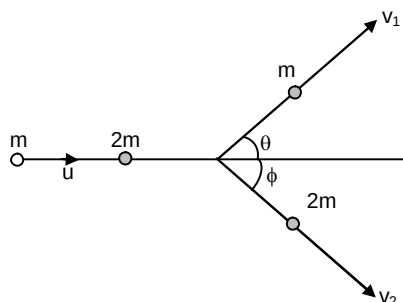
$$\lambda_2 = \frac{\lambda_1 \times T_1}{T_2} = \frac{9000 \times T_1}{3T_1} = 3000 \text{ \AA}$$

$$\text{Now, } \frac{hf}{\lambda_2} - W = eV_0 \text{ or } \frac{hf}{\lambda_2} - W = 13.6 \left(\frac{1}{2^2} - \frac{1}{3^2} \right)$$

Solving, we get, $W = 2.25 \text{ eV}$

15. 90.00

Sol.



$$mu = mv_1 \cos \theta + 2mv_2 \cos \phi$$

$$mv_1 \sin \theta = 2mv_2 \sin \phi$$

$$\frac{2}{3} \left(\frac{1}{2} mu^2 \right) = \frac{1}{2} 2mv_2^2$$

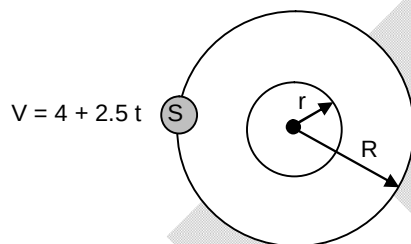
$$\frac{1}{3} \left(\frac{1}{2} mu^2 \right) = \frac{1}{2} mv_1^2$$

$$\text{Solving equations, } v_1 = v_2 = \frac{u}{\sqrt{3}}$$

$$\theta + \phi = 120^\circ \text{ and } \theta = 90^\circ$$

16. 1.25

Sol. The magnetic field at the center O due to the current in the larger loop is $B = \frac{\mu_0 I}{2R}$.



If r is the resistance per unit length, then

$$I = \frac{\text{potential difference}}{\text{resistance}} = \frac{4 + 2.5t}{2\pi R p}$$

$$\therefore B = \frac{\mu_0}{2R} \cdot \frac{4 + 2.5t}{2\pi R p}$$

$\therefore r \ll R$, so the field B can be taken almost constant over the entire area of the smaller loop.

$\therefore$ The flux linked with the smaller loop is

$$\phi = B \times \pi r^2 = \frac{\mu_0}{2R} \cdot \frac{4 + 2.5t}{2\pi R p} \cdot \pi r^2$$

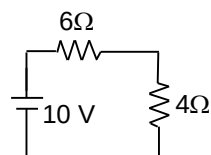
$$\text{Induced emf } e = \frac{d\phi}{dt} = \frac{\mu_0 r^2}{4R^2 p} \times 2.5$$

The corresponding current in the smaller loop is I' , then

$$\begin{aligned} I' &= \frac{e}{R} = \frac{\mu_0 r^2}{4R^2 p} \times 2.5 \times \frac{1}{2\pi r p} \\ &= \frac{2.5\mu_0 r}{8\pi R^2 p^2} = \frac{2.5 \times 4\pi \times 10^{-7} \times 0.1}{8\pi \times (1)^2 \times (10^{-4})^2} = 1.25 \text{ A} \end{aligned}$$

17. 1.00

Sol.



18. 4.00

Sol.

$$V_C - V_L = 120$$

$$V_R^2 + (V_C + V_L)^2 = E_y^2$$

$$V_R^2 + (120)^2 = (200)^2 \Rightarrow V_R = 160$$

$$I = \frac{V_R}{R} = \frac{160}{40} = 4A$$

Chemistry**PART – II****Section – A**

19. CD
Sol. In both fluorite and antifluorite, 100% TVs are occupied and Z_{eff} (number of formula unit) for both is 4.
20. BD
Sol. $\Delta_{\text{mixing}} V = 0$; $\Delta_{\text{mixing}} H = 0 \Rightarrow$ Look for ideal solutions.
21. ABC
Sol. Dust is an aerosol and not a solid sol.
22. ABC
Sol. Caro's acid H_2SO_5 contains peroxide group. Marshall's acid $\text{H}_2\text{S}_2\text{O}_8$ contains peroxo group. BaO_2 is barium peroxide.
Above three contains peroxo group hence oxygen state of O is -1 , whereas in K_2O , O.S. is -2 .
23. AC
Sol. $2\text{Na}_2\text{S}_2\text{O}_3 + \text{I}_2 \rightarrow \text{Na}_2\text{S}_4\text{O}_6 + 2\text{NaI}$
24. ABC
Sol. A. Fe^{2+} changes to brown-colored ring complex
B. $\text{NO} \rightarrow \text{NO}^{\oplus} + \text{e}^{\ominus}$
 $\text{Fe}^{2+} + \text{e}^{\ominus} \rightarrow \text{Fe}^{\oplus}$
C.

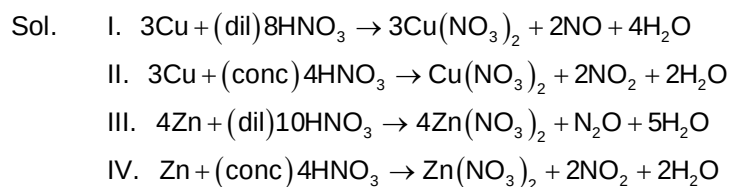
$$\text{Fe}^{2+} \Rightarrow 3d^6 \begin{array}{|c|c|c|c|c|c|} \hline \uparrow\downarrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ \hline \end{array}$$

$$\text{Fe}^{\oplus} \Rightarrow 3d^7 \begin{array}{|c|c|c|c|c|c|} \hline \uparrow\downarrow & \uparrow\downarrow & \uparrow & \uparrow & \uparrow & \uparrow \\ \hline \end{array}$$

$$n = 3$$

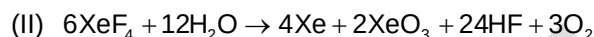
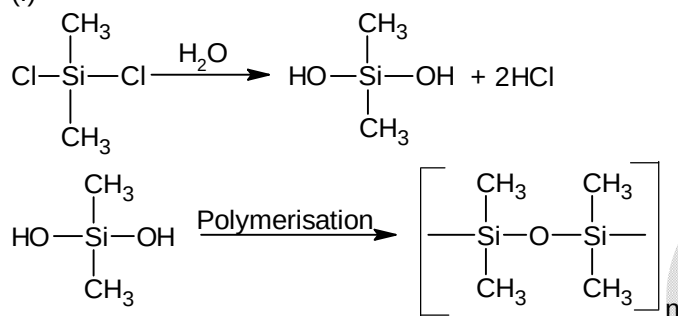
$$\mu_{\text{eff}} = \sqrt{n(n+2)} = \sqrt{15} = 3.87 \text{ BM}$$
 Thus, (A), (B) and (C) are true.
D. sp^3d^2 Hybridisation.
25. A
Sol. (I $\rightarrow$ PR) $\text{C}_2\text{H}_5\text{OH} + \text{H}_2\text{O}$ = Azeotropic mixture (at a particular composition) shows positive deviation.
(II $\rightarrow$ QT) $\text{C}_2\text{H}_5\text{Br} + \text{C}_2\text{H}_5\text{I}$ = Ideal solution; obeys Raoult's law.
(III $\rightarrow$ QT) $p_A = P_A^0 \chi_A$ = Ideal solution (with non-volatile solute)
(IV $\rightarrow$ PS) $\text{H}_2\text{O} + \text{H}_2\text{SO}_4$ = Azeotropic mixture (at a particular composition) shows negative deviation.
26. C
Sol. (P) = siderite = FeCO_3
(Q) = Malachite = $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$
(R) = Bauxite = $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$
(S) = Calamine = ZnCO_3
(T) = Argentite = Ag_2S

27. D

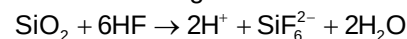


28. B

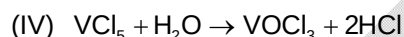
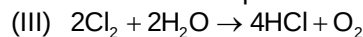
Sol. (I)



HF reacts with glass as shown:



This reaction is not possible with HCl, HBr and HI.



Section – B

29. 45.00

Sol. $d = \frac{Z \cdot M}{N_A \cdot V}$

$$\Rightarrow 1.5 \times 10^3 = \frac{6 \times M}{(6 \times 10^{23}) \times \left\{ \frac{12.5 \times 8.0 \times 3.0}{\times (10^{-9})^3} \right\}}$$

$$\Rightarrow M = 45 \text{ kg / mol}$$

30. 4000.00

Sol. For cubic void, $\frac{r}{a} = 0.732$

$$\Rightarrow a = \frac{366 / 2}{0.732} = 250 \text{ pm}$$

$$d = \frac{Z \cdot M}{N_A \cdot V} = \frac{1 \times 125}{(6 \times 10^{23}) \times (250 \times 10^{-10})^3} = \frac{100}{7.5}$$

$$\text{Now, } 300d = 300 \times \frac{100}{7.5} = 4000$$

31. 96.00

Sol. $\Delta T_f = i K_f \cdot m$

$$\Rightarrow 1.60 = 4.88 \times \frac{2/122}{26/1000} \times \left[1 + \alpha \left(\frac{1}{2} - 1 \right) \right]$$

$$\therefore \alpha = 0.96 \text{ or } 96\%$$

32. 50.00

Sol. Loss in weight of solution $\propto P$ Loss in weight of water $\propto (P^0 - P)$

$$\text{Now, } \frac{P^0 - P}{P} = \frac{n_1}{n_2} \Rightarrow \frac{0.01}{0.98} = \frac{\frac{1.25}{90} [1 + \alpha(3-1)]}{\frac{49}{18}}$$

$$\therefore \alpha = 0.50 \text{ or } 50\%$$

33. 25.00

Sol. Number of adsorbate molecules

$$= \frac{0.10 \times 10^{-3}}{0.25 \times 10^3} \times 6 \times 10^{23} = 2.4 \times 10^{17}$$

$$\therefore \text{Effective surface area} = \frac{0.06}{2.4 \times 10^{17}} = 25 \times 10^{-20} \text{ m}^2$$

34. 600.00

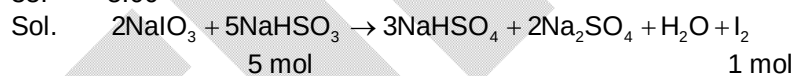
Sol. Mass of NaCl used

$$= (585 \times 1.2) \times \frac{1}{100} = 5.85 \times 1.2 \text{ gm}$$

$$\text{Moles of NaCl used} = \frac{5.85 \times 1.2}{58.5} = 0.12$$

$$\therefore \text{Coagulation value} = \frac{0.12 \times 10^3}{\frac{200}{1000}} = 600 \text{ millimole/litre}$$

35. 5.00



36. 2.00

Sol. Cr, Fe

Mathematics

PART – III

Section – A

37. BCD

Sol. $\therefore P(A_1) = \frac{9}{36} = \frac{1}{4}$, $P(A_2) = \frac{6}{36} = \frac{1}{6}$ & $P(A_3) = \frac{1}{12}$

So, (B) is correct.

$$P(A_1 \cap A_2) \neq P(A_1)P(A_2) = \frac{1}{6} \times \frac{1}{4} = \frac{1}{24}$$

So, (C) is correct.

$$P(A_3/A_1) = \frac{P(A_3 \cap A_1)}{P(A_1)} = \frac{1}{9}$$

So, (D) is correct.

38. ABCD

Sol. $\therefore 3I = 5A^2 - A^3$

$$\therefore A^3 - 5A^2 + 3I = 0$$

$$-x^3 + (3+x)x^2 + (3x-2y-3z+1)x + (-4z-y+3) = 0$$

$$\therefore x+3=5$$

$$2y+3z=7 \text{ \& } y+4z=6$$

$$\text{So, } x=2, y=2, z=1$$

39. ABCD

Sol. $\therefore Q = (2, 4, 4)$

$$\text{Area of } \triangle PQR = 3 \cot \theta$$

$$QR = \sqrt{6} \cot \theta \Rightarrow QR \sin \theta = \sqrt{6} \cos \theta$$

$$PR = \sqrt{6} \operatorname{cosec} \theta \Rightarrow \frac{\sqrt{18}}{PR} = \sqrt{3} \sin \theta$$

$$\therefore QR \sin \theta + \frac{\sqrt{18}}{PR} = \sqrt{6} \cos \theta + \sqrt{3} \sin \theta$$

$$\leq \sqrt{6+3} = 3$$

40. ACD

Sol. $\therefore p = 1 - \frac{1350}{2022} = 1 - \frac{225}{337} = \frac{112}{337}$

$$\text{So, } m = 112, n = 337 \Rightarrow \frac{m}{n} = \frac{112}{337}$$

$\therefore n$ must not be divisible by 2 or 3 or 337

$$\therefore \bar{x}_n = n(2) + n(3) + n(337) - n(6) - n(674) - n(1011) + n(2022)$$

$$= 1350$$

41. ABD

Sol. $\therefore \vec{a} \times \vec{b} = \vec{c}$ and $\vec{b} \times \vec{c} = \vec{a}$

$$\therefore \vec{a} = \vec{b} \times (\vec{a} \times \vec{b})$$

$$\Rightarrow \vec{a} = (\vec{b} \cdot \vec{b})\vec{a} - (\vec{b} \cdot \vec{a})\vec{b} \dots\dots (i)$$

$$\text{and } \vec{a} \cdot (\vec{a} \times \vec{b}) = \vec{a} \cdot \vec{c}$$

$$\Rightarrow \vec{a} \cdot \vec{c} = 0 \text{ \& } \vec{a} \cdot \vec{b} = 0 \text{ \& } \vec{c} \cdot \vec{a} = 0$$

$$\therefore |\vec{b}|^2 = 1 \text{ from (i)}$$

$$\text{So, (A) option } |\vec{a}|^2 + 0 + 0$$

$$[\vec{a} \vec{b} \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \vec{a} \cdot \vec{a} = |\vec{a}|^2$$

$$\& |\vec{a} + \vec{b} + \vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 0$$

$$= 2|\vec{a}|^2 + 1$$

42. BCD

$$\text{Sol. } \therefore \frac{dy}{dx} = 1 \text{ or } -\frac{x}{y}$$

$$\therefore y = x + 1 \text{ or } x^2 + y^2 = 25$$

43. C

$$\begin{aligned} \text{Sol. (I). } \vec{v} &= \lambda \vec{a} \times (\vec{a} \times \vec{b}) = \lambda \{(\vec{a} \cdot \vec{b})\vec{a} - (\vec{a} \cdot \vec{a})\vec{b}\} \\ &= \lambda \{-(\hat{i} + \hat{j} - \hat{k}) - 3(\hat{i} - \hat{j} + \hat{k})\} \\ &= \lambda \{-4\hat{i} + 2\hat{j} - 2\hat{k}\} = -2\lambda \{2\hat{i} - \hat{j} + \hat{k}\} \end{aligned}$$

$$\text{(II). Any such vector is } \vec{p} = \mu(\vec{v} \times \vec{a}) = -\lambda \{2\hat{i} - \hat{j} + \hat{k}\} \times (\hat{i} + \hat{j} - \hat{k}) = -6\lambda \mu (\hat{j} + \hat{k})$$

(III). Solve yourself

(IV). Solve yourself

44. A

Sol. Use homogeneous form.

45. D

$$\text{Sol. } A = [a_{ij}]_{4 \times 4}, a_{ij} = 1 \Rightarrow |A| = 0$$

$$A^2 = 4A \Rightarrow A^2 - 4A + I = I$$

$$A^3 = 16A$$

$$|A^2 - 4A + I| = 1$$

$$\text{and } A^2 - 4A + 3I = 3I$$

$$(A - 3I)(A - I) = 3I$$

$$\left(\frac{A}{3} - I\right) \cdot B = I$$

$$\frac{A}{3} - I = B^{-1}$$

$$\Rightarrow \frac{A}{3} = I + B^{-1} \Rightarrow \|B^{-1} + I\| = 0$$

46.

D

Sol.

1, 2, 3, 4, 5, 6, 7, 8, 9

$$n(S) = 9!$$

- (I) If the last digit from left is 9, then
 First two digits (from left) 18, 27, 36, 45 (8 ways)
 If the last digit is 8
 then first two (1, 7), (2, 6), (3, 5) (6 ways)
 If last is 7, then (1, 6), (2, 5), (3, 4) (6 ways)
 If last is 6, then (1, 5), (2, 4) (4 ways)
 If last is 5, then (1, 4), (2, 3) (4 ways)
 If last is 4, then (1, 3) (2 ways)
 If last is 3, then (1, 2) (2 ways)
32 ways

$$\text{Hence, } n(S) = \frac{32 \cdot 6!}{9!} = \frac{32}{9 \cdot 8 \cdot 7} = \frac{4}{63}$$

- (II) E-3: Unit place can be filled in 3, 4, ..., 9 = 7 ways
 $\therefore n(S) = 7 \cdot 8!$ (no restriction for others)

$$P(E-3) = \frac{7}{9}$$

- (III) E-1: Number of even $\begin{pmatrix} 2, 4, 6, 8 \\ 1, 3, 5, 7, 9 \end{pmatrix}$

$$n(A) = {}^4C_1 \cdot 8! \text{ (any one even on the last place and eight anywhere)}$$

$$P(E-1) = \frac{4 \cdot 8!}{9!} = \frac{4}{9}$$

Divisible by 6 $\Rightarrow$ divisible by 2 and 3.

Sum of all 9 digits is 45 and hence divisible by 3.

$\therefore$ Number must be even same as E-1.

- (IV) For first, fifth and ninth place select 3 number in 5C_3 ways and arrange in 3! Ways.

$$\therefore n(S) = {}^5C_3 \cdot 3! \cdot 6!; P(E-2) = \frac{{}^5C_3 \cdot 3! \cdot 6!}{9!} = \frac{5}{42}$$

Section – B

47.

144.00

Sol.

$$\begin{aligned} \det.(x^2 + x) &= |x^2 + x| \\ &= |x||x + I| \\ &= |A||A + I| = 144 \end{aligned}$$

48.

4.00

Sol.

$$y = x \sin(\ln x)$$

$$\frac{dy}{dx} = x \cdot \frac{\cos(\ln x)}{x} + \sin(\ln x)$$

$$\frac{dy}{dx} = \sin(\ln x) + \cos(\ln x)$$

$$x^2 \left(\frac{dy}{dx} \right)^2 = x^2 (\cos(\ln x) + \sin(\ln x))^2$$

$$x^2 (\cos(\ln x) + \sin(\ln x))^2 - (\lambda - 2)x^2 \sin^2 \ln x - (\lambda - 2)x^2$$

$$\sin(\ln x) \cos(\ln x) + 2x^2 \sin^2(\ln x) - x^2 = 0$$

$$\lambda - 2 = 2 \Rightarrow \lambda = 4$$

49. 79.00

Sol. $\therefore f'(x) = 1 + 2x^2 + \frac{2 \cdot 4}{3}x^4 + \frac{2 \cdot 4 \cdot 6}{3 \cdot 5}x^6 + \dots$

$$\Rightarrow f'(x) = 1 + x f(x) + x^2 f'(x)$$

$$\Rightarrow f(x) = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$$

$$\therefore \int_{1/\sqrt{2}}^{\sqrt{3}/2} \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx = \int_{\pi/4}^{\pi/3} dt = \frac{t^2}{2} \Big|_{\pi/4}^{\pi/3}$$

Put $t = \sin^{-1} x$

$$= \frac{1}{2} \cdot \frac{7\pi^2}{36}$$

$$= \frac{7\pi^2}{72}$$

50. 7.25

Sol. $\therefore f(x) = y = \left| \frac{5\pi}{2} - x \right| - \left| x - \frac{3\pi}{2} \right|$

$$\Rightarrow y = 4\pi - 2x$$

$$\therefore \text{Area} = \frac{1}{2} \cdot \frac{\pi}{2} \cdot \pi = \frac{\pi^2}{4} \Rightarrow k = 4$$

51. 3.00

Sol. Since $\vec{a}$ and $\vec{b}$ are orthogonal $\Rightarrow \vec{a} \cdot \vec{b} = 0$

$$\Rightarrow 4\sin^2 \theta + 4(\sqrt{3} - 1)\sin \frac{\theta}{2} \cos \frac{\theta}{2} - \sqrt{3} = 0$$

$$\Rightarrow 4\sin^2 \theta + 2(\sqrt{3} - 1)\sin \theta - \sqrt{3} = 0$$

$$\Rightarrow (2\sin \theta - 1)(2\sin \theta + \sqrt{3}) = 0$$

$$\Rightarrow \sin \theta = \frac{1}{2} \text{ or } \frac{-\sqrt{3}}{2}$$

and $\vec{c} = (1, 2, \cos 2\theta)$ makes an obtuse angle with z-axis.

$$\Rightarrow \vec{k} \cdot \vec{c} < 0 \Rightarrow \cos 2\theta < 0$$

Case I: When $\sin \theta = \frac{1}{2} \Rightarrow \cos^2 \theta = \frac{3}{4}$

$$\Rightarrow \cos 2\theta = \cos^2 \theta - \sin^2 \theta > 0 \text{ (Hence rejected.)}$$

$$\text{Case II: When } \sin \theta = \frac{-\sqrt{3}}{2} \Rightarrow \cos^2 \theta = \frac{1}{4}$$

$$\Rightarrow \cos 2\theta = \cos^2 \theta - \sin^2 \theta < 0$$

$$\sin \theta = \frac{-\sqrt{3}}{2}$$

$$\text{and } \sin \frac{\theta}{2} \geq 0$$

$$\Rightarrow \theta = 4m\pi + \frac{4\pi}{3}$$

$$\text{or } \theta = 4m\pi + \frac{5\pi}{3}.$$

52. 36.00

 Sol. (I) $C_1 \rightarrow C_1 - C_2$, then $C_2 \rightarrow C_2 - C_3$

$$\Rightarrow \begin{vmatrix} 1 & 4 & r-6 \\ 2 & 5 & 2r-11 \\ 3 & 6 & 3r-18 \end{vmatrix} = 0$$

 (II) $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 3R_1$

$$\begin{vmatrix} 1 & 4 & r-6 \\ 0 & -3 & 1 \\ 0 & -6 & 0 \end{vmatrix} = 6$$

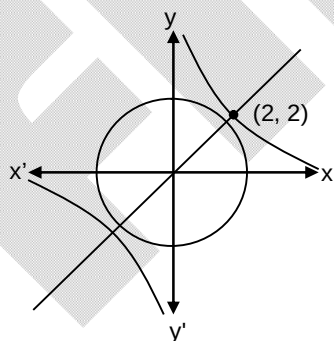
$$(III) \therefore \sum_{r=0}^5 T_r = \underbrace{6+6+\dots}_{6 \text{ times}} = 6 \times 6 = 36$$

53. 62.00

$$\text{Sol. } y \left(\frac{d^2 y}{dx^2} \right) = 2 \left(\frac{dy}{dx} \right)^2$$

$$\frac{y''}{y'} = \frac{2y'}{y}$$

$$\ln y' = 2 \ln y + \ln a$$



$$\frac{y'}{y^2} = a \Rightarrow \frac{dy}{y^2} = a dx$$

$$-\frac{1}{y} = ax + b$$

Since, curve is passing through (2, 2) and $\left(8, \frac{1}{2}\right)$

$$\therefore 2a + b = \frac{-1}{2} \quad \dots(i)$$

$$8a + b = -2 \quad \dots(ii)$$

Solving (i) and (ii)

$$a = -\frac{1}{4}, b = 0$$

$$\therefore C_1 : xy = 4$$

$$\text{Curve } C_2 \text{ is } x^2 + y^2 = 4$$

$$\therefore \text{SD between curves is } = 2\sqrt{2} - 2 = \sqrt{8} - 2$$

$$\therefore p^2 - q = 64 - 2 = 62$$

54. 2.09

$$\text{Sol. } [\vec{a} \vec{b} \vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & \cos \theta & \cos \theta \\ \cos \theta & 1 & \cos \theta \\ \cos \theta & \cos \theta & 1 \end{vmatrix}$$

$$= 2\cos^3 \theta - 3\cos^2 \theta + 1$$

$$= (2\cos \theta + 1)(1 - \cos \theta)^2$$

$$[\vec{a} \vec{b} \vec{c}]^2 \geq 0 \Rightarrow 1 + 2\cos \theta \geq 0$$

$$\Rightarrow \cos \theta \geq -\frac{1}{2} \Rightarrow \theta \leq \frac{2\pi}{3}$$

FIITJEE
ALL INDIA INTEGRATED TEST SERIES
JEE (Advanced)-2024
PART TEST – IV
PAPER –2
TEST DATE: 20-10-2023

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 04, 19 – 22, 37 – 40): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A (05 – 10, 23 – 28, 41 – 46): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	–1	In all other cases.

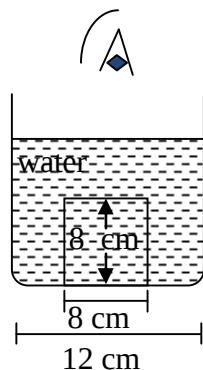
Physics

PART – I

Section – A (Maximum Marks: 12)

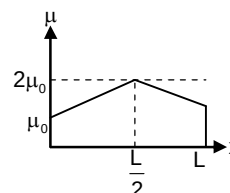
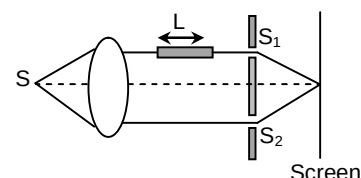
This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

1. A cylindrical vessel of diameter 12 cm contains $800\pi \text{ cm}^3$ of water. A cylindrical glass piece of diameter 8.0 cm and height 8.0 cm is placed in the vessel. If the bottom of the vessel under the glass piece is seen by the paraxial rays, locate its image. The index of refraction of glass is 1.50 and that of water is 1.33.



- (A) 9.2 cm above the bottom
(B) 12.5 cm above the bottom
(C) 7.1 cm above the bottom
(D) 7.1 cm above below the bottom

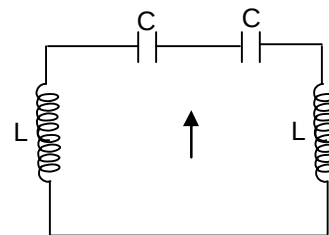
2. Figure shows a narrow –slit S illuminated by a monochromatic light of wavelength λ in a double –slit experiment. In the path of the rays reaching the upper slit S_1 , a tube of length L is interposed in which the index of refraction of the medium varies linearly as shown in figure. The position of the central maximum in the interference pattern on the screen was displaced by N fringes. Find the value of N in terms of μ_0 , L and λ .



- (A) $\frac{(1.5\mu_0 - 1)L}{\lambda}$
(B) $\frac{(\mu_0 - 1)L}{\lambda}$
(C) $\frac{3(\mu_0 - 1)L}{2\lambda}$
(D) $\frac{2(\mu_0 - 1)L}{3\lambda}$
3. If potential energy between a proton and an electron is given by $|U| = \frac{ke^2}{2R^3}$, where e is the charge of electron and R is the radius of atom, then radius of Bohr's orbit is given by (h = Planck's constant, k = constant)
- (A) $\frac{ke^2m}{h^2}$
(B) $\frac{6\pi^2 ke^2m}{n^2 h^2}$
(C) $\frac{2\pi ke^2m}{n h^2}$
(D) $\frac{4\pi^2 ke^2m}{n^2 h^2}$

4. The natural frequency of the circuit shown in figure is

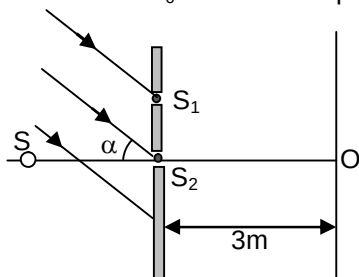
- (A) $\frac{1}{\sqrt{LC}}$
 (B) $\frac{1}{\sqrt{2LC}}$
 (C) $\frac{2}{\sqrt{LC}}$
 (D) None of these



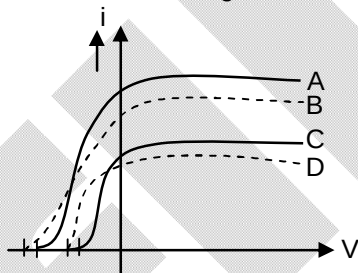
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. A parallel beam of light ($\lambda = 500 \text{ nm}$) is incident at an angle $\alpha = 30^\circ$ with the normal to the slit plane in Young's double-slit experiment. Assume that the intensity due to each slit at any point on the screen is I_0 . Point O is equidistant from S_1 and S_2 . The distance between slits is 1 mm . Then

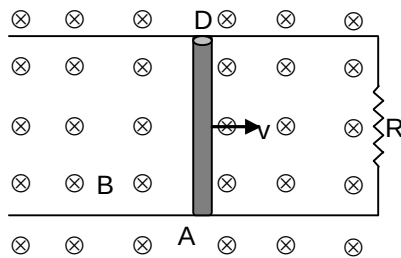


- (A) The intensity at O is $4I_0$
 (B) The intensity at O is zero
 (C) The intensity at a point on the screen 1 m below O is $4I_0$
 (D) The intensity at a point on the screen 1 m below O is zero
6. The figure shows the results of an experiment involving photoelectric effect. The graphs A, B, C and D relate to a light beam having different wavelengths. Select the correct alternative:

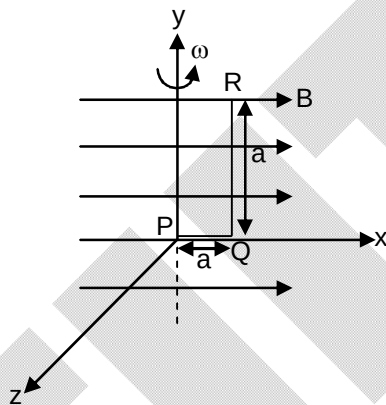


- (A) Beam B has highest frequency
 (B) Beam C has longest wavelength
 (C) Beam A has highest rate of photoelectric emission
 (D) Photoelectrons emitted by B have highest momentum
7. An X-ray tube is operated at 6.6 kV . In the continuous spectrum of the emitted X-rays, which of the following frequencies will be missing?
- (A) 10^{18} Hz
 (B) $1.5 \times 10^{18} \text{ Hz}$
 (C) $2 \times 10^{18} \text{ Hz}$
 (D) $2.5 \times 10^{18} \text{ Hz}$

8. The conductor AD moves to the right in a uniform magnetic field directed into the plane of the paper.



- (A) The free electron in AD will move toward A
 (B) D will acquire a positive potential with respect to A
 (C) A current will flow from A to D in AD in closed loop
 (D) The current in AD flows from lower to higher potential
9. The magnetic flux ϕ linked with a conducting coil depends on time as $\phi = 4t^n + 6$, where n is a positive constant. The induced emf in the coil is e .
 (A) If $0 < n < 1$, $e \neq 0$ and $|e|$ decreases with time.
 (B) If $n = 1$, e is constant
 (C) If $n > 1$, $|e|$ increases with time
 (D) If $n > 1$, $|e|$ decreases with time
10. In a region there exists a magnetic field B_0 along positive x-axis. A metallic wire of length $2a$, one side along x-axis and one side parallel of y-axis rotates about y-axis with a angular velocity. Then at the instant shown

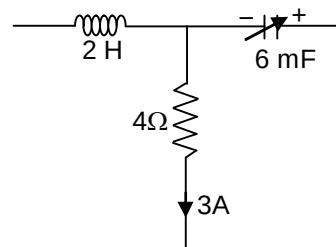
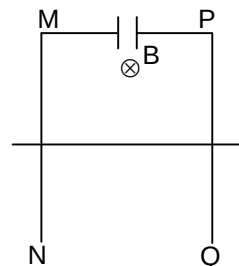


- (A) Potential difference across PQ is 0
 (B) Potential difference across PQ is $\frac{1}{2}B_0\omega a^2$
 (C) Potential difference across QR is $\frac{1}{2}B_0\omega a^2$
 (D) Potential difference across QR is $B_0\omega a^2$

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

11. An element of atomic number 9 emits K_{α} X-ray of wavelength λ . Find the atomic number of the element which emits K_{α} X-ray of wavelength 4λ .
12. In the spectrum of singly ionized helium, the wavelength of a line observed is almost the same as the first line of Balmer series of hydrogen. It is due to transition of electron from $n_1 = 6$ to $n_2 = X$. What is the value of X?
13. Find the quantum number 'n' corresponding to exciting state of He^+ ion if on transition to the ground state that ion emits two photons in succession with wavelength $1026.7 \text{ \AA}$ and $304 \text{ \AA}$. ($R = 1.096 \times 10^7 / \text{m}$)
14. Nuclei A and B convert into a stable nucleus C. Nucleus A is converted into C by emitting 2α – particle and 3β – particles. Nucleus B is converted into C by emitting one α – particle and 5β – particle. At time $t = 0$, nuclei of A are $4N_0$ and nuclei of B are N_0 . Initially, number of nuclei of C are zero. Half-life of A (into conversion of C) is 1 min and that of B is 2 min. Find the time (in minutes) at which rate of disintegration of A and B are equal.
15. The half-life of a radioactive nuclide is 20 h. It is found that the fraction $\left(\frac{1}{x}\right)$ of original activity remains after 40 hours? What is the value of x?
16. Figure shows a conducting rod of length $l = 10 \text{ cm}$, resistance R and mass $m = 100 \text{ mg}$ moving vertically downward due to gravity. Other parts are kept fixed. Magnetic field is $B = 1 \text{ T}$. MN and PQ are vertical, smooth, conducting rails. The capacitance of the capacitor is $C = 10 \text{ mF}$. The rod is released from rest. Find the maximum current (in mA) in the circuit.
17. A copper wire of length 2 m placed perpendicular to the plane of magnetic field $\vec{B} = (2\hat{i} + 4\hat{j}) \text{ T}$. If it moves with velocity $(4\hat{i} + 6\hat{j} + 8\hat{k}) \text{ m/sec}$. Calculate the magnitude of dynamic emf (in volt) across its ends.
18. Figure shows a part of a bigger circuit. The capacity of the capacitor is 6 mF and is decreasing at the constant rate 0.5 mF s^{-1} . The potential difference across the capacitor at the shown moment is changing as follows:



$$\frac{dV}{dt} = 2 \text{ Vs}^{-1}, \frac{d^2V}{dt^2} = \frac{1}{2} \text{ Vs}^{-2}$$

The current in the 4Ω resistor is decreasing at the rate of 1 mA s^{-1} . What is the potential difference (in mV) across the inductor at this moment?

Chemistry

PART – II

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

19. Chlorine reacts with excess of ammonia to form
(A) NH_4Cl (B) $\text{N}_2 + \text{HCl}$
(C) $\text{N}_2 + \text{NH}_4\text{Cl}$ (D) $\text{N}_2 + \text{NCl}_3$
20. The products of the chemical reaction between $\text{Na}_2\text{S}_2\text{O}_3$, Cl_2 and H_2O are
(A) S, HCl, Na_2SO_4 (B) S, HCl, Na_2S
(C) S, HCl, Na_2SO_3 (D) S, NaClO_3
21. Which of the following ions will finally give a black precipitate with Ag^+ ion?
(A) SO_3^{2-} (B) Br^-
(C) CrO_4^{2-} (D) $\text{S}_2\text{O}_3^{2-}$
22. Morphine narcotic are obtained from:
(A) Opium (B) Avena
(C) Datura (D) All of these

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

23. For the spinel structure (MgAl_2O_4), the correct statement is/are
(A) 50% OVs are occupied by ions.
(B) Al^{3+} is equally distributed in TVs and OVs.
(C) Oxide ions occupy ccp lattice.
(D) 12.5% TVs are occupied by ions.
24. 2 L of 1 molar solution of a complex salt $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ (M.W. = 266.5) at 27°C , shows an osmotic pressure of 98.52 atm. The solution is now treated with 1 L of 6 M AgNO_3 , which of the following are correct?
(A) Weight of AgCl precipitated is 861 g.
(B) The clear solution will show an osmotic pressure of 98.52 atm.
(C) The clear solution will show an osmotic pressure of 65.68 atm.
(D) 2 mol of $[\text{Cr}(\text{H}_2\text{O})_6(\text{NO}_3)_3]$ will be present in the solution.
25. The correct statement(s) about surface properties is(are)
(A) Adsorption is accompanied by decrease in enthalpy and decrease in entropy of the system
(B) The critical temperatures of ethane and nitrogen are 563 K and 126 K, respectively. The adsorption of ethane will be more than that of nitrogen on same amount of activated charcoal at a given temperature
(C) Cloud is an emulsion type of colloid in which liquid is dispersed phase and gas is dispersion medium
(D) Brownian motion of colloidal particles does not depend on the size of the particles but depends on viscosity of the solution

35. How many pairs of enantiomers are possible for $[M(AA)(BC)de]$?
where AA $\rightarrow$ symmetrical bidented ligand
BC $\rightarrow$ unsymmetrical bidented ligand and
d, e $\rightarrow$ monodentent ligand
36. An aqueous solution contains Hg^{2+} , Hg_2^{2+} , Pb^{2+} and Cd^{2+} . Out of these, how many ions will produce white precipitate with dilute HCl?

Mathematics**PART – III****Section – A (Maximum Marks: 12)**

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

37. Area by the curve $f(x) = \frac{x^2 - 4}{x^2 + 4}$ and line $y = 1$
- (A) π (B) 2π
(C) 3π (D) 4π
38. If A is a symmetric matrix such that $A = [a_{ij}]_{3 \times 3}$, all a_{ij} are decimal digits except zero, so number of matrices A for $\text{tr}(A) = 12$ is
- (A) $3(\underline{3})^2$ (B) $5(\underline{3})^2$
(C) $7(\underline{3})^2$ (D) $9(\underline{3})^2$
39. Two men X and Y have two independent target to shoot, both having probability $\frac{1}{3}$ and $\frac{3}{5}$ respectively. Then, probability of Y require more shots than A is
- (A) $\frac{1}{11}$ (B) $\frac{2}{11}$
(C) $\frac{3}{11}$ (D) $\frac{4}{11}$
40. If $|\vec{a}| = 2$, $|\vec{b}| = 5$, $|\vec{c}| = 8$, where $\vec{a}$, $\vec{b}$, $\vec{c}$ are vectors such that $[[\vec{a} \vec{b} \vec{c}]] = 80$, then value of $(\vec{b} + \vec{c}) \cdot \{(\vec{a} \times \vec{c}) \times (\vec{a} - \vec{c}) + 2\vec{b}\}$ has number of divisors
- (A) 9 (B) 11
(C) 12 (D) 24

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

41. If $f(x)$ is a differentiable function satisfying $\int_0^{2x} xf(t)dt + 2 \int_x^0 tf(2t)dt = 2x^4 - 2x^3$ for all $x \in \mathbb{R}$, then which of the following is/are correct?
- (A) $f(x) \geq -\frac{3}{4}$
(B) $f(|x|)$ is not differentiable at exactly one value of x
(C) Area bounded by $y = f(x)$ and x -axis is equal to 2
(D) $\lim_{x \rightarrow 1} \frac{f(x)}{(x-1)}$ exists and equal to 3

42. If the linear equations
 $8x + y + 4z = -2$
 $x + y + z = 0$
 $\lambda x - 3y = \mu$
 has infinitely many solutions, then
 (A) Vectors $(\lambda\hat{i} - \hat{j} + \mu\hat{k})$ and $((\lambda - 3)\hat{i} + \mu\hat{j} + 3\hat{k})$ are orthogonal to each other
 (B) Distance of a plane $8x + y + 4z + 2 = 0$ from $\left(\lambda, \mu, -\frac{1}{2}\right)$ is 4
 (C) Distance of plane $8x + y + 4z + 2 = 0$ from $\left(\lambda, \mu, -\frac{1}{2}\right)$ is $\left(\frac{\lambda}{\lambda - 1} - \mu\right)$
 (D) Equations $(a + b + c) = (\lambda - 2\mu)$ has 21 positive integral solutions for (a, b, c) triplets
43. If $x \in \mathbb{R}$ and $f(x) = \begin{vmatrix} x^2 + x & 2x - 1 & x + 3 \\ 3x + 1 & x^2 + 2 & x^3 - 3 \\ x - 3 & x^2 + 4 & 2x \end{vmatrix}$ is a polynomial of degree "7"
 $a_0x^7 + a_1x^6 + a_2x^5 + \dots + a_6x + a_7$, then
 (A) $\sum_{k=0}^{k=6} a_k = 111$
 (B) $\sum_{k=0}^{k=6} (-1)^k \cdot a_k = -11$
 (C) $\sum_{k=0}^{k=6} (-1)^k \cdot a_k = 53$
 (D) $\sum_{k=1}^{k=4} a_{2k-1} = 100$
44. If A and B are two events such that $P(B/A) = \frac{2}{5}$ and $P(A/B) = \frac{1}{7}$ and $P(A \cap B) = \frac{1}{9}$, then
 (A) $P(A' \cup B) = \frac{5}{6}$
 (B) $P(A' \cap B') = \frac{1}{18}$
 (C) $P(A - B) = \frac{1}{6}$
 (D) $P(A \cup B') = \frac{2}{3}$
45. Let $f : \mathbb{R} \rightarrow (0, 1)$ be a continuous function, then which of the following pair of vectors are linearly dependent for some $x \in (0, 1)$?
 (A) $\vec{a} = f(x)\hat{i} + 2\hat{j}$, $\vec{b} = x^2\hat{i} + 3\hat{j}$
 (B) $\vec{a} = f(x)\hat{i} + 3\hat{j}$, $\vec{b} = x^2\hat{i} + 2\hat{j}$
 (C) $\vec{a} = \left(\int_0^{1-x} f(t) dt\right)\hat{i} + 3\hat{j}$, $\vec{b} = x\hat{i} + 2\hat{j}$
 (D) $\vec{a} = \left(\int_0^{1-x} f(t) dt\right)\hat{i} + 2\hat{j}$, $\vec{b} = x\hat{i} + 3\hat{j}$
46. If $y = f(x)$ is a solution of $x^2 \frac{dy}{dx} = y^2 \cdot e^{\frac{1}{x}}$ ($x \neq 0$) and $\lim_{x \rightarrow 0^-} f(x) = 1$, then
 (A) $\lim_{x \rightarrow \infty} f(x) = 1$
 (B) Range of $f(x)$ is $(0, 1) - \left\{\frac{1}{2}\right\}$
 (C) $\lim_{x \rightarrow -\infty} f(x) = \frac{1}{2}$
 (D) $\int_0^e f(x) dx < \int_{-e}^0 f(x) dx$

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

47. Let three matrices $A = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$, then $t_r(A) + t_r\left(\frac{ABC}{2}\right) + t_r\left(\frac{A(BC)^2}{2}\right) + T_r\left(\frac{A(BC)^3}{2}\right) + \dots + \infty =$
48. Let $A_r = \begin{bmatrix} r & 3r-1 \\ 0 & \frac{1}{2^r} \end{bmatrix}$, then the value of $\lim_{n \rightarrow \infty} \sum_{r=1}^n \det(A_r)$ is equal to:
49. Let $\hat{u}$ and $\hat{v}$ are unit vectors and $\hat{w}$ is a vector such that $\hat{u} \times \hat{v} + \hat{u} = \hat{w}$ and $\hat{w} \times \hat{u} = \hat{v}$. Then the value of $|\vec{u} \vec{v} \vec{w}|$ is:
50. Find the number of points, whose perpendicular distance from yz, zx and xy-planes are in A.P. and whose distance from x, y and z-axes are $\sqrt{13}, \sqrt{10}$ and $\sqrt{5}$ respectively.
51. If $\begin{vmatrix} a^2 + \lambda^2 & ab + c\lambda & ca - b\lambda \\ ab - c\lambda & b^2 + \lambda^2 & bc + a\lambda \\ ca + b\lambda & bc - a\lambda & c^2 + \lambda^2 \end{vmatrix} \begin{vmatrix} \lambda & c & -b \\ -c & \lambda & a \\ b & -a & \lambda \end{vmatrix} = (1 + a^2 + b^2 + c^2)^3$, find λ .
52. If $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)} \forall x, y \in \mathbb{R}$ and $f'(1)$ exists, and area under the curve $f(x)$ bounded by x-axis, $x = 0$ and $x = 1$ is $\frac{1}{3}$, then find $\lim_{n \rightarrow \infty} \sum_{r=1}^n e^{r/n} f\left(\frac{\sqrt{r}}{n}\right)$.
53. If area of the region given by inequalities $0 \leq y \leq \sqrt{4-x^2}$ and $x \leq y+x \leq 2$ is 'A' square units, then $[A]$ is (where $[A]$ denotes greatest integer function)
54. A person is known to speak truth 3 out of 4 times. He throws a dice and responds that it is a six. If the probability that it is actually a six is P, find the ten's digit in 1000P.

FIITJEE
ALL INDIA INTEGRATED TEST SERIES
JEE (Advanced)-2024
PART TEST – IV
PAPER –2
TEST DATE: 20-10-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. C
Sol. Given $r = 6$ cm, $r_1 = 4$ cm, $h_1 = 8$ cm
Let h = final height of water column
The volume of the cylindrical water column after the glass piece is put will be
$$\pi r^2 h = 800\pi + \pi r_1^2 h_1$$
$$r^2 h = 800 + r_1^2 h_1$$
$$(6)^2 h = 800 + (4)^2 \cdot 8$$
$$h = \frac{800 + 128}{36} = \frac{928}{36} = 25.7 \text{ cm}$$

There are two shifts due to glass block as well as water
Total shift = $(2.66 + 4.44) = 7.1$ cm above the bottom.

2. A
Sol. Optical path difference,
$$\Delta x = 2 \int_0^{L/2} \mu dx - L = 2\mu_0 \int_0^{L/2} \left(1 + \frac{2x}{L}\right) dx - L$$
$$= \frac{3}{2} \mu_0 L - L = [(1.5)\mu_0 - 1]L$$

Now, $\Delta x = N'\lambda$
$$\Rightarrow N = \frac{\Delta x}{\lambda} = \frac{(1.5\mu_0 - 1)L}{\lambda}$$

3. B
Sol. $U = -\frac{ke^2}{2R^3}, F = -\frac{dU}{dR} = -\frac{3ke^2}{2R^4}$
But, $F = \frac{mv^2}{R} \Rightarrow \frac{mv^2}{R} = \frac{3ke^2}{2R^4}$

Also, $mvR = \frac{nh}{2\pi}$

Solve to get $R = \frac{6\pi^2 ke^2 m}{n^2 h^2}$

4. A

Sol. $\omega = \frac{1}{\sqrt{L_{eq} C_{eq}}} = \frac{1}{\sqrt{\frac{2LC}{2}}} = \frac{1}{\sqrt{LC}}$

5. ABCD

Sol. Δx at O = d [path difference is maximum at O]

So, if $d = \frac{5\lambda}{2}$, O will be a minima. If $d = \lambda$, O will be a maxima.

If $d = \frac{5\lambda}{2}$, O will be a minima and hence intensity is minimum.

If $d = 4.8\lambda$, then total 10 minima can be observed on the screen, 5 above O and 5 below O, which correspond to

$$\Delta x = \pm \frac{\lambda}{2}, \pm \frac{3\lambda}{2}, \pm \frac{5\lambda}{2}, \pm \frac{7\lambda}{2}, \pm \frac{9\lambda}{2}$$

6. ABCD

Sol. Stopping potential $\propto$ frequency $\equiv \frac{1}{\text{wavelength}}$

Saturation current $\propto$ rate of photoelectron emission

Also, $K_{\max} = hv - \phi$, $p = \sqrt{2mK}$

7. CD

Sol. $V = 6.6 \text{ kV} = 6600 \text{ V}$

Now, $f_{\max} = \frac{eV}{h} = \frac{1.6 \times 10^{-19} \times 6600}{6.6 \times 10^{-34}}$

8. ABCD

Sol. Use concept of motional emf.

9. ABC

Sol. $\phi = 4t^n + 6$

$$\frac{d\phi}{dt} = 4nt^{n-1}$$

$$|e| = 4nt^{n-1}, |e| = \frac{4n}{t^{1-n}}$$

10. AD

Sol. PQ is parallel to magnetic field, hence no emf is induced in it. QR is $\perp$ perpendicular to magnetic field. Induced emf = $Bvl = B_0 \omega aa = B_0 \omega a^2$.

Section – B

11. 5

Sol. For K_{α} X-ray, $(Z-1)^2 \lambda = \text{constant}$.Hence, $(9-1)^2 \lambda = (Z-1)^2 (4\lambda)$

$$(Z-1)^2 = \frac{64}{4} = 16$$

$$Z-1 = 4$$

$$\text{Or } Z = 5$$

12. 4

Sol. For the first line of Balmer series of hydrogen

$$\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = \frac{5R}{36} \Rightarrow \lambda = \frac{36}{5R}$$

For singly ionized helium $Z = 2$

$$\frac{1}{\lambda'} = 4R \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$$

$$\text{Given } \lambda' = \lambda \Rightarrow 4R \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right) = \frac{5R}{36}$$

$$\Rightarrow n_2 = 4$$

13. 6

Sol. According to conservation of energy

$$\frac{hc}{\lambda_1} + \frac{hc}{\lambda_2} = RchZ^2 \left(\frac{1}{1^2} - \frac{1}{n^2} \right)$$

$$n = 6$$

14. 6

Sol. We have to find the time at which $\lambda_A N_A = \lambda_B N_B$

$$\left(\frac{\ln 2}{T_A} \right) (4N_0 e^{-\lambda_A t}) = (N_0) \left(\frac{\ln 2}{T_B} \right) (e^{-\lambda_B t})$$

$$e^{(\lambda_A - \lambda_B)t} = 8$$

$$(\lambda_A - \lambda_B)t = \ln 8 = 3(\ln 2)$$

$$\left(\frac{\ln 2}{1} - \frac{\ln 2}{2} \right) t = 3 \ln(2) \Rightarrow t = 6 \text{ min}$$

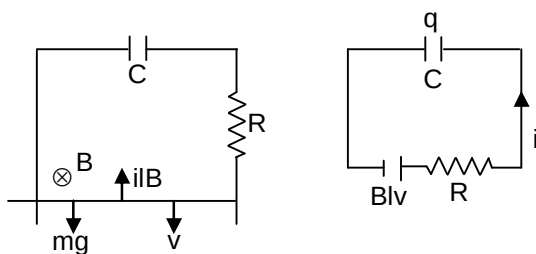
15. 4

Sol. We have $\frac{t}{t_{1/2}} = \frac{40 \text{ hours}}{20 \text{ hours}} = 2$

$$\text{Thus, } A = \frac{A_0}{2^{t/t_{1/2}}} = \frac{A_0}{2^2} = \frac{A_0}{4}$$

So one fourth of the original activity will remain after 40 hours.

16. 5
Sol.



By Newton's law, $mg - ilB = m \frac{dv}{dt}$ (i)

Using KVL $Blv = iR + \frac{q}{C}$ (ii)

Differentiating equation (ii) w.r.t. time, we get

$$Bl \frac{dv}{dt} = R \frac{di}{dt} + \frac{i}{C} \quad \text{....(iii)}$$

Eliminating $\frac{dv}{dt}$ from equations (i) and (iii), we get

$$\begin{aligned} mg - ilB &= \frac{m}{Bl} \left[R \frac{di}{dt} + \frac{i}{C} \right] \\ \Rightarrow mg Bl - iB^2 l^2 &= m \left(R \frac{di}{dt} + \frac{mi}{C} \right) \quad \text{....(iv)} \end{aligned}$$

I will be maximum when $\frac{dv}{dt} = 0$. Use this in equation (iv)

$$\Rightarrow mg BlC = i(B^2 l^2 C + m)$$

$$\Rightarrow i_{\max} = \frac{mgBlC}{m + B^2 l^2 C}$$

17. 8
Sol.

Dynamic emf

$$e_d = -\vec{l} \cdot (\vec{v} \times \vec{B}) = \vec{l} \cdot (\vec{B} \times \vec{v})$$

$$(\vec{B} \times \vec{v}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 4 & 0 \\ 4 & 6 & 8 \end{vmatrix} = \hat{i}(32-0) - \hat{j}(16-0) + \hat{k}(12-16)$$

$$= 32\hat{i} - 16\hat{j} - 4\hat{k}$$

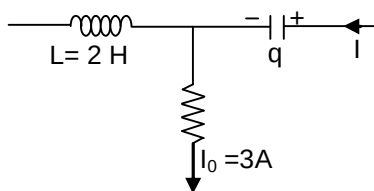
$$e_d = (2\hat{k}) \cdot (32\hat{i} - 16\hat{j} - 4\hat{k}) = -8 \text{ volt}$$

18. 4

Sol.

$$q = CV, \quad I = \frac{dq}{dt} = C \frac{dv}{dt} = V \frac{dC}{dt}, \quad I_L = I_0 - I$$

$$\frac{dI_L}{dt} = \frac{dI_0}{dt} - \left[C \frac{d^2V}{dt^2} + \frac{dV}{dt} + \frac{dC}{dt} + V \frac{d^2C}{dt^2} \right]$$



$$= -1 \times 10^{-3} - \left[6 \times 10^{-3} \times \frac{1}{2} + 2(-0.5 \times 10^{-3}) + 2(-0.5 \times 10^{-3}) + 0 \right]$$

$$= -2 \times 10^{-3} \text{ A/s}$$

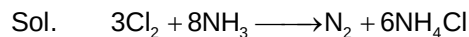
$$V_L = L \frac{dI}{dt} = 2 \times 2 \times 10^{-3} = 4 \times 10^{-3} \text{ V} = 4 \text{ mV}$$

Chemistry

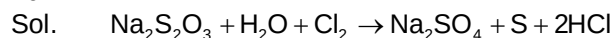
PART – II

Section – A

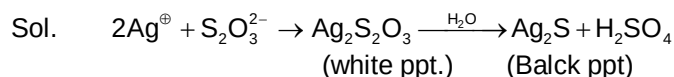
19. C



20. A



21. D



22. A

Sol. Morphine narcotics are sometimes referred to as opiates, since they are obtained from the opium poppy.

23. ACD

Sol. For spinel structure, O^{2-} ions form fcc arrangement.

$\therefore$ Number of TV = 8, Number of OV = 4

$$\text{Number of } \text{Mg}^{2+} \text{ ions} = \frac{1}{8} \times \text{TV} = \frac{1}{8} \times 8 = 1$$

$$\text{Number of } \text{Al}^{3+} \text{ ions} = \frac{1}{2} \times \text{OV} = \frac{1}{2} \times 4 = 2$$

So, 50% OVs are occupied and $1/8^{\text{th}}$, i.e., 12.5% TVs are occupied.

24. ACD

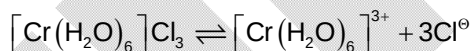


$$\pi = i \text{CRT}$$

$$98.52 = i \times 1 \times 0.0821 \times 300$$

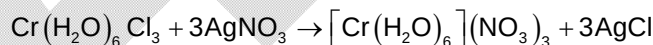
$$\therefore i = 4$$

$\therefore \text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ can be written as,



$$\begin{array}{ccc} 1 & 0 & 0 \\ 1-\alpha & \alpha & 3\alpha \end{array}$$

3 mol of AgNO_3 will react with 1 mol of $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$.



$$\begin{array}{cccc} \text{Moles} & 2 \times 1 = 2 & 1 \times 6 = 6 & 0 & 0 \\ & 0 & 0 & 2 & 6 \end{array}$$

$\therefore$ Moles of AgCl formed = 6

Weight of AgCl formed = $6 \times 143.5 = 861\text{g}$

$$\pi = \text{CRT} \times (1 + 3\alpha) = \frac{2}{3} \times 0.0821 \times 300 \times 4 = 65.68 \text{ atm}$$

25. AB

Sol. [Statements (A) and (B) are correct]

In adsorption process both ΔH and ΔS is $-ve$. Higher the critical temperature of a gas higher the extent of adsorption. Cloud is not an emulsion. [Statement (C) is wrong] Brownian motion depends on the size of the particles. [Statement (D) is also wrong].

26. ABC

Sol. Option (D) is wrong as Azurite is $Cu_3(CO_3)_2(OH)_2$

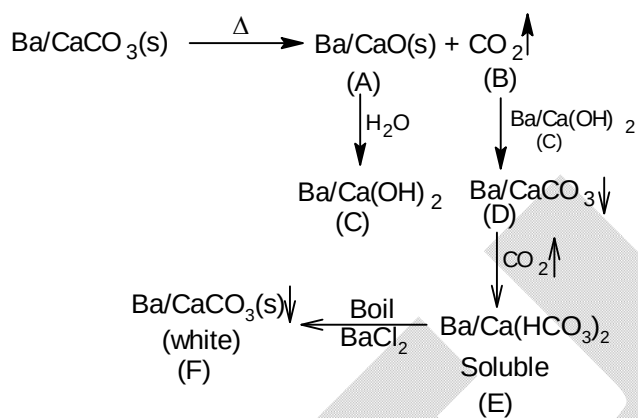
27. BCD

Sol. The number of unpaired electrons of the ions in the given choices are

 $Mn^{2+} (n = 5); Ni^{2+} (n = 2), Co^{2+} (n = 3) \text{ and } Fe^{2+} (n = 4)$

28. AC

Sol.

**Section – B**

29. 7

Sol. For ZnS structure, (Z_{eff} of ZnS = 4)Number of $B^\ominus = 4/\text{unit cell}$ (corner + face center)Number of $A^\oplus = 4/\text{unit cell}$ (in alternate TVs)Number of $B^\ominus$ ions removed = 4 (Two from each axis passing through the face center)

$$\times \frac{1}{2} (\text{per face center share}) = 2$$

Number of B^- ions left = $4 - 2 = 2/\text{unit cell}$ Number of Z^{2-} ions entering in place of $B^\ominus = 1$.[To maintain electrical neutrality, $2B^\ominus = 1Z^{2-}$]Formula = $A_4B_2Z_1$

$$\therefore x + y + z = 4 + 2 + 1 = 7$$

30. 2

Sol. Let $\pi_1 = 200 \text{ mm}; T_1 = 283$

$$\pi_2 = 105.3; T_2 = 298$$

$$\text{Now, } \pi = \frac{n}{V} RT$$

$$\text{At } T_1, 200 = \frac{n}{V_1} R \times 283 \quad \dots(i)$$

$$\text{At } T_2, 105.3 = \frac{n}{V_2} \times R \times 298 \quad \dots(ii)$$

Dividing Eq. (i) by Eq. (ii), we get

$$\frac{200}{105.3} = \frac{V_2}{V_1} \times \frac{283}{298} \Rightarrow V_2 = 2V_1$$

31. 3

Sol. Galena (PbS), sphalerite (Zn, Fe)S, chalcocite (Cu₂S).

32. 4

Sol. $\text{XeF}_4, \text{BrF}_4^\ominus \cdot [\text{Cu}(\text{NH}_3)_4]^{2+}, [\text{PtCl}_4]^{2-}$

33. 3

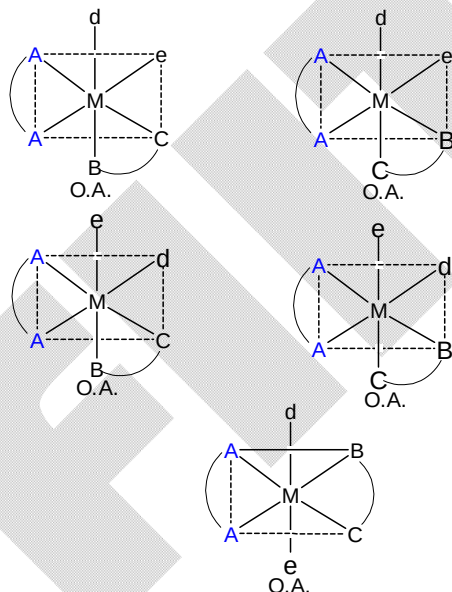
Sol. Volume of Cl₂ = $\frac{49}{100} \times 20 \times \frac{22.4}{71} = 3 \text{ L}$

34. 7

Sol. $\text{MnO}_4^\ominus, \text{Cr}_2\text{O}_7^{2-}, \text{CrO}_4^{2-}$ (they have 3d⁰ configuration yet they are colored due to charge transfer theory) $\text{Mn}^{3+}(3d^4), \text{Cu}^{2+}(3d^9), \text{Fe}^{2+}(3d^5)$, all are colored due to the presence of unpaired electrons. Colorless ions are: $\text{Sc}^{3+}(3d^0), \text{Ti}^{4+}(3d^0), \text{Zn}^{2+}(3d^{10})$

35. 5

Sol.



Five pairs of enantiomers are possible.

36. 2

Sol. $\text{Hg}_2^{2+}, \text{Pb}^{2+}$ given white ppt. of HgCl_2 and PbCl_2 with dil. HCl

Mathematics**PART – III****Section – A**

37. D

Sol. $\therefore 4 \times \int_{-1}^1 \sqrt{\frac{1+x}{1-x}} dx = 4 \times 2 \times \frac{\pi}{2} = 4\pi$

38. C

Sol. $\therefore (1, 2, 9); (1, 3, 8); (1, 4, 7); (1, 5, 6); (2, 3, 7); (2, 4, 6); (3, 4, 5)$

For $\text{tr}(A) = 12$, so ${}^7C_1 \times ({}^3P_2)$ are cases.

39. B

Sol. Let n shots for X .

So, probability $p_1 = \left(\frac{2}{3}\right)^{n-1} \cdot \frac{1}{3} = \frac{1}{2} \left(\frac{2}{3}\right)^n$

and m shots for Y

so, probability $p_2 = \left(\frac{2}{5}\right)^{m-1} \cdot \frac{3}{5}$

$$\sum_{m=n+1}^{\infty} p_2 = \left(\frac{2}{5}\right)^n$$

So, $P(m > n)$

$$\sum_{n=1}^{\infty} \frac{1}{2} \cdot \left(\frac{4}{15}\right)^n = \frac{2}{11}$$

40. C

Sol. $\therefore [(\vec{a} \times \vec{b}) \cdot \vec{c}] = 80 \Rightarrow \vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular vectors.

$\therefore$ Value of $(\vec{b} + \vec{c}) \cdot \{(\vec{a} \times \vec{c}) \times (\vec{a} - \vec{c}) + 2\vec{b}\}$ is equal to $306 = 3^2 \times 2 \times 17$

So, 12 divisors.

41. ABD

Sol. $x \cdot \int_0^{2x} f(t) dt + 2 \int_x^0 t f(2t) dt = 2x^4 - 2x^3$

$$\Rightarrow x(2f(2x)) + 1 \cdot \int_0^{2x} f(t) dt - 2x \cdot f(2x) = 8x^3 - 6x^2$$

$$\Rightarrow \int_0^{2x} f(t) dt = 8x^3 - 6x^2$$

$$\Rightarrow f(2x) \cdot 2 = 24x^2 - 12x$$

$$\Rightarrow f(2x) = 12x^2 - 6x$$

$$\Rightarrow f(x) = 12 \cdot \frac{x^2}{4} - 6 \cdot \frac{x}{2}$$

$$\Rightarrow f(x) = 3x^2 - 3x$$

42. ACD

Sol. $\therefore \begin{vmatrix} 8 & -2 & 4 \\ 1 & 0 & 1 \\ \lambda & \mu & 0 \end{vmatrix} = \begin{vmatrix} 8 & 1 & 4 \\ 1 & 1 & 1 \\ \lambda & -3 & 0 \end{vmatrix} = 0$

$\Rightarrow \lambda = 4 \text{ \& } \mu = -2$

So, (A) is correct as $4\hat{i} - \hat{j} - 2\hat{k}$ and $\hat{i} - 2\hat{j} + 3\hat{k}$ are perpendicular.

(B) is not correct as $\frac{10}{3}$ distance.

(C) is correct as $\left(\frac{4}{3} + 2\right) = \frac{10}{3}$

(D) is correct as $a + b + c = 8$, so ${}^{8-1}C_{3-1} = {}^7C_2 = 21$

43. ACD

Sol. $\therefore f(1) = 132 \text{ \& } f(0) = 21 = a_7$

$\therefore$ (A) is correct, $(132 - 21) = 111$

$f(-1) = -a_0 + a_1 - a_2 + a_3 - \dots - a_6 + a_7$

$-32 = -(a_0 - a_1 + a_2 - a_3 + \dots + a_6) + 21$

$\therefore \sum_{k=0}^{k=6} (-1)^k \cdot a_k = 53$

$f(1) + f(-1) = 100$

44. ABC

Sol. $\therefore P(A \cap B) = \frac{1}{9}$

$\therefore P(B/A) = \frac{2}{5}$

$\Rightarrow \frac{P(B \cap A)}{P(A)} = \frac{2}{5}$

$\Rightarrow \frac{1/9}{P(A)} = \frac{2}{5}$

$\Rightarrow P(A) = \frac{5}{18}$

$\& P(B) = \frac{7}{9}$

$\therefore P(A' \cup B) = 1 - P(A \text{ only}) = 1 - \frac{1}{6} = \frac{5}{6}$

45. BCD

Sol. $\therefore 2f(x) - 3x^2 = 0$

For (B), let $g(x) = 2f(x) - 3x^2$

$g(0) > 0 \text{ \& } g(1) < 0$

Similarly for (C) and (D).

46. BCD

Sol. $\therefore \frac{dy}{y^2} = \frac{e^x}{x^2} dx \Rightarrow -\frac{1}{y} = e^{-\frac{1}{x}} + c$

$$\Rightarrow y = \frac{1}{\frac{1}{1+e^x}}$$

So, $\lim_{x \rightarrow \pm\infty} f(x) = \frac{1}{1+1} = \frac{1}{2}$

and $y = \frac{1}{2} \Rightarrow e^{\frac{1}{x}} = 1 \Rightarrow x$ has no solution

So, range is $(0, 1) - \left\{\frac{1}{2}\right\}$.

Section – B

47. 6

Sol. $BC = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix} \Rightarrow BC = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$

$$t_r(A) + t_r\left(\frac{A}{2}\right) + t_r\left(\frac{A}{2^2}\right) + \dots$$

$$= t_r(A) + \frac{1}{2}t_r(A) + \frac{1}{2^2}t_r(A) + \dots$$

$$= \frac{t_r(A)}{1 - \left(\frac{1}{2}\right)} = 2t_r(A)$$

$$= 2(2+1) = 6$$

48. 2

Sol. $A_r = \frac{r}{2^r}$

Let $\lim_{n \rightarrow \infty} \sum_{r=1}^n dt(A_r) = S$

$$\therefore S = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \frac{4}{2^4} + \dots \quad \dots(i)$$

$$\Rightarrow \frac{1}{2}S = \frac{1}{2^2} + \frac{2}{2^3} + \frac{3}{2^4} + \dots \quad \dots(ii)$$

On eqns. (ii) – (i),

$$\Rightarrow \frac{S}{2} = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots$$

$$\Rightarrow S = 2$$

49. 1

 Sol. Given that $\vec{u} \times \vec{v} + \vec{u} = \vec{w}$ and $\vec{w} \times \vec{u} = \vec{v}$ -

$$\Rightarrow (\vec{u} \times \vec{v} + \vec{u}) \times \vec{u} = \vec{v}$$

$$\Rightarrow (\vec{u} \times \vec{v}) \times \vec{u} = \vec{v}$$

$$\Rightarrow \vec{v} - (\vec{u} \cdot \vec{v})\vec{u} = \vec{v}$$

$$\Rightarrow (\vec{u} \cdot \vec{v})\vec{u} = 0 \Rightarrow (\vec{u} \cdot \vec{v}) = 0$$

$$\begin{aligned} \text{Now, } [\vec{u} \vec{v} \vec{w}] &= \vec{u} \cdot (\vec{u} \times \vec{w}) = \vec{u} \cdot \{\vec{v} \times (\vec{u} \times \vec{v} + \vec{u})\} \\ &= \vec{u} \cdot \{\vec{v} \times (\vec{u} \times \vec{v}) + \vec{v} \times \vec{u}\} \\ &= \vec{u} \cdot \{\vec{v}^2 \vec{u} - (\vec{u} \cdot \vec{v})\vec{v} + \vec{v} \times \vec{u}\} \\ &= \vec{v}^2 \vec{u}^2 = 1 \end{aligned}$$

50. 8

Sol. As perpendicular distance are in A.P.

 Let $|a-d|, |a|, |a+d|$ be these distances.

So, the coordinates of point P are

$$[\pm(a-d), \pm a, \pm(a+d)]$$

Now, distance from P of x-axis, y-axis and z-axis are respectively,

$$2a^2 + d^2 - 2ad = 5 \quad \dots(i)$$

$$2a^2 + 2d^2 = 10 \quad \dots(ii)$$

$$2a^2 + d^2 + 2ad = 13 \quad \dots(iii)$$

Solving eqns. (i) and (iii), we get

$$4ad = 8.$$

Substituting in eqn. (ii)

$$\frac{4}{d^2} + d^2 = 5$$

$$\Rightarrow d^2 = 1 \text{ or } 4$$

$$\Rightarrow d = \pm 1, a = \pm 2$$

 So, points are $(\pm 1, \pm 2, \pm 3)$. (There will be 8 such points)

51. 1

Sol. We observe that the elements in the perfector are the cofactors of the corresponding elements of the post factor.

Hence L. H. S

$$\begin{aligned} &= \begin{vmatrix} \lambda & c & -b \\ -c & \lambda & a \\ b & -a & \lambda \end{vmatrix} = [\lambda(\lambda^2 + a^2 + b^2 + c^2)]^3 \\ &= (1 + a^2 + b^2 + c^2)^3 \Rightarrow \lambda = 1 \end{aligned}$$

52. 1

 Sol. If $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)} \forall x, y \in \mathbb{R}$

$$\Rightarrow f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

$$\Rightarrow f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - 1}{h} \quad \dots (i) \quad (\because f(1) = 1)$$

$$\text{Now, } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f\left(\frac{x+h}{x}\right) - 1}{h} \cdot f(x)$$

$$= f(x) \lim_{h \rightarrow 0} \frac{f\left(1 + \frac{h}{x}\right) - 1}{x \cdot \frac{h}{x}}$$

$$= \frac{f(x)}{x} f'(1) \text{ from equation (i)}$$

$$\Rightarrow f'(x) = \frac{f(x)}{x} \cdot k \text{ where, } f'(1) = k$$

$$\Rightarrow f(x) = x^k + c \text{ for } x = 1, f(x) = 1 \Rightarrow c = 0$$

$$\Rightarrow f(x) = x^k$$

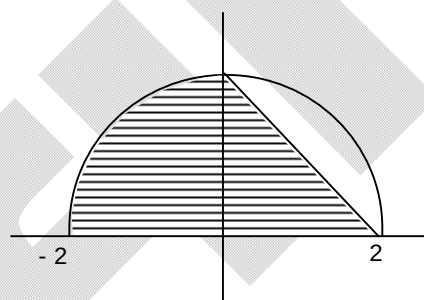
$$\text{Now, Area} = \int_0^1 f(x) dx = \frac{1}{3}$$

$$= \left[\frac{x^{k+1}}{k+1} \right]_0^1 = \frac{1}{3} \Rightarrow k = 2$$

$$\text{So, } f(x) = x^2$$

$$\text{Now, } \lim_{n \rightarrow \infty} \sum_{r=1}^n e^{r/n} f\left(\frac{\sqrt{r}}{n}\right) \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(e^{r/n} \cdot \frac{r}{n} \cdot \frac{1}{n} \right) = \int_0^1 e^x \cdot x \cdot dx = 1$$

53. 5
Sol.



$$A = \pi + 2$$

$$\therefore [A] = 5$$

54. 7
Sol. Let

E = the event that A reports it is six

E_1 : six turns up when a dice is thrown.

E_2 : six does not turn up when a dice is thrown.

Probability that six actually showed up if A reported that of was a six $= P\left(\frac{E_1}{E}\right)$

$$P\left(\frac{E_1}{E}\right) = \frac{P(E_1 \cap E)}{P(E)} = \frac{P(E_1) \cdot P\left(\frac{E_1}{E}\right)}{P(E_1) \cdot P\left(\frac{E_1}{E}\right) + P(E_2) \cdot P\left(\frac{E_2}{E}\right)}$$

$P\left(\frac{E}{E_1}\right)$ = Probability that the person reports six when it actually showed

= Probability that he speak truth $= \frac{3}{4}$

$P\left(\frac{E}{E_2}\right)$ = Probability that the person reports six when it did not show six

= Probability that he lies $= 1 - \frac{3}{4} = \frac{1}{4}$

Also, $P(E) = \frac{1}{6}$ and $P(E_2) = \frac{5}{6}$

$$P\left(\frac{E_1}{E}\right) = \frac{\frac{1}{6} \cdot \frac{3}{4}}{\frac{1}{6} \cdot \frac{3}{4} + \frac{5}{6} \cdot \frac{1}{4}} = \frac{\frac{1}{8}}{\frac{1}{8} + \frac{5}{24}} = \frac{3}{8}$$

$$\Rightarrow 1000P = 1000 \cdot \frac{3}{8} = 375$$

FIITJEE

ALL INDIA INTEGRATED TEST SERIES

PART TEST – IV

JEE (Main)-2024

TEST DATE: 15-09-2023

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

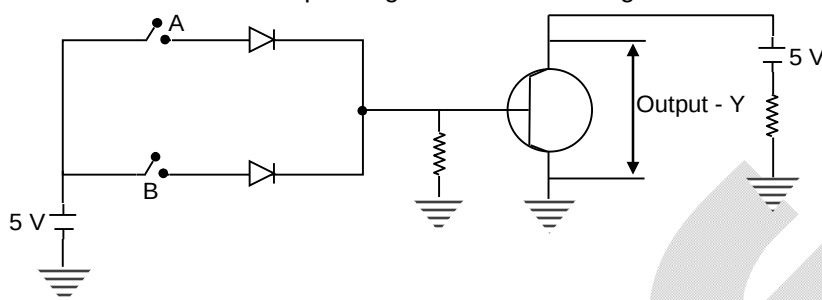
PART – A

SECTION – A

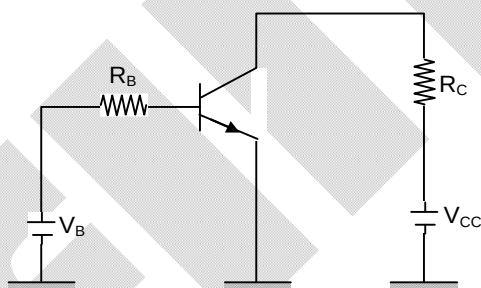
(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

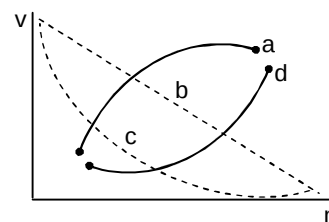
1. Boolean relation at the output stage-Y for the following circuit is:



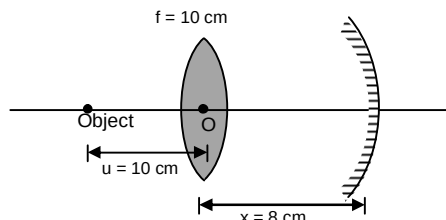
- (A) $A + B$
 (B) $\bar{A} + \bar{B}$
 (C) $\bar{A} \cdot \bar{B}$
 (D) $A \cdot B$
2. Given below are two statements:
 Statement-I : Potentials of both ends are same in an unbiased PN diode are same.
 Statement-II : In a solar cell P side is at higher potential than N side, upon incidence of solar light.
 (A) Statement-I is true but Statement-II is false
 (B) Statement-I is false but Statement-II is true
 (C) Both Statement-I and Statement-II are true
 (D) Both statement-I and Statement-II are false
3. A common emitter amplifier circuit built using an npn transistor, is shown in the figure. Its dc current gain is 250, $R_C = 1\text{ k}\Omega$ and $V_{CC} = 10\text{ V}$. What is the minimum base current for V_{CE} to reach saturation?



- (A) $100\mu\text{A}$
 (B) $7\mu\text{A}$
 (C) $40\mu\text{A}$
 (D) $10\mu\text{A}$
4. Which of the following curves may represent the speed of the electron in a hydrogen atom as a function of the principal quantum number n ?

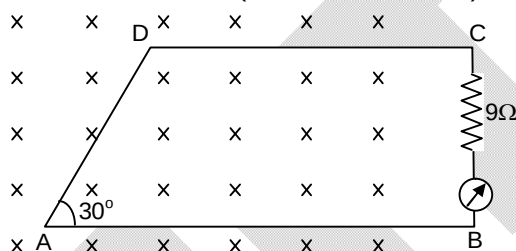


5. In one average-life,
 (A) half the active nuclei decay (B) less than half the active nuclei decay
 (C) more than half the active nuclei decay (D) Insufficient information
6. An object is placed at 10 cm from the lens. If the final image is formed at the optical centre of lens, then focal length of mirror will be



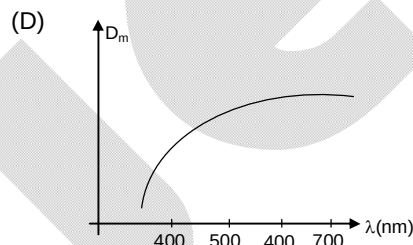
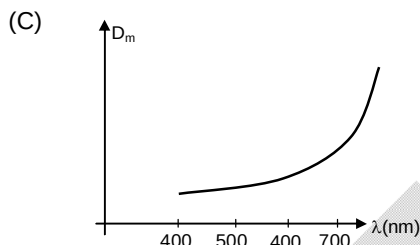
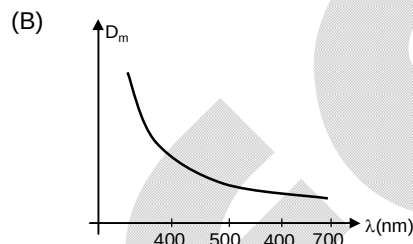
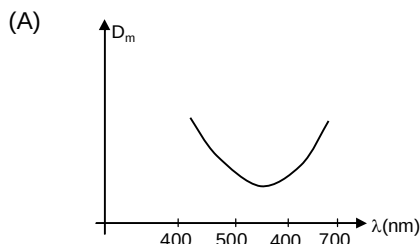
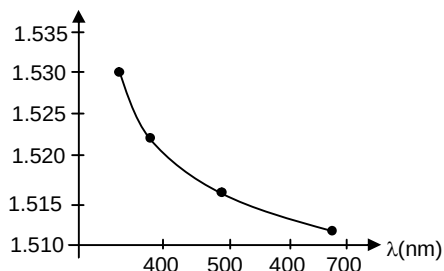
- (A) +10 cm (B) -8 cm
 (C) +8 cm (D) -10 cm
7. Energy levels A, B and C of a certain atom corresponding to increase values of energy, i.e., $E_A < E_B < E_C$. If λ_1, λ_2 and λ_3 are the wavelengths of radiations corresponding to transition C to B, B to A and C to A, respectively. Then, which of the following statements/relations is correct?
- (A) $\lambda_3 = \lambda_1 + \lambda_2$ (B) $\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$
 (C) $\lambda_1 + \lambda_2 + \lambda_3 = 0$ (D) $\lambda_3^2 = \lambda_1^2 + \lambda_2^2$

8. A loop ABCD is moving with velocity v towards right. The magnetic field is 2 T. Loop is connected to a resistance of 9Ω . If steady current of 2A flows in the loop, then value of v if loop has resistance of 4Ω is (Given, $AD = 30$ cm)



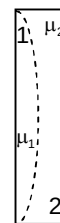
- (A) 87 m/s (B) 50 m/s
 (C) 33.33 m/s (D) 60 m/s
9. In a double slit experiment, a glass plate of refractive index 1.5 of thickness t is introduced in the path of one of the interfering beams of wavelength λ . The position of central maxima has got shifted by $\frac{2\lambda D}{d}$. Then, what would be the thickness of plate?
- (A) 4λ (B) 2λ
 (C) λ (D) 3λ

10. The variation of refractive index of a crown glass thin prism with wavelength of the incident light is shown. Which of the following graphs is the correct one, if D_m is the angle of minimum deviation?



11. In a photoelectric experiment, increasing the intensity of incident light:
- Increases the number of photons incident and the K.E. of the ejected electrons remains unchanged.
 - Increases the frequency of photons incident and the K.E. of the ejected electrons remains unchanged.
 - Increase the number of photons incident and also increases the K.E. of the ejected electrons.
 - Increases the frequency of photons incident and increases the K.E. of the ejected electrons.
12. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion-A: An electron microscope can achieve better resolving power than an optical microscope
Reason-R: The de-Broglie's wavelength of the electrons emitted from an electron gun is much less than wavelength of visible light
 In the light of the above statements, choose the correct answer from the options given below:
- A** is true but **R** is false
 - A** is false but **R** is true
 - Both **A** and **B** are true but **R** is NOT the correct explanation of **A**
 - Both **A** and **R** are true and **R** is the correct explanation of **A**

13. One plano-convex and one plano-concave thin lens of same radius of curvature 'R' but of different materials are joined side by side as shown in the figure. If the refractive index of the material of 1 is μ_1 and that of 2 is μ_2 , then the focal length of the combination is:



- (A) $\frac{R}{2 - (\mu_1 - \mu_2)}$ (B) $\frac{2R}{\mu_1 - \mu_2}$
 (C) $\frac{R}{2(\mu_1 - \mu_2)}$ (D) $\frac{R}{\mu_1 - \mu_2}$
14. Two plane mirrors are inclined to each other such that a ray of light incident on the first mirror (M_1) and parallel to the second mirror (M_2) is finally reflected from the second mirror (M_2) parallel to the first mirror (M_1). The angle between the two mirrors will be:
 (A) 90° (B) 45°
 (C) 75° (D) 60°

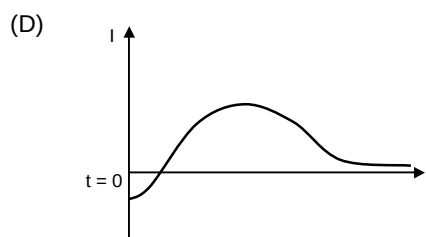
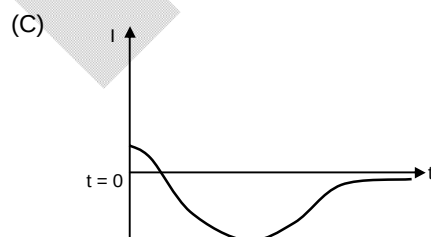
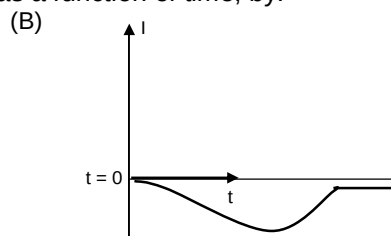
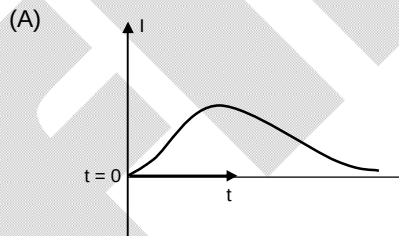
15. A plane polarized monochromatic EM wave is travelling a vacuum along z-direction such that at $t = t_1$ it is found that the electric field is zero at a spatial point z_1 . The next zero that occurs in its neighborhood is at z_2 . The frequency of the electromagnetic wave is:

- (A) $\frac{3 \times 10^8}{|z_2 + z_1|}$ (B) $\frac{6 \times 10^8}{|z_2 + z_1|}$
 (C) $\frac{1.5 \times 10^8}{|z_2 - z_1|}$ (D) $\frac{1}{t_1 + \frac{|z_2 - z_1|}{3 \times 10^8}}$

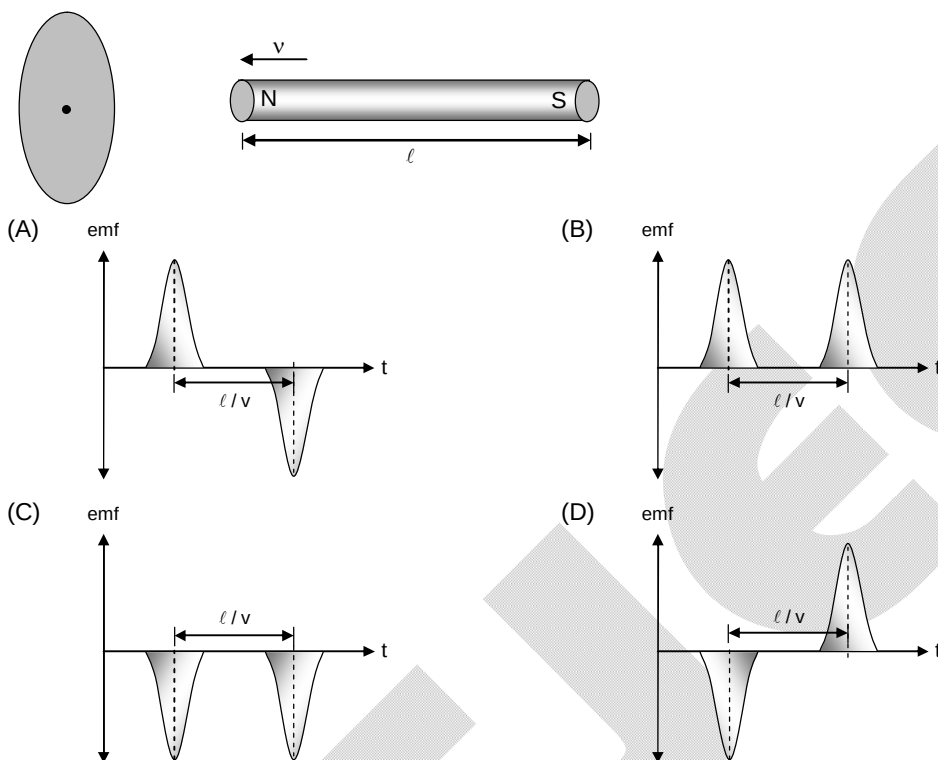
16. An alternating current is given by the equation $i = i_1 \sin \omega t + i_2 \cos \omega t$. The rms current will be:

- (A) $\frac{1}{\sqrt{2}}(i_1 + i_2)^2$ (B) $\frac{1}{\sqrt{2}}(i_1 + i_2)$
 (C) $\frac{1}{\sqrt{2}}(i_1^2 + i_2^2)^{1/2}$ (D) $\frac{1}{2}(i_1^2 + i_2^2)^{1/2}$

17. A very long solenoid of radius R is carrying current $I(t) = kte^{-\alpha t}$ ($k > 0$), as a function of time ($t \geq 0$). Counterclockwise current is taken to be positive. A circular conducting coil of radius 2R is placed in the equatorial plane of the solenoid and concentric with the solenoid. The current induced in the outer coil is correctly depicted, as a function of time, by:



18. A bar magnet is passing through a conducting loop of radius R with velocity v . The radius of the bar magnet is such that it just passes through the loop. The induced e.m.f. in the loop can be represented by the approximate curve: (Consider emf to be positive if clockwise when viewed from initial position of magnet)



19. An aeroplane, with its wings spread 10 m, is flying at a speed of 180 km/h in a horizontal direction. The total intensity of earth's field at that part is $2.5 \times 10^{-4} \text{ Wb/m}^2$ and the angle of dip is 60° . The emf induced between the tips of the plane wings will be _____.
 (A) 62.50 mV (B) 54.125 mV
 (C) 88.3 mV (D) 108.25 mV
20. An electromagnetic wave of intensity 50 Wm^{-2} enters in a medium of refractive index ' n ' and $\mu_r = 1$ without any loss. The ratio of the magnitudes of electric fields, and the ratio of the magnitudes of magnetic fields of the wave before and after entering into the medium are respectively, given by:
 (A) $\left(\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}\right)$ (B) $\left(\sqrt{n}, \frac{1}{\sqrt{n}}\right)$
 (C) $(\sqrt{n}, \sqrt{n})$ (D) $\left(\frac{1}{\sqrt{n}}, \sqrt{n}\right)$

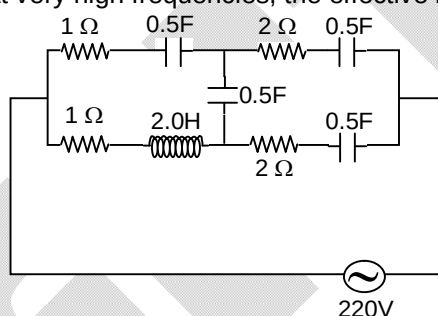
SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. Mobility of electrons in a semiconductor is defined as the ratio of their drift velocity to the applied electric field. If, for an n-type semiconductor, the density of electrons is 10^{18} m^{-3} and their mobility is $1.6 \text{ m}^2/(\text{V.s})$ then the resistivity of the semiconductor (since it is an n-type semiconductor contribution of holes is ignored) is close to _____ Ωm .

22. A single slit of width b is illuminated by a coherent monochromatic light of wavelength λ . If the second and fourth minima in the diffraction pattern at a distance 1 m from the slit are at 3 cm and 6 cm respectively from the central maximum, what is the width (in cm) of the central maximum? (i.e. distance between first minimum on either side of the central maximum):
23. An unpolarized light beam is incident on the polarizer of a polarization experiment and the intensity of light beam emerging from the analyzer is measured as 12 Lumens. Now, if the analyzer is rotated around the horizontal axis. (direction of light) by 30° in clockwise direction, the intensity of emerging light will be _____ Lumens.
24. A compound microscope consists of an objective lens of focal length 1 cm and an eye piece of focal length 5 cm with a separation of 10 cm. The distance between an object and the objective lens, at which the strain on the eye is minimum is $\frac{n}{4}$ cm. The value of n is:
25. When radiation of wavelength λ is incident on a metallic surface, the stopping potential of ejected photoelectrons is 4.8 V. If the same surface is illuminated by radiation of double the previous wavelength, then the stopping potential becomes 1.6 V. The ratio of threshold wavelength of metal to λ is
26. An unstable heavy nucleus at rest breaks into two nuclei which move away with velocities in the ratio of 8 : 27. If radius of lighter nucleus is $4r$ and radius of heavier nucleus is mr , find m .
27. The thickness at the centre of a plano convex lens is 3 mm and the diameter is 6 cm. If the speed of light in the material of the lens is $2 \times 10^8 \text{ ms}^{-1}$. The focal length of the lens is $6p$ cm. Find p .
28. The electric field in an electromagnetic wave is given by $E = (50 \text{ NC}^{-1}) \sin \omega(t - x/c)$. The energy contained in a cylinder of volume V is $5.5 \times 10^{-12} \text{ J}$. The value of V is $100n \text{ cm}^3$. The value of n is (Given $\epsilon_0 = 8.8 \times 10^{-12} \text{ C}^2 - \text{N}^{-1} \text{ m}^{-2}$)
29. At very high frequencies, the effective impedance of the given circuit will be _____ Ω .



30. From the given data, the amount of energy required to break the nucleus of aluminium ${}_{13}^{27}\text{Al}$ is $x \times 10^{-3} \text{ J}$. Find $\frac{x}{9}$. Mass of neutron = 1.00866 u, Mass of proton = 1.00726 u, Mass of Aluminium nucleus = 2.18846 u. (Assume 1 u corresponds to x J of energy) (Round off to the nearest integer)

Chemistry**PART – B****SECTION – A****(Single Choice Answer Type)**

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

31. For a solution of 0.849 g of mercurous chloride in 50 g of $\text{HgCl}_2(\text{l})$ the freezing point depression is 1.24°C . K_f for HgCl_2 is 34.3. What is the state of mercurous chloride in HgCl_2 ?
($\text{Hg} - 200, \text{Cl} - 35.5$)
(A) as Hg_2Cl_2 molecules (B) as HgCl molecules
(C) as Hg^+ and Cl^- ions (D) as Hg_2^{2+} and Cl^- ions
32. 20 moles of liquid A are mixed with 20 moles of liquid B to form an ideal binary solution. Calculate moles of A in liquid state when half of the solution has vaporized:
(Given : $P_A^\circ = 100$ torr, $P_B^\circ = 121$ torr)
(A) $\frac{11}{21}$ (B) $\frac{110}{21}$
(C) $\frac{200}{21}$ (D) $\frac{220}{21}$
33. Let the height of hcp unit cell is 'h'. The height of octahedral voids from the base is:
(A) $\frac{h}{2}$ (B) $\frac{h}{3}, \frac{2h}{3}$
(C) $\frac{h}{8}, \frac{7h}{8}$ (D) $\frac{h}{4}, \frac{3h}{4}$
34. The unit cell cube length for LiCl (just like NaCl structure) is $5.14 \text{ \AA}$. Assuming anion-cation contact, the ionic radius for chloride ion is:
(A) $1.815 \text{ \AA}$ (B) $2.8 \text{ \AA}$
(C) $3.8 \text{ \AA}$ (D) $4.815 \text{ \AA}$
35. Which of the following sols is positively charged?
(A) Arsenious sulphide (B) Aluminium hydroxide
(C) Ferric hydroxide (D) Silver iodine in silver nitrate solution
36. What is the $[\text{Fe}^{2+}]$ in a cell at 25°C for which $E = -0.458 \text{ V}$ with a standard hydrogen electrode?
 $\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Fe}(\text{s}) \quad E^\circ = -0.440 \text{ V}$
(A) 0.246 M (B) 0.496 M
(C) 2.01 M (D) 4.06 M
37. In $\text{H}_2 - \text{O}_2$ fuel cell the reaction occurring at cathode is:
(A) $2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^- \rightarrow 4\text{OH}^-$ (B) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}(\text{l})$
(C) $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$ (D) $\text{H}^+ + \text{e}^- \rightarrow \frac{1}{2}\text{H}_2$

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46. $\text{MnO}_2 + \text{KOH} \xrightarrow{\text{O}_2} \text{X}$. X must be:
 (A) KMnO_4 (B) K_2MnO_4
 (C) MnO (D) Mn_3O_4
47. The incorrect statement for lanthanides among the following statements is:
 (A) 4f and 5d orbitals are so close in energy that it is very difficult to locate the exact position of electrons in lanthanides
 (B) Most common stable oxidation state is +3
 (C) Tripositive lanthanide ions have characteristic colour depending on nature of group with which they combine to form compounds
 (D) Some lanthanide ions absorb colour either in infrared or ultraviolet region of electromagnetic spectrum
48. $\text{Cr}_2\text{O}_7^{2-} + 4\text{H}_2\text{O}_2 + 2\text{H}^+ \xrightarrow[\text{Solvent}]{\text{Organic}} 2 \text{CrO}_5 + 5\text{H}_2\text{O}$
- In above reaction, 1-pentanol is recommended. Dimethyl ether is not recommended for general use owing to its:
 (A) highly non-flammable character
 (B) highly inflammable character
 (C) highly poisonous character
 (D) none of the above
49. Blister copper is refined by stirring molten impure metal with green logs of wood because such a wood liberates hydrocarbon gases (like CH_4). This process X is called _____ and the metal contains impurities of Y is _____.
 (A) X = cupellation, Y = CuO_2
 (B) X = poling, Y = Cu_2O
 (C) X = poling, Y = CuO
 (D) X = cupellation, Y = CuO
50. Which of the following statement is incorrect regarding the substance known as Plumbago or black lead?
 (A) It is good conductor of electricity
 (B) Its formula is $\text{PbO} \cdot 2\text{PbO}_2$
 (C) It leaves black mark on paper
 (D) It is very soft and slippery

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. Find the number of species which would undergo disproportionation in alkaline medium.
 $\text{NO}_2, \text{Cr}_2\text{O}_7^{2-}, \text{Cl}_2, \text{O}_6, \text{Br}_2, \text{P}_4, \text{S}_8, \text{Al}_2, \text{O}_3$
52. How many of the following reactions would produce phosphine gas?
 (a) $\text{PH}_4\text{Cl} + \text{NaOH} \rightarrow$ (b) $\text{P}_4 + \text{NaOH} \xrightarrow{\Delta}$
 (c) $\text{Ca}_3\text{P}_2 + \text{H}_2\text{O} \rightarrow$ (d) $\text{H}_3\text{PO}_3 \xrightarrow{\Delta}$
 (e) $\text{PCl}_5 + \text{H}_2\text{O} \rightarrow$ (f) $\text{H}_3\text{PO}_2 \xrightarrow{\Delta}$
 (g) $\text{P}_2\text{Cl}_4 + \text{H}_2\text{O} \rightarrow$ (h) $\text{AlP} + \text{H}_2\text{O} \xrightarrow{\Delta}$
 (i) $\text{P}_4\text{O}_{10} + \text{H}_2\text{O} \rightarrow$

53. When mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4 then compound X is formed. What is the oxidation state of central atom of X?
54. Number of ions which gives blue color in aqueous state:
 $V^{+4}, Ni^{+2}, Ti^{+3}, Co^{+2}, Fe^{+3}, Cu^{+2}$
55. Find out the number of minerals given below which contain iron as Fe(II).
Hematite, Magnetite, Limonite, Siderite, Chromite, Wolframite
56. How many of the following metals is extracted by self reduction process?
Pb, Au, Mg, Sn, Fe, Al, Ag, Hg, Na, Zn, Cu, Ca
57. Elevation in boiling point of 0.25 M aq. Solution of weak acid HX is 0.14. What is molarity of H^+ ion in solution?
 $HX_{aq} \rightleftharpoons H_{aq}^+ + X_{aq}^-$
Given: $K_b(H_2O) = 0.5$
Molality = Molarity
[Given : Answer as $[H^+] \times 100$]
58. 100 ml of 1 M $AgNO_3$ solution is electrolysed using Pt electrode by passing a current of 9.65 A for 100 sec. Calculate pH of solution after electrolysis.
59. The reduction potential of hydrogen electrode at $25^\circ C$ when placed in a buffer solution is found to be -0.413 V. The pH of the buffer is:
60. If number of nearest neighbours, next nearest (2^{nd} nearest) neighbor and next to next nearest (3^{rd} nearest) neighbours are x, y and z respectively for body centered cubic unit cell, then calculate value of $\frac{xy}{z}$.

Mathematics

PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

61. If a, b, c are non-zero real numbers and $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = 0$, then $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$ is equal to
 (A) 1
 (B) -1
 (C) 0
 (D) None of these
62. If ' ω ' is a non-real cube root of unit, then $\begin{vmatrix} 1 & 1+i+\omega^2 & \omega^2 \\ 1-i & -1 & \omega^2-1 \\ -1 & -i+\omega-1 & -1 \end{vmatrix}$, where $i = \sqrt{-1}$, is equal to
 (A) 1
 (B) i
 (C) ω
 (D) 0
63. If $x < 1$, then value of $\begin{vmatrix} x^2+2 & 2x+1 & 1 \\ 2x+1 & x+2 & 1 \\ 3 & 3 & 1 \end{vmatrix}$ is
 (A) Non-negative
 (B) Non-positive
 (C) Negative
 (D) Positive
64. If $\vec{a} = x\hat{i} + (x-1)\hat{j} + \hat{k}$ and $\vec{b} = (x+1)\hat{i} + \hat{j} + a\hat{k}$ always make an acute angle with each other for every value of $x \in \mathbb{R}$, then
 (A) $a \in (-\infty, 2)$
 (B) $a \in (2, \infty)$
 (C) $a \in (-\infty, 1)$
 (D) $a \in (1, \infty)$
65. $\hat{i} \times (\vec{a} \times \hat{i}) + \hat{j} \times (\vec{a} \times \hat{j}) + \hat{k} \times (\vec{a} \times \hat{k})$ is always equal to
 (A) $\vec{a}$
 (B) $-\vec{a}$
 (C) $2\vec{a}$
 (D) $-2\vec{a}$
66. Distance of $P(\vec{p})$ from the line $\vec{r} = \vec{a} + \lambda\vec{b}$, is
 (A) $\left| (\vec{a} - \vec{p}) + \frac{((\vec{p} - \vec{a}) \cdot \vec{b})\vec{b}}{|\vec{b}|^2} \right|$
 (B) $\left| (\vec{b} - \vec{p}) + \frac{((\vec{p} - \vec{a}) \cdot \vec{b})\vec{b}}{|\vec{b}|^2} \right|$
 (C) $\left| (\vec{a} - \vec{p}) + \frac{((\vec{p} - \vec{b}) \cdot \vec{b})\vec{b}}{|\vec{b}|^2} \right|$
 (D) None of these

67. If $\vec{a} = \vec{b} + \vec{c}$, $\vec{b} \times \vec{d} = \vec{0}$, $\vec{c} \cdot \vec{d} = 0$, then the vector $\frac{\vec{d} \times (\vec{a} \times \vec{d})}{|\vec{d}|^2}$ is always equal to
- (A) $\vec{a}$ (B) $\vec{d}$
(C) $\vec{b}$ (D) $\vec{c}$
68. $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{j} - \hat{k}$, $\vec{c}$ be a vector such that $\vec{a} \cdot \vec{c} = 3$, $\vec{a} \times \vec{c} = \vec{b}$, then
- (A) $\vec{c} = \frac{1}{2}(\hat{i} + 4\hat{j} + \hat{k})$ (B) $\vec{c} = \frac{1}{3}(-\hat{i} + 8\hat{j} + \hat{k})$
(C) $\vec{c} = \frac{1}{2}(4\hat{i} + \hat{j} + \hat{k})$ (D) $\vec{c} = \frac{1}{3}(5\hat{i} + 2\hat{j} + 2\hat{k})$
69. If $\vec{a}$ and $\vec{b}$ are two non-collinear vectors, then $|\vec{a} + \vec{b} + (\vec{a} \times \vec{b})|^2 + (1 - \vec{a} \cdot \vec{b})^2$ is always equal to
- (A) $(1 + |\vec{a}|^2)(1 + |\vec{b}|^2)$ (B) $(1 - |\vec{a}|^2)(1 - |\vec{b}|^2)$
(C) $(1 + |\vec{a}|^2)(1 - |\vec{b}|^2)$ (D) $(1 - |\vec{a}|^2)(1 + |\vec{b}|^2)$
70. The probabilities of happening the events A and B are $\frac{1}{4}$ and $\frac{1}{2}$ respectively. If the probability of happening of A and B simultaneously is $\frac{7}{50}$, then probability of neither A nor B happening, is equal to
- (A) $\frac{39}{100}$ (B) $\frac{1}{4}$
(C) $\frac{11}{100}$ (D) None of these
71. A and B play a game where each is asked to select a number from 1 to 25. If the number selected by A and B match, both of them win a prize. The probability that they win their third prize on 5th game is equal to
- (A) $\frac{6 \cdot (24)^2}{(25)^5}$ (B) $\frac{6 \cdot (21)^2}{(25)^5}$
(C) $\frac{(24)^2}{(25)^5}$ (D) $\frac{(21)^2}{(25)^5}$
72. A real estate man has eight master keys to open several new homes. Only one master key will open any given home. If 40% of these homes are usually left unlocked, the probability that the real estate man can get into a specific home, if it is given that he selected 3 keys randomly before leaving his office, is equal to
- (A) $\frac{5}{8}$ (B) $\frac{3}{8}$
(C) $\frac{3}{4}$ (D) $\frac{1}{4}$

73. If $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 4 \\ 2 & 3 \end{bmatrix}$ and $A \cdot B \cdot C = \begin{bmatrix} 4 & 8 \\ 3 & 7 \end{bmatrix}$, then C equals to
- (A) $\begin{bmatrix} \frac{9}{11} & \frac{5}{8} \\ \frac{1}{22} & \frac{1}{6} \end{bmatrix}$ (B) $\begin{bmatrix} -\frac{9}{11} & -\frac{5}{3} \\ \frac{1}{22} & \frac{1}{6} \end{bmatrix}$
- (C) $\begin{bmatrix} \frac{9}{11} & \frac{5}{3} \\ \frac{1}{22} & -\frac{1}{6} \end{bmatrix}$ (D) None of these
74. Area enclosed by the curve $(y - \sin^{-1} x)^2 = x - x^2$, is equal to
- (A) $\frac{\pi}{2}$ sq. units (B) $\frac{\pi}{4}$ sq. units
- (C) $\frac{\pi}{8}$ sq. units (D) None of these
75. The tangent at any arbitrary point P(x, y) on the curve $y = f(x)$, meets the x and y-axis at the points A and B respectively. If $PA : PB = 2 : 1$, then curve is
- (A) $xy^2 = c$ (B) $xy = c^2$
- (C) $x^2y = c$ (D) None of these
76. Solution of the differential equation $\frac{xdx + ydy}{ydx - xdy} = \frac{x - \cos^2(x^2 + y^2)}{y^3}$ is
- (A) $\cot(x^2 + y^2) = \frac{x^2}{y^2} + c$ (B) $\tan(x^2 + y^2) = \frac{y^2}{x^2} + c$
- (C) $\tan(x^2 + y^2) = \frac{x^2}{y^2} + c$ (D) $\cot(x^2 + y^2) = \frac{y^2}{x^2} + c$
77. General solution of $(xy \sin xy + \cos xy)ydx + (xy \sin xy - \cos xy)x dy = 0$, is
- (A) $y \cdot \sin(xy) = cx$ (B) $x \cdot \sec(xy) = cy$
- (C) $y \cdot \tan(xy) = cx$ (D) $y \cdot \tan(xy) = cx$
78. Orthogonal trajectories of the family of parabolas $y^2 = ax$, is
- (A) $x^2 - 2y^2 = C$ (B) $x^2 + 2y^2 = C$
- (C) $y^2 - 2x^2 = C$ (D) $y^2 + 2x^2 = C$
79. The equation of a plane passing through (1, 2, -3), (0, 0, 0) and perpendicular to the plane $3x - 5y + 2z = 11$, is
- (A) $3x + y + \frac{5}{3}z = 0$ (B) $4x + y + 2z = 0$
- (C) $3x - y + \frac{z}{3} = 0$ (D) $x + y + z = 0$

80. The distance of the point $(-1, 2, 6)$ from the line $\frac{x-2}{6} = \frac{y-3}{3} = \frac{z+4}{-4}$, is equal to
 (A) 7 units (B) 9 units
 (C) 10 units (D) 12 units

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. If $2ax - 2y + 3z = 0$, $x + ay + 2z = 0$ and $2x + az = 0$ have a non-trivial solution, then
82. If $a \neq p, b \neq q, c \neq r$ and $\begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix} = 0$, then the value of $\frac{p}{p-a} + \frac{p}{q-b} + \frac{r}{r-c}$ is equal to
83. A, B, C and D are any four points in the space. If $|\overline{AB} \times \overline{CD} + \overline{BC} \times \overline{AD} + \overline{CA} \times \overline{BD}| = \lambda \Delta_{ABC}$, where Δ_{ABC} is the area of triangle ABC, then λ is equal to
84. For any four vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$, the expression $(\vec{b} \times \vec{c}) \cdot (\vec{a} \cdot \vec{d}) + (\vec{c} \times \vec{a}) \cdot (\vec{b} \times \vec{d}) + (\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d})$ is always equal to
85. $\overline{OA} = \vec{a}, \overline{OC} = \vec{b}, \overline{OB} = 10\vec{a} + 2\vec{b}$ where O, A and C are non-collinear points. Let p denotes the area of quadrilateral OABC and q denotes the area of parallelogram with OA and OC as adjacent sides. If $p = \lambda q$, then λ is
86. Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}, \vec{b} = \hat{i} + \hat{j}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = |\vec{c}|, |\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $\vec{a} \times \vec{b}$ and $\vec{c}$ is $\frac{\pi}{6}$. Then $2|(\vec{a} \times \vec{b}) \times \vec{c}|$ is equal to
87. A set is constructed random from a group of 6 boys and 4 girls. It is known that in this group exactly one girl and one boy belong to same parents. The probability that the selected group of children have no blood relations, is equal to
88. Area bounded by the curves $4y = |x^2 - 4|$ and $y + |x| = 7$, is equal to k sq. units, then k is
89. A point 'P' moves in xy plane in such a way that $[|x|] + [|y|] = 1$ where $[.]$ denotes the greatest integer function. Area of the region representing all possible positions of the point 'p' is equal to (in sq. units)
90. Order of differential equation, representing the family of circles having constant radius, is

FIITJEE

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PART TEST – IV

JEE (Main)-2024

TEST DATE: 15-09-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. C
Sol. In figure shown if any switch is closed base of transistor will become forward biased and output potential difference becomes zero so truth table is as shown below which is for

$$Y = \bar{A} \cdot \bar{B}$$

A	B	Y
0	0	1
1	0	0
0	1	0
1	1	0

2. B
Sol. N side is at higher potential than P side in unbiased PN diode
In a solar cell upon separation of e-h pair in depletion region, electron reach N side and hole reach P side, then P is at higher potential than N-side.

3. C
Sol. At saturation

$$V_{CE} = 0$$

$$\Rightarrow i_C = \frac{V_{CC}}{R_C} = \frac{10}{1000} = 0.01 \text{ A}$$

De current gain of transistor is

$$\beta = \frac{i_C}{i_B}$$

$$\Rightarrow i_B = \frac{i_C}{\beta} = \frac{0.01}{250} = 4 \times 10^{-5} \text{ A} = 40 \mu\text{A}$$

4. C

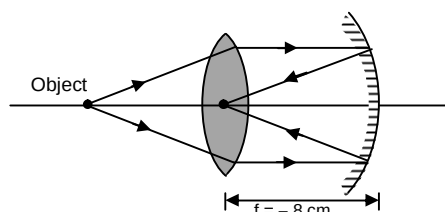
 Sol. $V \propto \frac{1}{n}$

5. C

Sol. Average life > Half life

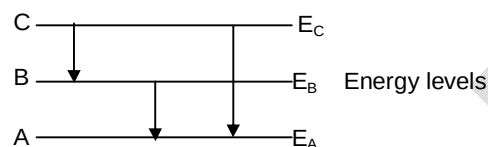
6. B

Sol. From the ray diagram,



7. B

Sol. Consider the energy levels of atom



From the figure

$$(E_B - E_A) + (E_C - E_B) = E_C - E_A$$

$$\Rightarrow \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2} = \frac{hc}{\lambda_3}$$

$$\Rightarrow \lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$$

8. A

 Sol. When a straight conductor moves with velocity v perpendicular to magnetic field then emf induced is given by $V = Bvl$. Here, it will be $V = Bv(l \sin \theta)$

 Potential difference is $V = Bv(l \sin \theta)$

$$V = 2 \times v \times \frac{30}{100} \times \frac{1}{2} = 0.3v$$

$$[\because \theta = 30^\circ]$$

$$\text{Now, current} = \frac{V}{R} \Rightarrow 2 = \frac{0.3v}{13\Omega}$$

$$\Rightarrow v = 86.66 \text{ m/s} = 87 \text{ m/s}$$

9. A

 Sol. Due to insertion of glass plate of refractive index μ of thickness t the optical path becomes μt instead of t . So, net increase in path travelled by wave is, $\mu t - t = (\mu - 1)t$.

 This increase would shift the fringe by a distance of $(\mu - 1)t \times \frac{D}{d}$

Now given shift is $(\mu - 1)t \frac{D}{d} = \frac{2\lambda D}{d}$

$$\Rightarrow (\mu - 1)t = 2\lambda$$

$$\Rightarrow t = \frac{2\lambda}{\mu - 1} = \frac{2\lambda}{1.5 - 1} = 4\lambda$$

10. B

Sol. As for thin prism, minimum deviation is $D_m = (\mu - 1)A$

So on increasing the wavelength, μ decreases & hence, D_m decreases.

11. A

Sol. The number of photons incident per second on a surface depends on the intensity of light as

$$N = \frac{IA}{h\nu}$$

Thus with increase in light intensity, number of photons increases but it does not change the kinetic energy of photoelectrons as frequency is not changing.

12. D

Sol. Resolving power $\propto \frac{1}{\lambda}$

As the wavelength of electron is much less than visible light, its resolving power will be much more.

13. D

Sol. By lens maker's formulas, focal lengths of the two lenses are

$$f_1 = \frac{R}{\mu_1 - 1} \text{ and } f_2 = -\frac{R}{\mu_2 - 1}$$

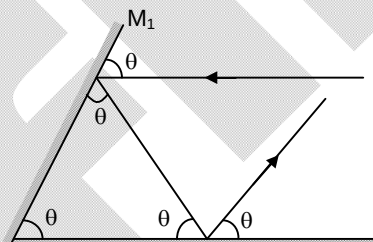
For combination, we use

$$\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{\mu_1 - 1}{R} - \frac{\mu_2 - 1}{R} = \frac{\mu_1 - \mu_2}{R}$$

$$\Rightarrow f_{eq} = \frac{R}{\mu_1 - \mu_2}$$

14. D

Sol.



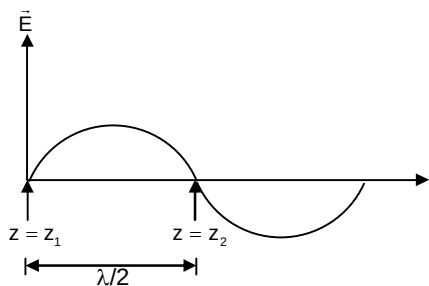
Assuming angles between two mirrors be θ as per geometry, a triangle sum of angles of is

$$3\theta = 180^\circ$$

$$\Rightarrow \theta = 60^\circ$$

15. C

Sol. As z_1 and z_2 are two neighboring points where $EF = 0$



As we have $|(z_2 - z_1)| = \frac{\lambda}{2}$

or $\lambda = c / \nu$

$$\Rightarrow |z_2 - z_1| = \frac{c}{2\nu}$$

$$\Rightarrow \nu = \frac{c}{2|z_2 - z_1|} = \frac{3 \times 10^8}{2|z_2 - z_1|}$$

$$\Rightarrow \nu = \frac{1.5 \times 10^8}{|z_2 - z_1|}$$

16.

Sol. C For the given AC current, RMS value can be calculated by taking square root of average of squares, given as

$$i = i_1 \sin \omega t + i_2 \sin(\omega t + 90)$$

$$i^2 = i_1^2 \sin^2 \omega t + i_2^2 \cos^2 \omega t + 2i_1 i_2 \sin \omega t \cos \omega t$$

$$\langle i^2 \rangle = i_1^2 \left(\frac{1}{2} \right) + i_2^2 \left(\frac{1}{2} \right) + 0$$

$$i_{\text{RMS}} = \sqrt{\langle i^2 \rangle} = \sqrt{\frac{i_1^2 + i_2^2}{2}}$$

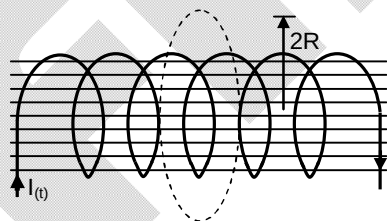
17.

Sol. B Current in solenoid

$$I_{(t)} = k t e^{-\alpha t}$$

Flux through solenoid is

$$\phi_{(t)} = B.A = (\mu_0 n I_{(t)}) \times \pi R^2$$



(same flux will pass through coil of radius 2R) emf induced in outer coils is

$$e = - \frac{d\phi_{(t)}}{dt}$$

$$\Rightarrow e = -\mu_0 n \pi R^2 k [e^{-\alpha t} + t(-\alpha e^{-\alpha t})]$$

$$\Rightarrow e = -C e^{-\alpha t} (1 - \alpha t)$$

Current induced in coil is

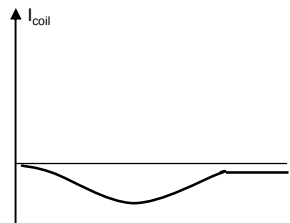
$$I_{\text{coil}} = \frac{e}{R_{\text{coil}}}$$

$$\Rightarrow I_{\text{coil}} = -\frac{C}{R} e^{-\alpha t} (1 - \alpha t)$$

$$\text{at } t = 0 \rightarrow I_{\text{coil}} \rightarrow 0$$

$$\text{at } t = \infty \rightarrow I_{\text{coil}} \rightarrow 0$$

$$I_{\text{coil}} \rightarrow -ve$$



18. D

Sol. As Magnet is approaches the loop, magnetic flux increases through the loop and rate of increment of flux also increases so emf increases with time. When magnet enters into the loop then at this moment emf is maximum and negative and when magnet is in middle of coil, at this moment emf is zero. Now again flux starts changing however polarity of emf get reversed. When magnet is coming out of the other side from the loop again at this moment emf is maximum and in opposite direction and as it gradually recedes away emf becomes zero. Hence option (D) is most appropriate for this situation.

19. D

Sol. Motional EMF induced across the plane wings is given as

$$e = Bv\ell \sin \theta$$

$$\Rightarrow e = (2.5 \times 10^{-4} \text{ T}) \left(180 \times \frac{5}{18} \text{ m/s} \right) (10 \text{ m}) \sin 60^\circ$$

$$\Rightarrow e = 108.25 \times 10^{-3} \text{ V}$$

20. B

Sol. Speed of EM wave in free space is

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

In a medium speed is given as

$$v = \frac{1}{\sqrt{k \epsilon_0 \mu_0}} \text{ [For transparent medium } \mu \approx \mu_0 \text{]}$$

$$\Rightarrow \frac{c}{v} = \sqrt{k} = n \text{ (refractive index of medium)}$$

EM wave intensity is given as

$$I = \frac{1}{2} \epsilon_0 E_0^2 c = \frac{1}{2} \epsilon_0 k E^2 v$$

$$\Rightarrow E_0^2 c = k E^2 v$$

$$\Rightarrow \frac{E_0^2}{E^2} = \frac{kv}{c} = \frac{n^2}{n}$$

$$\Rightarrow \frac{E_0}{E} = \sqrt{n}$$

For magnetic field we can use

$$\frac{B_0^2 c}{2\mu_0} = \frac{B^2 v}{2\mu_0}$$

$$\Rightarrow \frac{B_0}{B} = \frac{1}{\sqrt{n}}$$

SECTION – B

21. 4

Sol. Resistivity of a semiconductor is given as

$$\rho = \frac{1}{\sigma} = \frac{1}{ne\mu_e}$$

$$\Rightarrow \rho = \frac{1}{10^{19} \times 1.6 \times 10^{-19} \times 1.6}$$

$$\Rightarrow \rho = 0.4 \Omega \text{m}$$

22. 3

Sol. For first minima in diffraction pattern we use

$$b \sin \theta = n\lambda \quad \dots(1) \quad [n = 1]$$

The angular position of second minimum from screen centre is given as

$$\tan \theta_2 = \sin \theta_2 = \theta_2 = \frac{3 \text{cm}}{100 \text{cm}}$$

$$\Rightarrow \theta_2 = \frac{3 \text{ cm}}{1 \text{ m}} = 0.03$$

From equation (1), for 2nd minima, we use

$$b \sin \theta_2 = 2\lambda$$

$$\Rightarrow \theta_2 = \frac{2\lambda}{b} = 0.03$$

Angular width of central maxima in diffraction pattern is given as

$$2\theta_1 = \frac{2\lambda}{b} = 0.03$$

Width of central maxima is

$$w = 2\theta_1 D = 0.03 \times 1 \text{m} = 0.03 \text{m} = 3 \text{cm}$$

23. 9

Sol. The intensity of emerging light from analyzer after it is rotated by 30° is given as

$$I = \frac{I_0}{2} \cos^2 30^\circ$$

$$\Rightarrow I = 12 \left(\frac{\sqrt{3}}{2} \right)^2 = 12 \times \frac{3}{4} = 9$$

24. 5

Sol. Given focal lengths of objective and eyepiece are

$$f_o = 1 \text{ cm}$$

$$f_e = 5 \text{ cm}$$

$$\ell = 10 \text{ cm (Tube length)}$$

For final image at infinity (for least strain one eye), image produced by objective must be at focal point of eyepiece i.e., 5 cm from it. For objective we use

$$f_o = +1 \text{ cm}$$

$$V_o = +5 \text{ cm}$$

$$\Rightarrow u = \frac{V_o f_o}{V_o - f_o} = \frac{5 \times 1}{4} = 1.25 \text{ cm} = \frac{n}{4}$$

$$\Rightarrow n = 5$$

25. 4

Sol. By photoelectric effect equation, we have

$$KE = h\nu - \phi$$

$$\Rightarrow eV = \frac{hc}{\lambda} - \phi$$

For first case, we use

$$4.8 = \frac{hc}{\lambda_0} - \phi \quad \dots(1)$$

For second, case, we use

$$1.6 = \frac{hc}{2\lambda_0} \quad \dots(2)$$

From equation (1) and (2)

$$\phi = \frac{hc}{4\lambda_0}$$

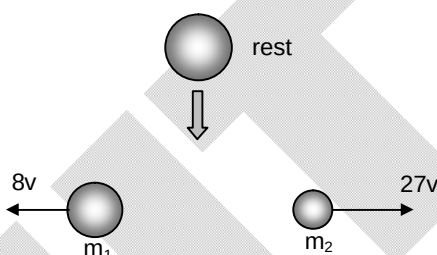
For threshold wavelength λ_{th} we have

$$\phi = \frac{hc}{\lambda_{th}}$$

$$\Rightarrow \frac{hc}{4\lambda_0} = \frac{hc}{\lambda_{th}}$$

$$\Rightarrow \lambda_{th} = 4\lambda_0$$

26. 6

Sol. Formula for fermi radius is $R = R_0 (A)^{1/3}$ 

By conservation of linear momentum we use

$$m_1(8v) = m_2(27v)$$

$$8m_1 = 27m_2$$

If R_1 and R_2 are the radii of nuclei of masses m_1 and m_2 , we have

$$\frac{R_1}{R_2} = \left(\frac{m_1}{m_2} \right)^{1/3}$$

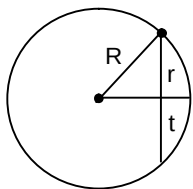
$$\Rightarrow \frac{R_1}{R_2} = \left(\frac{27}{8} \right)^{1/3} = \frac{3}{2}$$

27. 5

Sol. Using the construction diagram below for the lens, the radius of lens can be related with its thickness and radius of curvature as

$$R^2 = r^2 + (R - t)^2$$

$$\Rightarrow R^2 = r^2 + R^2 + t^2 - 2Rt \quad \dots(1)$$



For small thickness neglecting t^2 in equation-(1), we get

$$R = \frac{r^2}{2t}$$

Using lens maker's formula, we have

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R} - \frac{1}{\infty} \right) = \frac{\mu - 1}{R}$$

$$\Rightarrow f = \frac{R}{\mu - 1} = \frac{r^2}{2t(\mu - 1)}$$

$$\Rightarrow f = \frac{(3 \times 10^{-2})^2}{2 \times 3 \times 10^{-3} \times \left(\frac{3}{2} - 1 \right)}$$

$$\Rightarrow f = \frac{9 \times 10^{-4}}{6 \times 10^{-3} \times 1} \times 2 = 0.3 \text{ m} = 30 \text{ cm}$$

28.

5

Sol. Energy density in electromagnetic wave is given as

$$u_e = \frac{1}{2} \epsilon_0 E^2$$

Energy confined in the given volume can be calculated as

$$\text{Energy} = \frac{1}{2} \epsilon_0 E^2 \times \text{volume}$$

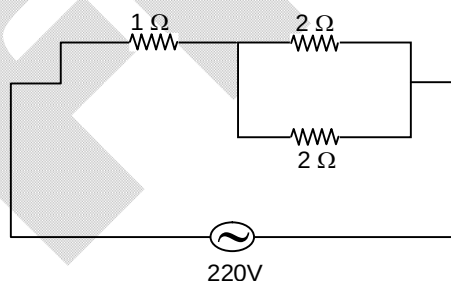
$$\Rightarrow 5.5 \times 10^{-12} = \frac{1}{2} \times 8.8 \times 10^{-12} \times 50^2 \times V$$

$$\Rightarrow V = 500 \text{ cm}^3$$

29.

2

Sol. At very high frequencies i.e. at $\omega \rightarrow \infty$, capacitive reactance can be taken as zero and inductive reactance can be considered as open circuit so the effective circuit can be drawn as shown in figure below



Thus equivalent resistance in this case can be calculated as

$$Z = 1 + \frac{2 \times 2}{2 + 2} = 2 \Omega$$

30. 3

Sol. Binding energy of Al nucleus is given as

$$BE = \Delta m C^2$$

Mass defect is calculated as

$$\Delta m = [13 \times 1.0026 + 14 \times 1.00866 - 27.18846]$$

$$\Delta m = 0.02716 = 27.16 \times 10^{-3} \text{ u} \approx 27 \text{ u}$$

$$\Rightarrow BE = 27 \times 10^{-3} \text{ J}$$

Chemistry**PART – B****SECTION – A**

31. A

Sol. $1.24 = i \times 34.3 \times \left[\frac{0.849}{\frac{235.5}{50}} \right] \times 1000$

$$i = 0.5$$

Hence, dimerization occurs to form Hg_2Cl_2 .

32. D

Sol. **1st method :** when half the solution has vapourised, A being less volatile will have more than 10 moles in solution. Only (D) option satisfies.

2nd method : $P = \sqrt{P_A^0 \cdot P_B^0} \Rightarrow 110 \text{ torr}$

$$110 = \frac{x}{20} \times 100 + \frac{(20-x)}{20} \times 121$$

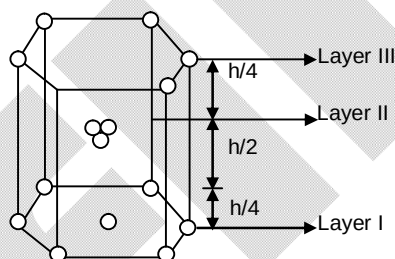
$$\Rightarrow x = \frac{220}{21}$$

33. D

Sol. Height of crystal lattice is h so effective height of one layer be $\frac{h}{2}$. Octahedral voids is exactly

between two layers so its height from base is $\left(\frac{\frac{h}{2}}{2} \right) = \frac{h}{4}$ and from symmetry, we can say height of

second octahedral voids will be $\frac{h}{4} + \frac{h}{2} = \frac{3h}{4}$.



34. A

Sol. $r_c + r_a = \frac{a}{2} = \frac{5.14}{2}$

For NaCl type structure,

$$\frac{r_c}{r_a} = 0.414, r_c = 0.414r_a$$

$$\text{So, } 0.414r_a + r_a = \frac{a}{2}$$

$$1.414r_a = 2.57 \text{ \AA}$$

$$r_a = \frac{2.57}{1.414} = 1.815 \text{ \AA}$$

35. D
Sol. Theoretical.

36. A
Sol. $\text{Fe} + 2\text{H}^+ \rightarrow \text{Fe}^{2+} + \text{H}_2$
 $E = +0.458 \text{ V}$
 $\Rightarrow 0.458 = +0.440 - \frac{0.059}{2} \log[\text{Fe}^{2+}]$
 $\Rightarrow [\text{Fe}^{2+}] = 0.245 \text{ M}$

37. A
Sol. Anode: $\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$
 Cathode : $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$

38. B
Sol. $\Delta S = nF \left(\frac{dE_{\text{cell}}}{dT} \right) = 2 \times 96500 \times 3.4 \times 10^{-4} = 65.62$
 $\text{H}_2 + \text{Hg}_2\text{Cl}_2 \rightarrow 2\text{H}^+ + 2\text{Cl}^- + 2\text{Hg}; (n = 2)$

39. A
Sol. Theoretical.

40. B
Sol. Protective power $\propto \frac{1}{\text{Gold number}}$

41. A
Sol. Theoretical.

42. D
Sol. $\left[\text{Cu}(\text{NH}_3)_4 \right]^{+2} \text{SO}_4 \xrightarrow{\text{H}_2\text{S}} \text{CuS} \downarrow (\text{Black})$
 $\Rightarrow \text{dsp}^2$, square planar (one unpaired electron)

43. C
Sol. Theoretical.

44. D
Sol. Theoretical.

45. B
Sol. $\text{Na}_2\text{Cr}_2\text{O}_7$ is deliquescent, so cannot be accurately weighed.
46. B
Sol. $2\text{MnO}_2 + 4\text{KOH} + \text{O}_2 \rightarrow 2\text{K}_2\text{MnO}_4 + 2\text{H}_2\text{O}$
47. C
Sol. All but one (Lu^{3+}), lanthanide ions show absorptions in the visible or near UV region of the spectrum.
48. B
Sol. Theoretical.
49. B
Sol. Theoretical.
50. B
Sol. Theoretical.

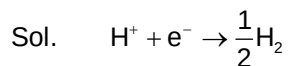
SECTION – B

51. 5
Sol. Non-metals undergo disproportionation in alkaline medium. $[\text{NO}_2, \text{Cl}_2\text{O}_6, \text{Br}_2, \text{P}_4, \text{S}_8]$
52. 6
Sol. (a), (b), (c), (d), (f), (h)
53. 6
Sol. In this reaction, CrO_2Cl_2 is formed in which Cr is +6.
54. 2
Sol. $\text{V}^{+4}, \text{Cu}^{+2}$
55. 4
Sol. Magnetite, Siderite, Chromite, Wolframite, $\left(\overset{+2}{\text{Fe}} \overset{+6}{\text{WO}_4} \right)$
56. 3
Sol. Pb, Cu, Hg
57. 3
Sol. $\Delta T_b = i \times K_b \times m$
 $0.14 = i \times 0.5 \times 0.25$
 $i = 1.12$
 $\alpha = 0.12$
 $[\text{H}^+] = 0.25 \times 0.12$
 Ans. $= 100 \times 0.25 \times 0.12 = 3$
58. 1
Sol. Number of equivalent of $\text{H}^+ = \frac{9.65 \times 100}{96500} = 0.01$

$$[H^+] = \frac{0.01}{0.1} = 0.1$$

$$\text{pH} = 1$$

59. 7



$$E = +\frac{0.059}{1} \log[H^+]$$

$$-0.413 = -0.059 \text{ pH}$$

$$\Rightarrow \text{pH} = 7$$

60. 4

Sol. $x = 8; y = 6; z = 12$

Mathematics

PART – C

SECTION – A

61. B

Sol. Applying $C_1 \rightarrow \frac{1}{a}C_1, C_2 \rightarrow \frac{1}{b}C_2, C_3 \rightarrow \frac{1}{c}C_3$, we get

$$\Delta = \begin{vmatrix} 1+\frac{1}{a} & \frac{1}{b} & \frac{1}{c} \\ \frac{1}{a} & 1+\frac{1}{b} & \frac{1}{c} \\ \frac{1}{a} & \frac{1}{b} & 1+\frac{1}{c} \end{vmatrix} abc$$

Applying $C_1 \rightarrow C_1 + C_2 + C_3$ we get,

$$\Delta = \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \begin{vmatrix} 1 & \frac{1}{b} & \frac{1}{c} \\ 1 & 1+\frac{1}{b} & \frac{1}{c} \\ 1 & \frac{1}{b} & 1+\frac{1}{c} \end{vmatrix} (abc)$$

Applying $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$

$$\Delta = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \begin{vmatrix} 1 & \frac{1}{b} & \frac{1}{c} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= abc \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + 1\right)$$

Now, $\Delta = 0$

$$\Rightarrow \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = -1$$

62. D

Sol. Applying $R_1 \rightarrow R_1 - (R_2 - R_3)$, we get

$$\Delta = \begin{vmatrix} 0 & i + \omega^2 + \omega & 0 \\ 1-i & -1 & \omega^2 - 1 \\ -1 & -1 + \omega - 1 & -1 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & 0 & 0 \\ 1-i & -1 & \omega^2 - 1 \\ -1 & -1 + \omega - 1 & -1 \end{vmatrix}$$

$$= 0$$

63. C

Sol. $R_1 \rightarrow R_1 - R_2, R_2 \rightarrow R_2 - R_3$, reduces the determinant to

$$\begin{vmatrix} x^2-1 & x-1 & 0 \\ 2x-2 & x-1 & 0 \\ 3 & 3 & 1 \end{vmatrix}$$

$$= (x^2-1)(x-1) - 2(x-1)^2$$

$$= (x^2-1)(x+1-2)$$

$$= (x-1)^3$$

This is clearly negative for $x < 1$

64. B

Sol. $\vec{a} \cdot \vec{b} = (x\hat{i} + (x-1)\hat{j} + \hat{k}) \cdot ((x+1)\hat{i} + \hat{j} + a\hat{k})$

$$= x(x+1) + x-1 + a$$

$$= x^2 + 2x + a - 1$$

$$\vec{a} \cdot \vec{b} > 0 \forall x \in \mathbb{R}$$

$$\Rightarrow x^2 + 2x + a - 1 > 0 \forall x \in \mathbb{R}$$

$$\Rightarrow 4 - 4(a-1) < 0$$

$$\Rightarrow a > 2$$

65. C

Sol. $\hat{i} \times (\vec{a} \times \hat{i}) = (\hat{i} \cdot \hat{i})\vec{a} - (\vec{a} \cdot \hat{i})\hat{i} = \vec{a} - (\vec{a} \cdot \hat{i})\hat{i}$

Similarly, $\hat{j} \times (\vec{a} \times \hat{j}) = \vec{a} - (\vec{a} \cdot \hat{j})\hat{j}$

And $\hat{k} \times (\vec{a} \times \hat{k}) = \vec{a} - (\vec{a} \cdot \hat{k})\hat{k}$

$$\Rightarrow \hat{i} \times (\vec{a} \times \hat{i}) + \hat{j} \times (\vec{a} \times \hat{j}) + \hat{k} \times (\vec{a} \times \hat{k})$$

$$= 3\vec{a} - ((\vec{a} \cdot \hat{i})\hat{i} + (\vec{a} \cdot \hat{j})\hat{j} + (\vec{a} \cdot \hat{k})\hat{k})$$

$$= 2\vec{a}$$

66. A

Sol. Let $Q(\vec{q})$ be the foot of altitude drawn

From $P(\vec{p})$ to the line $\vec{r} = \vec{a} + \lambda\vec{b}$,

$$\Rightarrow (\vec{q} - \vec{p}) \cdot \vec{b} = 0 \text{ and } \vec{q} = \vec{a} + \lambda\vec{b}$$

$$\Rightarrow (\vec{a} + \lambda\vec{b} - \vec{p}) \cdot \vec{b} = 0$$

$$\Rightarrow (\vec{a} - \vec{p}) \cdot \vec{b} + \lambda|\vec{b}|^2 = 0$$

$$\Rightarrow \lambda = \frac{(\vec{p} - \vec{a}) \cdot \vec{b}}{|\vec{b}|^2}$$

$$\Rightarrow \vec{q} - \vec{p} = \vec{a} + \frac{((\vec{p} - \vec{a}) \cdot \vec{b})\vec{b}}{|\vec{b}|^2} - \vec{p}$$

$$\Rightarrow |\vec{q} - \vec{p}| = \left| (\vec{a} - \vec{p}) + \frac{((\vec{p} - \vec{a}) \cdot \vec{b}) \vec{b}}{|\vec{b}|^2} \right|$$

67. D

 Sol. $\vec{a} = \vec{b} + \vec{c}, \vec{b} \times \vec{d} = \vec{0}, \vec{c} \cdot \vec{d} = 0$

$$\Rightarrow \vec{a} \times \vec{d} = \vec{b} \times \vec{d} + \vec{c} \times \vec{d} = \vec{c} \times \vec{d}$$

$$\Rightarrow \vec{d} \times (\vec{a} \times \vec{d}) = \vec{d} \times (\vec{c} \times \vec{d})$$

$$= (\vec{d} \cdot \vec{d}) \vec{c} - (\vec{c} \cdot \vec{d}) \vec{d} = |\vec{d}|^2 \vec{c}$$

$$\Rightarrow \frac{\vec{d} \times (\vec{a} \times \vec{d})}{|\vec{d}|^2} = \vec{c}$$

68. D

 Sol. $\vec{a} \times \vec{c} = \vec{b}$

$$\Rightarrow \vec{a} \times (\vec{a} \times \vec{c}) = \vec{a} \times \vec{b}$$

$$\Rightarrow (\vec{a} \cdot \vec{c}) \vec{a} - (\vec{a} \cdot \vec{a}) \vec{c} = \vec{a} \times \vec{b}$$

$$\vec{a} \cdot \vec{a} = |\hat{i} + \hat{j} + \hat{k}|^2 = 3,$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{vmatrix} = -2\hat{i} + \hat{j} + \hat{k}$$

$$\Rightarrow 3\vec{a} - 3\vec{c} = \vec{a} \times \vec{b} = -2\hat{i} + \hat{j} + \hat{k}$$

$$\Rightarrow 3\vec{c} = 3\hat{i} + 3\hat{j} + 3\hat{k} + 2\hat{i} - \hat{j} - \hat{k}$$

$$= 5\hat{i} + 2\hat{j} + 2\hat{k}$$

$$\Rightarrow \vec{c} = \frac{1}{3}(5\hat{i} + 2\hat{j} + 2\hat{k})$$

69. A

$$\begin{aligned} \text{Sol. } & |\vec{a} + \vec{b} + (\vec{a} \times \vec{b})|^2 \\ &= (\vec{a} + \vec{b} + (\vec{a} \times \vec{b})) \cdot (\vec{a} + \vec{b} + (\vec{a} \times \vec{b})) \\ &= \vec{a} \cdot \vec{a} + 2\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{b} + |\vec{a} \times \vec{b}|^2 \\ \Rightarrow & |\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2 + |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 \\ \text{Now, } & (1 - \vec{a} \cdot \vec{b})^2 = 1 + (\vec{a} \cdot \vec{b})^2 - 2(\vec{a} \cdot \vec{b}) \\ \Rightarrow & |\vec{a} + \vec{b} + (\vec{a} \times \vec{b})|^2 + (1 - \vec{a} \cdot \vec{b})^2 \\ &= 1 + |\vec{a}|^2 + |\vec{b}|^2 + |\vec{a}|^2 |\vec{b}|^2 \\ &= (1 + |\vec{a}|^2)(1 + |\vec{b}|^2) \end{aligned}$$

70. A

Sol. $P(A) = \frac{1}{4}, P(B) = \frac{1}{2}, P(A \cap B) = \frac{7}{50}$

$$\begin{aligned} \text{Now, } P(\bar{A} \cap \bar{B}) &= P(\bar{A}) + P(\bar{B}) - P(\bar{A} \cup \bar{B}) \\ &= 1 - P(A) + 1 - P(B) - (1 - P(A \cap B)) \\ &= 1 - P(A) - P(B) + P(A \cap B) \\ &= 1 - \frac{1}{4} - \frac{1}{2} + \frac{7}{50} \\ &= \frac{39}{100} \end{aligned}$$

71. A

Sol. Probability of winning the prize in single game

$$= \frac{25}{25^2} = \frac{1}{25}$$

In this case first 4 games, must result in exactly two prizes and 5th game must result in prize.
Thus, required probability

$$\begin{aligned} &= {}^4C_2 \left(\frac{1}{25}\right)^2 \cdot \left(\frac{24}{25}\right)^2 \cdot \frac{1}{25} \\ &= \frac{6(24)^2}{(25)^5} \end{aligned}$$

72. A

Sol. E_1 : Specific home is locked. E_2 : Specific home is unlocked.

A: Real estate man get in to the home.

$$P(E_2) = \frac{40}{100} = \frac{2}{5}, P(E_1) = \frac{3}{5}$$

$$P(A) = P(E_1) \cdot P\left(\frac{A}{E_1}\right) + P(E_2) \cdot P\left(\frac{A}{E_2}\right)$$

$$= \frac{3}{5} \cdot \frac{{}^7C_2}{{}^8C_3} + \frac{2}{5} \cdot 1$$

$$= \frac{25}{40} = \frac{5}{8}$$

73. B

Sol. $AB = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} -1 & 4 \\ 2 & 3 \end{bmatrix}$

$$= \begin{bmatrix} 1 \cdot -1 - 2 \cdot 2 & 1 \cdot 4 - 2 \cdot 3 \\ 3 \cdot -1 + 0 \cdot 2 & 3 \cdot 4 + 0 \cdot 3 \end{bmatrix}$$

$$= \begin{bmatrix} -5 & -2 \\ -3 & 12 \end{bmatrix}$$

$$\Rightarrow \text{adj.}(AB) = \begin{bmatrix} 12 & 3 \\ 3 & -5 \end{bmatrix}^T$$

$$= \begin{bmatrix} 12 & 2 \\ 3 & -5 \end{bmatrix}$$

$$\text{Also, } |AB| = -60 - 6 = -66$$

$$\Rightarrow (AB)^{-1} = \frac{\text{adj}(AB)}{|AB|}$$

$$= -\frac{1}{66} \begin{bmatrix} 12 & 2 \\ 3 & -5 \end{bmatrix}$$

$$\Rightarrow C = (AB)^{-1} \begin{bmatrix} 4 & 8 \\ 3 & 7 \end{bmatrix}$$

$$= -\frac{1}{66} \begin{bmatrix} 12 & 2 \\ 3 & -5 \end{bmatrix} \begin{bmatrix} 4 & 8 \\ 3 & 7 \end{bmatrix}$$

$$= -\frac{1}{66} \begin{bmatrix} 48+6 & 96+14 \\ 12-15 & 24-35 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{9}{11} & -\frac{5}{3} \\ \frac{1}{22} & \frac{1}{6} \end{bmatrix}$$

74. B

Sol. Clearly $x \leq 1$ and $x - x^2 \geq 0 \Rightarrow x \in [0, 1]$. Curve has two branches whose equations are

$y = \sin^{-1} x \pm \sqrt{x - x^2}$, and it is defined only in between $[0, 1]$.

Area enclosed by the curve,

$$\Delta = \int_0^1 (y_1 - y_2) dx$$

$$= 2 \int_0^1 \sqrt{x - x^2} dx$$

$$2 \int_0^1 \sqrt{\frac{1}{4} - \left(x - \frac{1}{2}\right)^2} dx$$

$$= 2 \left(\frac{2x-1}{4} \sqrt{x-x^2} \Big|_0^1 + \frac{1}{8} \sin^{-1}(2x-1) \Big|_0^1 \right)$$

$$= \frac{\pi}{4} \text{ sq. units}$$

75. A

Sol. Equation of tangent at $P(x, f(x))$ is

$$y - f(x) = f'(x)(X - x)$$

$$\Rightarrow A \equiv \left(x - \frac{f(x)}{f'(x)}, 0 \right), B \equiv (0, f(x)f'(x))$$

Now, since $PA : PB = 2 : 1$

$$\Rightarrow 3x = x - \frac{f(x)}{f'(x)}$$

$$\Rightarrow \frac{f'(x)}{f(x)} + \frac{1}{2x} = 0$$

$$\Rightarrow 2 \frac{d'f(x)}{f(x)} + \frac{dx}{x} = 0$$

$$\Rightarrow 2 \cdot \ln|f(x)| + \ln|x| = c'$$

$$\Rightarrow xy^2 = c$$

76. C

Sol. $\frac{xdx + ydy}{ydx - xdy} = \frac{x \cdot \cos^2(x^2 + y^2)}{y^3}$

$$\Rightarrow \sec^2(x^2 + y^2) \cdot (xdx + ydy) = \frac{x}{y} \cdot d\left(\frac{x}{y}\right)$$

$$\Rightarrow \frac{1}{2} \tan(x^2 + y^2) = \frac{x^2}{2y^2} + c$$

77. B

Sol. $(xy \sin xy + \cos xy)ydx + (xy \sin xy - \cos xy)x dy = 0$

$$\Rightarrow xy \cdot \sin xy (ydx + xdy) + \cos xy (ydx - xdy) = 0$$

$$\Rightarrow \tan xy \cdot d(xy) + \frac{dx}{x} - \frac{dy}{y} = 0$$

$$\Rightarrow \ln|\sec xy| + \ln\left|\frac{x}{y}\right| = c'$$

$$\Rightarrow x \sec(xy) = cy$$

78. D

Sol. $y^2 = ax$

$$\Rightarrow 2y \cdot \frac{dy}{dx} = a$$

$$\Rightarrow y = 2x \frac{dy}{dx}$$

Thus D.E. of orthogonal trajectory is

$$y = -2x \cdot \frac{dy}{dx}$$

$$\Rightarrow ydy + 2xdx = 0$$

$$\Rightarrow \frac{y^2}{2} + x^2 = c'$$

$$\Rightarrow y^2 + 2x^2 = c$$

79. D

Sol. Let the required plane be

$$ax + by + cz = 0$$

We have

$$3a - 5b + 2c = 0, a + 2b - 3c = 0$$

$$\Rightarrow \frac{a}{15-4} = \frac{b}{2+9} = \frac{c}{6+5}$$

$$\Rightarrow a : b : c = 11 : 11 : 11$$

Thus plane is

$$x + y + z = 0$$

80. A

Sol. Any point on the line is

$$P \equiv (6r_1 + 2, 3r_1 + 3, -4r_1 - 4)$$

Direction ratio of the line segment PQ,

Where $Q \equiv (-1, 2, 6)$, are

$$6r_1 + 3, 3r_1 + 1, -4r_1 - 10$$

If 'P' be the foot of altitude drawn from Q to the given line, then

$$6(6r_1 + 3) + 3(3r_1 + 1) + 4(4r_1 + 10) = 0$$

$$\Rightarrow r_1 = -1.$$

$$\text{Thus, } P \equiv (-4, 0, 0)$$

$$\therefore \text{ Required distance} = \sqrt{9 + 4 + 36}$$

$$= 7 \text{ units.}$$

SECTION - B

81. 2

Sol. For no trivial solution, we must have
$$\begin{vmatrix} 2a & -2 & 3 \\ 1 & a & 2 \\ 2 & 0 & a \end{vmatrix} = 0$$

$$\Rightarrow 2a(a^2 - 0) + 2(a - 4) + 3(0 - 2a) = 0$$

$$\Rightarrow 2a^3 + 2a - 8 + 0 - 6a = 0$$

$$\Rightarrow 2a^3 - 4a - 8 = 0$$

$$\Rightarrow a^3 - 2a - 4 = 0$$

$$\Rightarrow a^3 - 2a^2 + 2a^2 - 4a + 2a - 4 = 0$$

$$\Rightarrow a^2(a - 2) + 2a(a - 2) + 2(a - 2) = 0$$

$$\Rightarrow (a - 2)(a^2 + 2a + 2) = 0$$

$$\Rightarrow a = 2$$

82. 2

Sol. $R_1 \rightarrow R_1 - R_2, R_2 \rightarrow R_2 - R_3$ reduces the determinant to,

$$\begin{vmatrix} p-a & p-q & 0 \\ 0 & q-b & c-r \\ a & b & r \end{vmatrix} = 0$$

$$\Rightarrow (p-a)((q-b)r + (r-c)b) + (q-b)((r-c)a) = 0$$

$$\Rightarrow (p-a)((q-b)(r-c+c) - (r-c)(q-b-q)) - (q-b)(r-c)(p-a-p) = 0$$

$$\Rightarrow (p-a)(q-b)(r-c) + c(p-a)(q-b) - (p-a)(q-b)(r-c) + q(p-a)(r-c)$$

$$- (r-c) - (p-a)(q-b)(r-c) + p(q-b)(r-c) = 0$$

Dividing out by $(p-a)(q-b)(r-c)$ we get

$$\begin{aligned}\frac{c}{r-c} + \frac{q}{q-b} + \frac{p}{p-a} &= 1 \\ \Rightarrow 1 + \frac{c}{r-c} + \frac{q}{q-b} + \frac{p}{p-a} &= 2 \\ \Rightarrow \frac{c}{r-c} + \frac{q}{q-b} + \frac{p}{p-a} &= 2\end{aligned}$$

83. 4

Sol. Let P.V. of A, B, C and D be $\vec{a}, \vec{b}, \vec{c}$ and $\vec{0}$.

$$\Rightarrow \overrightarrow{AB} \times \overrightarrow{CD} = (\vec{b} - \vec{a}) \times -\vec{c}$$

$$\overrightarrow{BC} \times \overrightarrow{AD} = (\vec{c} - \vec{b}) \times -\vec{a}$$

$$\text{And } \overrightarrow{CA} \times \overrightarrow{BD} = (\vec{a} - \vec{c}) \times -\vec{b}$$

$$\begin{aligned}\Rightarrow \overrightarrow{AB} \times \overrightarrow{CD} + \overrightarrow{BC} \times \overrightarrow{AD} + \overrightarrow{CA} \times \overrightarrow{BD} \\ = \vec{c} \times \vec{b} + \vec{a} \times \vec{c} + \vec{a} \times \vec{c} + \vec{b} \times \vec{a} - \vec{a} \times \vec{b} + \vec{c} \times \vec{b} \\ = 2(\vec{c} \times \vec{b} + \vec{b} \times \vec{a} + \vec{a} \times \vec{c}) \\ = 2(\vec{c} \times (\vec{b} - \vec{a}) - \vec{a} \times (\vec{b} - \vec{a})) \\ = 2((\vec{c} - \vec{a}) \times (\vec{b} - \vec{a})) \\ = 2(\overrightarrow{AC} \times \overrightarrow{AB}) \\ \Rightarrow |\overrightarrow{AB} \times \overrightarrow{CD} + \overrightarrow{BC} \times \overrightarrow{AD} + \overrightarrow{CA} \times \overrightarrow{BD}| \\ = 4 \left| \frac{1}{2} (\overrightarrow{AC} \times \overrightarrow{AB}) \right| \\ = 4\Delta_{ABC}\end{aligned}$$

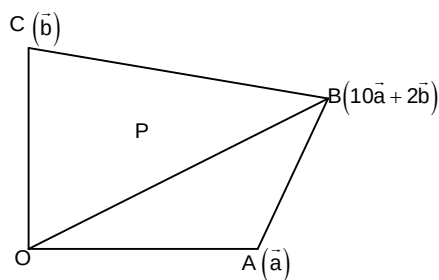
84. 0

$$\begin{aligned}\text{Sol. } (\vec{b} \times \vec{c}) \cdot (\vec{a} \times \vec{d}) &= (\vec{b} \cdot \vec{a})(\vec{c} \cdot \vec{d}) - (\vec{b} \cdot \vec{d})(\vec{c} \cdot \vec{a}) \\ (\vec{c} \times \vec{a}) \cdot (\vec{b} \times \vec{d}) &= (\vec{c} \cdot \vec{b})(\vec{a} \cdot \vec{d}) - (\vec{c} \cdot \vec{d})(\vec{a} \cdot \vec{b}) \\ (\vec{a} \cdot \vec{b}) \cdot (\vec{c} \times \vec{d}) &= (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{d}) - (\vec{a} \cdot \vec{d})(\vec{b} \cdot \vec{c}) \\ \Rightarrow (\vec{b} \times \vec{c}) \cdot (\vec{a} \times \vec{d}) + (\vec{c} \times \vec{a}) \cdot (\vec{b} \times \vec{d}) + (\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) &= 0\end{aligned}$$

85. 6

$$\text{Sol. } q = |\vec{a} \times \vec{b}|$$

$$\begin{aligned}p &= \frac{1}{2} (|\vec{a} \times (10\vec{a} + 2\vec{b})| + |\vec{b} \times (10\vec{a} + 2\vec{b})|) \\ &= \frac{1}{2} (|2\vec{a} \times \vec{b}| + |10\vec{b} \times \vec{a}|) \\ &= 6|\vec{a} \times \vec{b}| \\ \Rightarrow p &= 6q \\ \Rightarrow \lambda &= 6\end{aligned}$$



86. 3

Sol. $\left| (\vec{a} \times \vec{b}) \times \vec{c} \right| = |\vec{a} \times \vec{b}| |\vec{c}| \sin \frac{\pi}{6}$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 1 & 1 & 0 \end{vmatrix} = 2\hat{i} - 2\hat{j} + \hat{k}$$

$$\Rightarrow |\vec{a} \times \vec{b}| = 3$$

$$\text{Since } |\vec{c} - \vec{a}|^2 = 3$$

$$\Rightarrow |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{a} \cdot \vec{c} = 8$$

$$\Rightarrow |\vec{c}|^2 + 9 - 2|\vec{c}| = 8$$

$$\Rightarrow |\vec{c}|^2 - 2|\vec{c}| + 1 = 0$$

$$\Rightarrow |\vec{c}| = 1$$

$$\Rightarrow \left| (\vec{a} \times \vec{b}) \times \vec{c} \right| = 3 \cdot 1 \cdot \frac{1}{2} = \frac{3}{2}$$

$$\Rightarrow 2 \left| (\vec{a} \times \vec{b}) \times \vec{c} \right| = 3$$

87. 7

Sol. Let the constructed set be $\{n, n+1, n+2, \dots, n+14\}$.

If 'n' is even, then

Total number of odd numbers in the set = 7.

If 'n' is odd, then total number of odd

Numbers in the set = 8.

Thus, required probability = p

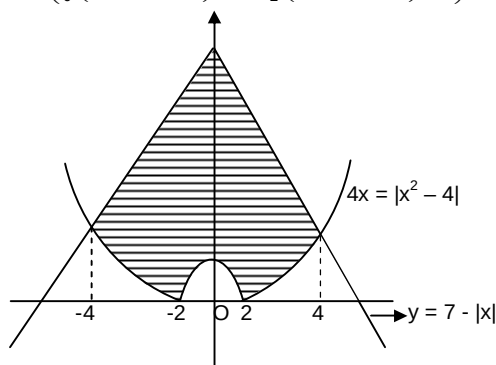
$$= \frac{1}{2} \left(\frac{{}^7C_2}{{}^{15}C_2} + \frac{{}^8C_2}{{}^{15}C_2} \right) = \frac{7}{30} \Rightarrow 30p = 7$$

88. 32

Sol. Required area

$$= 2 \left(\int_0^2 \left(7 - x - \left(\frac{4 - x^2}{4} \right) \right) dx + \int_2^4 \left(7 - x - \left(\frac{x^2 - 4}{4} \right) \right) dx \right)$$

$$= 2 \left(\int_0^2 \left(6 - x - \frac{x^2}{4} \right) dx + \int_2^4 \left(8 - x - \frac{x^2}{4} \right) dx \right)$$



$$= 2 \left(6x - \frac{x^2}{2} + \frac{x^3}{12} \right)_0^2 + 2 \left(8x - \frac{x^2}{2} - \frac{x^3}{12} \right)_2^4$$

$$= 32 \text{ sq. units} \Rightarrow k = 32.$$

89. 8

Sol. Clearly, $\lceil |x| \rceil = 1, \lceil |y| \rceil = 0$

$$\Rightarrow 1 \leq |x| < 2, 0 \leq |y| < 1$$

$$\Rightarrow x \in (-2, -1] \cup [1, 2), y \in (-1, 1)$$

$$\text{Or } \lceil |x| \rceil = 0, \lceil |y| \rceil = 1$$

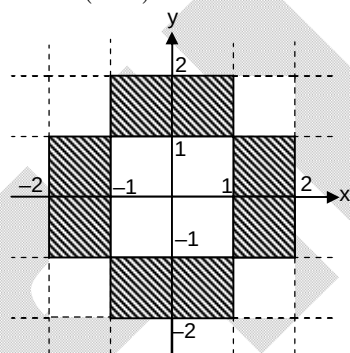
$$\Rightarrow 0 \leq x < 1$$

$$\Rightarrow x \in (-1, 1), y \in (-2, -1] \cup [1, 2)$$

$$\text{Or } 0 \leq x < 1$$

$$\Rightarrow 0 \leq x < 1$$

$$\Rightarrow x \in (-1, 1), y \in (-2, -1] \cup [1, 2)$$



$$\text{Area of required region}$$

$$= 4(2-1)(1-(-1))$$

$$= 8 \text{ sq. units}$$

90. 2

Sol. If center be (h, k) and radius be 'r' then its equation is,

$$(x-h)^2 + (y-k)^2 = r^2.$$

Since it has two parameters, thus order of corresponding D.E. will be 2.